13
Dental Biomaterials

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This chapter discusses dental biomaterials with the practicing dentist and dental student in mind. As such, it is not meant to be an in-depth review of materials that would be of interest to material scientists. The chapter discusses dental biomaterials from the perspective of the clinician trying to make a decision about, for example, what composite resin material, dental adhesive system, or impression material is best for his or her practice in order to provide optimum care for patients.

These decisions are often important because, in addition to clinical time, materials themselves are very expensive and the cost of premature failure of a restoration and subsequent replacement is considerable. Most materials are brought to the marketplace with little or no clinical testing. Practicing dentists are basically doing the clinical testing for manufacturers. Understanding the basic nature and composition of dental materials can allow clinicians to make educated decisions related to new materials and avoid catastrophic outcomes (Fig. 13.1).

Contemporary dental practitioners are expected to practice "evidence-based" dentistry.[1-4] The evidence-based diagram (Fig. 13.2) shows the top of the hierarchy is filtered information from meta-analyses and systematic reviews of published randomized controlled trials (RCTs). However, the reality is that it takes at least 10 years from the introduction of a new material before 5-year RCTs are published. Most currently published (2017) systematic reviews are of limited value to the clinician, because the clinical trials included in the reviews are usually very weak and poorly designed.[5] Thus in the absence of good "evidence," it is essential that dentists understand the composition of commonly used dental materials and the functions of each component of those materials.

Dental materials have a number of important physical and mechanical properties that will be described in the next section. One of the most important properties is "technique sensitivity." A material is described as technique sensitive when different operators get different outcomes when using that material. Materials that are low in technique sensitivity are desirable, and manufacturers strive to develop materials that are not technique sensitive. A good example of this is seen with dental adhesives. Dental adhesives have evolved over many years and as many as seven generations have been described. For many years, the fourth generation of dental adhesives was identified as the best system, but dentists have had variable results with it, as it required three separate steps (etch, prime, bond) and involved the vague concept of "moist" dentin. These materials were indeed "technique sensitive." Subsequent generations attempted to simplify the bonding procedure and make it less technique sensitive, but in fact made it less effective. Dentists who understood the fourth generation of bonding agents and who were meticulous in their technique achieved excellent clinical results. Those who did not understand these materials and/or were less meticulous in their technique got less successful outcomes.

One material that is very low in technique sensitivity is silver amalgam (Fig. 13.3). A short time after placement of a silver amalgam restoration, oral fluids penetrate the microscopic space between the amalgam and tooth structure.[6] This process, called percolation, results in the deposition of corrosion products and effectively seals the interface between the amalgam and the tooth. This process of "self-sealing" occurs irrespective of the skill and knowledge of the operator, which makes silver amalgam one of the least technique sensitive materials.

The discussion on technique sensitivity leads to the concept that there are two parts to the successful use of dental materials: material selection and material manipulation. Dentists must select the most appropriate restorative material based on knowledge of the disease process, knowledge of materials appropriate for dealing with the dental defect, and knowledge of expected outcomes of the proposed intervention. Appropriate material manipulation of the selected material requires knowledge about the material and meticulous attention to details such as isolation, bonding technique, incremental buildup, curing time, matrix selection, and many other variables. Thus, both material selection and proper manipulation of the selected material are critical to clinical success.

Dentists have variable approaches to their adoption of new materials and procedures. Some are very conservative and patiently wait for high quality of evidence prior to using a new material or technique. Others, described as "early adopters," eagerly utilize every new material or technique the instant it comes on market, and others are between the two extremes. There is room in the profession for all three groups, but it is clear that early adopters must have an excellent knowledge and understanding of both the selection and the manipulation of dental materials.

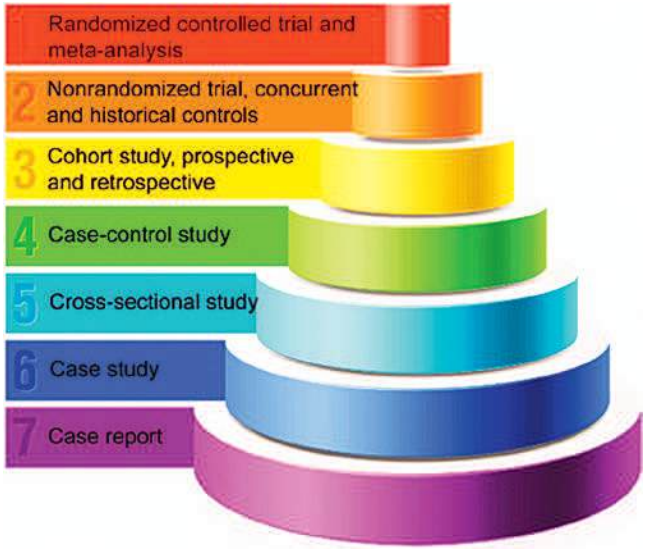
THE BOTTOM LINE

1. Understanding the composition and nature of dental materials is important to obtain optimum clinical outcomes.
2. Materials are brought to the market with almost no clinical testing.

[a]Drs. Bayne and Thompson were inactive authors in this edition.



• **Fig. 13.1** Fractured porcelain veneer.



• **Fig. 13.2** Evidence-based medicine pyramid.



• **Fig. 13.3** Amalgam restorations.

3. Premature clinical failures of dental restorations are extremely costly to dental practices.
4. The evidence base for restorative materials is not robust.
5. The relative technique sensitivity of a material is an important attribute.
6. Both material selection and material manipulation are important.

Physical, Mechanical, and Optical Properties

A number of important physical and mechanical properties of dental materials can be identified. This section will discuss some of these properties and illustrate how they might be important to the long-term clinical success of a restoration. The discussion will also attempt to answer the question, "Does a comparison of physical properties of dental materials help predict clinical success or failure?"[2] This section is not meant to be an exhaustive description of all of the physical and mechanical properties that can be evaluated in dental materials. The reader is referred to any of the excellent contemporary textbooks on dental materials for that discussion. Rather, it is intended to be a brief description of several properties of dental materials that might be critical for clinical success.

One important point to be considered when evaluating studies on physical and mechanical properties is that materials should be tested in the mode in which they are used. Many in vitro studies fail to do this. One example is dental cements. Clinically cements are used in thin films between 15 and 100 microns. It is physically impossible to test a sample of this dimension, so tests on dental cements are done on samples that are not clinically relevant. Thus any in vitro data generated that is related to physical and mechanical properties of dental cements should not be used to infer significant clinical implications.

The following is an incomplete list and description of physical and mechanical properties of dental materials and some examples of why they are important.

Dimensional Change. Dimensional change is the percent shrinkage or expansion of a material, which usually occurs as a result of the setting reaction. One example is with composite resin. Basically composite resin materials consist of inert filler particles in a matrix of resin, usually mostly bisphenol A glycidyl methacrylate (Bis-GMA). The resin material undergoes a polymerization reaction as it sets, with individual monomers joining to become long-chain polymers that then cross-link to form the polymer matrix. During the process, the material undergoes polymerization shrinkage. With contemporary hybrid composite resin materials, the shrinkage is between 2.4% and 2.8%. If not accounted for in the placement technique, this amount of shrinkage can lead to clinically significant contraction gaps between the margin of the preparation and the composite resin restoration. This can result in microleakage, postoperative sensitivity, and recurrent caries.

Another example is with elastomeric impression materials. These also set with a polymerization reaction that results in shrinkage. This shrinkage can be controlled with use of custom impression trays that provide a 1- to 2-mm cross-sectional thickness of the material and proper use of tray adhesives, which ensures the direction of shrinkage is toward the tray.[7] Use of stock trays results in excessive thicknesses of impression material that may result in undesirable distortion of the impression.

Thermal Coefficient of Expansion (COE). The COE is defined as the amount a material expands per unit length if heated 1 degree higher. Waxes have the highest COE of all dental materials. When the lost wax process is used, wax patterns should be invested as soon as possible after margination to ensure that there is no distortion of the wax pattern due to normal fluctuations in room temperature (Table 13.1).

This property can be very important when dealing with materials that have cores with veneering materials. An example is metal-ceramic crowns. The feldspathic porcelain veneer, which provides excellent esthetic result with such restorations, is very brittle and must be supported with a rigid material (metal alloy).

TABLE 13.1 Linear Thermal Coefficient of Expansion of Dental Materials in the Temperature Range 20°C to 50°C

Material	Coefficient ($\times 10^6$/°C)
Human teeth	8–15
Ceramics	8–14
Glass ionomer base	10–11
Gold alloys	12–15
Dental amalgam	22–28
Composites	25–68
Unfilled acrylics and sealants	70–100
Inlay wax	300–1000

TABLE 13.2 Thermal Conductivity of Different Materials Used in Dentistry

Material	Thermal Conductivity (cal/sec/cm^2[°C/cm])
Unfilled acrylics	0.0005
Zinc oxide–eugenol cement	0.0011
Human dentin	0.0015
Human enamel	0.0022
Composites	0.0025
Ceramic	0.0025
Zinc phosphate cement	0.0028
Dental amalgam	0.055
Gold alloys	0.710

When fabricating metal-ceramic restorations the crowns must be heated to a high temperature several times to sinter the veneering ceramic. Then the units must be cooled to bench temperature. If the COEs of the metal alloy and veneering porcelains are not closely matched, stresses can build up in the veneering porcelain that result in cracking or fracture of the ceramic. This can occur at the time of fabrication or later in clinical service. Generally the COE of the metal alloy is reduced with the addition of platinum, and the COE of the porcelain is increased with the addition of sodium. Whatever the mechanism used, it is clear the COEs of both the core and veneering material must be closely matched (Fig. 13.4).

It is often thought that the COEs of restorative materials must be matched with the COE of tooth structure. If they are not matched, the ingestion of hot beverages (e.g., coffee) and cold foods and beverages (e.g., ice cream and beer) could result in uneven expansion and contraction of the restoration and tooth, resulting in microleakage. The reality is foods and beverages do not surround the tooth/restoration complex for a long enough time frame to effect any significant change in the temperature of the tooth restoration.

Thermocycling of tooth restoration samples is frequently done during in vitro studies, and many incorrectly believe this is intended to mimic intraoral conditions. Samples are alternately placed in water at 5°C and 55°C for a specified dwell time. This is generally done to artificially age the samples and to stress the bond being tested. This thermocycling takes advantage of the differing COEs of the restorative material and tooth structure to stress the bond and accelerate the failure rate to shorten the necessary testing time.

Percolation. The process of fluids entering the microscopic space between a restorative material and the wall of the cavity preparation is called percolation. This is not a beneficial process with materials such as composite resins, but works wonderfully with materials such as silver amalgam. When percolation occurs with amalgam, corrosion products build up at the tooth-restoration interface and effectively seal it, usually 6 weeks after placement of the amalgam. With modern high copper amalgams with reduced gamma-2 phase, this phenomenon occurs but takes slightly longer to create an effective seal (Fig. 13.5; Table 13.2).

Thermal Conductivity. Thermal conductivity is the rate at which thermal changes are conducted through a material. Materials with high thermal conductivity are good conductors of hot and cold, and could be a factor in postoperative sensitivity. This is the rationale for placing bases and/or liners with cavity preparations that are close to the pulp. Bases and liners have low thermal conductivity and thus are efficient insulators (see Table 13.2).

There is speculation by some that these bases and liners are not needed because dentin itself is a good insulator. However, often the thickness of dentin over the pulp may be minimal, so it is usually recommended that the thickness of remaining dentin plus the thickness of the base or liner should be at least 2 mm.



• **Fig. 13.4** Metal-ceramic restorations.



• **Fig. 13.5** Teeth numbers 12, 13, 14, 15 restored with amalgam restorations. Percolation = corrosion product = seal.

It is interesting to note that amalgam is an excellent thermal conductor whereas composite resin is an effective insulator; however, the incidence of postoperative sensitivity is significantly higher with composite resin restorations than with amalgam restorations. This demonstrates that the phenomenon of postoperative sensitivity is multifactorial and complex, and not due to thermal conductivity alone.

Layered zirconia restorations experience a rate of veneer chipping about eight times that of metal-ceramic restorations. The etiology of this chipping is multifactorial, but one of the potential causes is the thermal conductivity of the zirconia core. The thermal conductivity is extremely low, and zirconia can be considered an insulator. The consequence of this is that when a layered zirconia restoration is removed from the sintering oven, the core cools very slowly and the veneering ceramic cools past its glass transition temperature while the core continues to cool and shrink. The net result of this interaction is that stress is built up in the veneering ceramic, which is later released via clinical chipping of the veneering material. New firing cycles involving slower cooling rates seem to have reduced the incidence of chipping in recent years.[8]

Creep. Creep is defined as time dependent plastic deformation of a material under static load or constant stress. This property is important with silver amalgam. Conventional low copper amalgam materials have high creep values and exhibit poor marginal integrity over time as the amalgam creeps into a thin layer and then breaks off creating marginal ditching. Contemporary high copper amalgams have low creep values and maintain good marginal integrity as a function of time.

Creep is one of the few properties that has a demonstrated ability to predict clinical performance. Amalgam materials with low creep values perform well clinically when compared with amalgams with high creep values.

Electrical Properties. The three major electrical properties in dentistry are galvanism, tarnish, and corrosion. Galvanism occurs whenever different metals are in contact with one another in the presence of an electrolyte (saliva). Clinically, this can occur when a gold crown is placed adjacent to an existing amalgam restoration or vice versa. Gold crowns adjacent to metal-ceramic crowns can have a similar result as there are over 277 different commercially available porcelain bonding alloys. The clinical symptoms resemble an "electrical shock" or the presence of a metallic taste. These symptoms typically are self-limiting and are not considered serious problems (Fig. 13.6).

Tarnish is a surface phenomenon resulting in unesthetic discoloration of dental alloys, similar to tarnish of commonly used silverware (Fig. 13.7). It does not result in physical breakdown of the alloy. It is important to note that the term "noble metal" connotes that the metal is resistant to tarnish in the oral cavity. The major noble metals used in dentistry are gold, platinum, and palladium. Others are rhodium, ruthenium, iridium, and osmium. Alloys should contain a minimum 45% noble metal to prevent intraoral tarnishing. Chromium is a metal that prevents tarnish in base metal alloys. One metal that is precious but not noble is silver.

Corrosion is chemical or electrochemical dissolution of metals in oral fluids. The varying pHs in the oral environment create the possibility of corrosion, although in fact it is a rare occurrence. Corrosion of prefabricated metal endodontic dowels with fractured endodontically treated teeth has been seen occasionally. In these cases it is believed that fracture of the endodontically treated tooth occurred first, allowing contact of oral fluids with the post resulting in corrosion. The other possibility is that the corrosion occurred first followed by fracture due to the production of corrosion products (Fig. 13.8).

Solubility. Solubility is the relative tendency to dissolve in oral fluids. This was a major problem with the first direct esthetic restorative material, silicate cement. This cement was a powder



• **Fig. 13.6** The generation of electrical currents due to contact of dissimilar metals in an electrolyte is known as galvanism.



• **Fig. 13.7** Tarnish of silverware. Dental alloys may undergo the same discoloration in the oral environment.



• **Fig. 13.8** Example of corrosion due to electrochemical dissolution in water.

liquid system, the powder being ground alumino-silicate glass and the liquid phosphoric acid. When mixed, it formed a paste that was inserted into the prepared cavity and when set and finished provided a restoration suitable for class III restorations that was expected to provide only 3 to 5 years of service. The reason for this short time of service was that silicate cements were soluble in oral fluids. An unexpected benefit of this problem was that as the alumino-silicate glass dissolved, it released high concentrations of fluoride. When the partially dissolved restoration was removed for replacement, recurrent caries was never found due to this high level of fluoride release (Fig. 13.9).

Conventional nonresin-based dental cements had variable levels of solubility; for example, zinc-oxide eugenol (ZOE) cements that are often used for provisional restorations are quite soluble in oral fluids. From a practical standpoint, this means that provisional restorations should not be cemented for more than 6 weeks if ZOE is used. At that time, the provisional restorations should be removed, cleaned, and recemented.

The relative solubility of conventional nonresin cements made it necessary to provide precisely fitting definitive restorations. Cast gold restorations provide an extremely long time of service cemented with zinc-phosphate cement, providing they have precision of fit. Poorly fitting cast gold restorations provide a short term of service as they are simply chunks of metal in a sea of cement.

Solubility of dental cements is typically measured with a 24-hour in vitro solubility test where a sample of cement is immersed in a solute for 24 hours and the amount of dissolved cement is calculated.[9] Long-term (6 months) in vivo solubility tests are preferred. These types of studies indicate that there are small differences between cements in long-term solubility, but importantly find there are significant differences among patients in regard to solubility. Some patients dissolve considerable amounts of cement, while other patients demonstrate minimal solubility. The causes of this phenomenon are not known.

Glass ionomer cements perform poorly in 24-hour in vitro solubility testing because they are quite soluble in the first 24 hours after mixing. However, they perform very well in 6-month in vivo testing, and in fact are the best of the conventional nonresin cements in that parameter. This means they must be protected from saliva immediately after cementation using excellent isolation and coating with petroleum jelly or varnish.

Optical Properties. Optical properties of bulk materials include interactions with visible light that involve reflection, refraction, absorption (and fluorescence), or transmission. The radiation typically involves different intensities for different wavelengths (or energies) over the range of interest (spectrum). Any of these interactive events can be measured using a relative scale or an absolute scale. When the electromagnetic radiation is visible light, the amount of reflection can be measured in relative terms as gloss or in absolute terms as percent reflection. Visible light absorption can be measured in absolute terms as percent absorption (or transmission) for every wavelength (in the visible spectrum). Color is a perception by an observer of the distribution of wavelengths. The same color sensation may be produced by different absorption spectra (metamerism). An individual's eye is capable of sensing dominant wavelength, luminous reflectance (intensity), and excitation purity. Variations among individuals' abilities to sense these characteristics give rise to varying perceptions of color.

The quality of color is measured by the Commission Internationale de l'Eclairage system as tristimulus values and reported as color differences (ΔL^*, Δa^*, and Δb^*) compared with standard conditions.



• **Fig. 13.9** Class III cavity restored with silicate cement after 3 years. Notice the poor resistance to solubility and color change. No evidence of recurrent carries.

Radiation of another wavelength may be preferentially absorbed (e.g., x-rays). Composites that contain lithium, barium, strontium, or other good x-ray absorbers may appear radiopaque (radiodense) in dental radiographs. Materials that are good absorbers (for whatever form of radiation) are described as opaque.

The appearance of a dental restoration is a combination of events of surface reflection, absorption, and internal scattering. The scattering simply may deflect the path of the radiation during transmission (refraction), or it may reflect the radiation internally from varying depths back out of a solid to the observer (translucency). Enamel naturally displays a high degree of translucency; translucency is a desirable characteristic for restorative materials attempting to mimic enamel. A tooth that is isolated and kept from any contact with oral moisture (saliva) soon develops a transient whiter appearance. Refractive index is the angle of changed path for a standard wavelength of light energy under standard conditions (Fig. 13.10).

Compressive Strength, Tensile Strength, Diametral Tensile Strength, Flexural Strength, and Fracture Toughness. All of these are mechanical properties of materials that are tested utilizing nonclinical samples and thus are only useful when comparing the tested material to a known standard. Compressive strength tests use cylindrical samples. Tensile tests can be done with certain materials such as wires but cannot be used with materials such as dental cements. Diametral tensile strength tests use discs and apply the force to an edge. Because these tests do not test the material in the mode in which it is used, data obtained from the tests are of little value in predicting clinical performance but may be useful in comparing a new product to a standard (Fig. 13.11).

Flexural strength testing has become popular for evaluating new ceramic materials. Many of the newer ceramics such as lithium disilicate and monolithic zirconia have vastly improved flexural strength compared to feldspathic porcelain or leucite reinforced ceramic. However, that data in itself does not automatically ensure improved clinical performance, because ceramic crowns do not fracture because of a lack of strength. They fracture because of inherent defects called Griffith flaws, which are incorporated into the restorations during fabrication. These defects propagate under occlusal stress and due to static fatigue, and eventually result in catastrophic failure. These newer materials may well provide improved clinical performance because the techniques used in fabrication (pressing and machining) result in fewer inherent defects, and the phase transformation capability of zirconia prevents defects from propagating after initiation.

Fracture toughness determination may be one of the most valuable in vitro tests because it evaluates failure initiating from a predetermined defect. It is possible that materials with improved fracture toughness may show improved clinical performance.

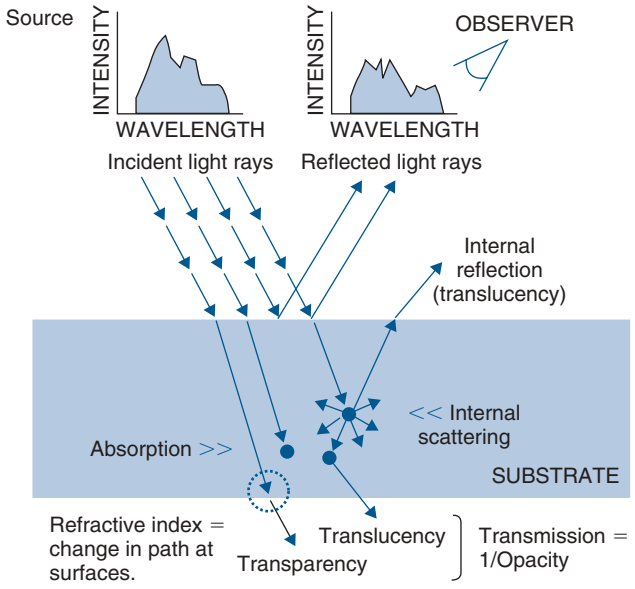
Elastic Modulus. Elastic modulus is a measure of the relative stiffness or rigidity of a material (Fig. 13.12). The elastic modulus is the ratio of stress over strain and is the slope of the straight-line portion of a stress-strain diagram. It is a measure of how the material will deform when placed under stress. A material with a high elastic modulus will be stiff; a material with a low elastic modulus will be flexible.

Ceramic materials have a very high elastic modulus and are very brittle. These materials are frequently bonded to a core material to support the ceramic veneer and prevent fracture. It is critical that the core material have an elastic modulus equal to or higher than that of the ceramic material, because if the core material flexes, the ceramic will fracture.

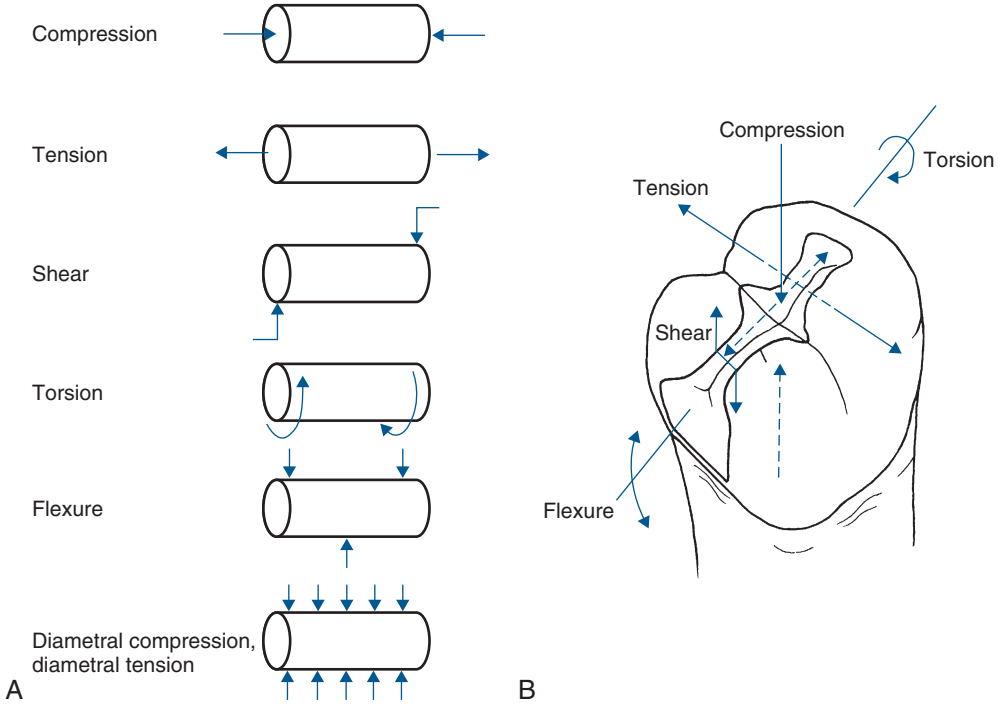
Another area where the modulus of elasticity of a material is important is in the restoration of noncarious cervical lesions (NCCLs). One potential etiology of NCCLs is tooth flexure leading to cervical breakdown. Traditionally, materials with a low elastic modulus such as microfilled composite resins and resin modified glass ionomers have been more successful in restoring NCCLs than stiff materials such as hybrid composite resins.[10] Contemporary adhesive systems are now highly filled and, because they have a low elastic modulus, can act as stress breakers during tooth flexure, permitting successful restoration with stiffer composite resins.

Load-to Failure Testing Versus Fatigue Testing. The traditional approach to testing the strength of dental restorations is load-to-failure testing. With load-to-failure testing, a restoration is fabricated and cemented/bonded to the prepared tooth or die and then loaded with a testing machine and the forces increased until the restoration fractures. If the load at which the restorations fail exceeds the maximum bite force in the area of the restoration, it is then deemed that materials will provide excellent clinical performance. Data generated from load-to-failure testing can be very misleading and do not provide predictive information on clinical performance.

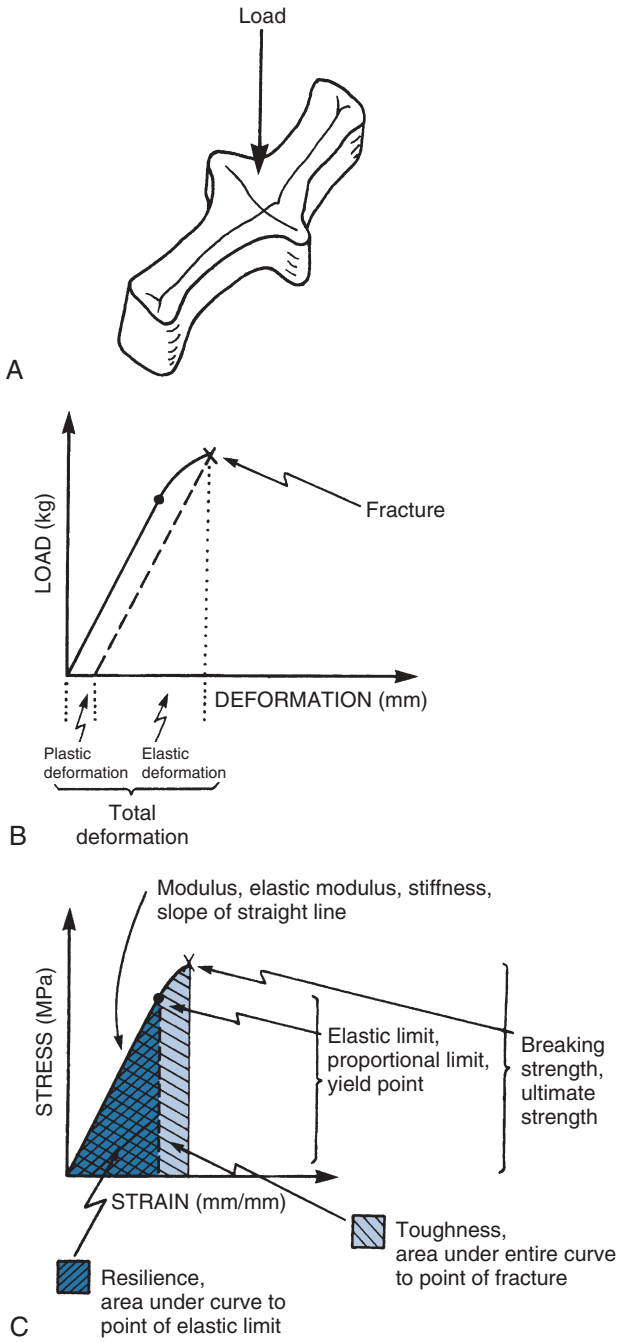
Several problems with load-to-failure testing can be identified. Clinically, teeth and restorations generally fail by defect propagation. Load-to-failure testing does not take this into account. The loads at which failure occurs with load-to-failure testing are much higher than the loads at which clinical failure occurs.[11] The failures with



• **Fig. 13.10** Schematic summary of interactions of electromagnetic radiation with materials. The color perceived by the observer is the result of several interactions between substrate and incoming radiation producing reflection, internal scattering, absorption, fluorescence, and transmission.



• **Fig. 13.11** Examples of directions of loading. A, Uniaxial loading of cylinder. B, Uniaxial loading of a mesiooccusal amalgam restoration.



• **Fig. 13.12** Schematic summary of mechanical properties with respect to amalgam restoration in function. A, Occlusal loading of Class I amalgam restoration. B, Load/deformation curve describing behavior of amalgam. C, Normalization of load/deformation curve to stress-strain curve with important characteristics of curve indicated. (Mechanical responses depend on temperature and strain rate involved.)

load-to-failure testing initiate at the occlusal surface of the restoration being tested while clinically restorations fail by defect propagation from defects at the tooth cement surface. Finally, the fracture pattern of failed restorations with load-to-failure testing results in the restoration fracturing in multiple fragments. Clinically, failure usually occurs with the restoration breaking into two pieces.

One major argument against load-to-failure testing is that restorations do not fail because the patient bites down with incredible force



• **Fig. 13.13** Eight-chamber chewing simulator.

on a restoration, but rather they fail by defect propagation due to low occlusal forces over a long period of time. In order to make in vitro testing more clinically related, fatigue testing using cyclic loading machines has become increasingly popular (Fig. 13.13).

Fatigue testing using cyclic loading is considerably more relevant than load-to-failure testing because it much more closely mimics the clinical situation. Fortunately, many major research organizations have made the considerable investment in cyclic loading machines and more and more fatigue studies are appearing in the peer reviewed literature. It is important to note that dental fatigue testing is in its infancy and many factors need to be considered when attempting to bring the study environment closer to the clinical situation.

Some of the variables that need to be considered include loading force, number of cycles, loading frequency, and antagonist material. Whether or not to include a simulated periodontal ligament is an important consideration. Whether to fatigue under water or dry is also a consideration. With ceramic materials, it is critical to fatigue under water as it has been proven that ceramic materials fail at half the load under water as in air. This is because of the presence of static fatigue with ceramic materials under water or saliva. The use of thermocycling, the type of abutment or die material, the use of vertical plus lateral movements, and the use of anatomic versus flat samples are all important considerations when designing a fatigue study.

The bottom line is that load-to-failure studies have no clinical relevance whatsoever and should not be published. While recognizing that designing appropriate fatigue studies is challenging, it is clear that these types of studies are going to play an increasingly important role in dental materials research.

Wettability. Wettability is a measure of the affinity of a liquid for a solid as indicated by the spreading of a sessile drop (Fig. 13.14).

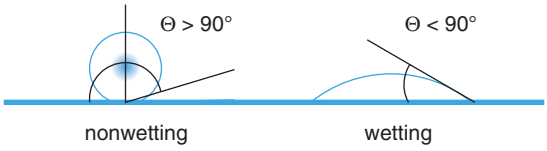
This might be best explained by the state of a raindrop on a freshly waxed car. It will bead up, indicating a high contact angle and low wettability. Similarly, placing a drop on air abraded versus polished metal will result in a spreading drop (low contact angle; good wettability) on the former and a beaded drop (high contact angle; poor wettability) on the latter.

Early versions of addition reaction silicone impression material were extremely hydrophobic and difficult to pour. Modern

elastomeric impression materials contain nonionic surfactants, which render them more hydrophilic and considerably easier to pour because the surface is more wettable and forms a lower contact angle with liquids.

Hardness Testing. The surface hardness of a material is often of interest and is determined by use of some form of indenter. Basically the indenting tool is dropped onto the surface of the material with a specified load, and the size of the defect created on the material is measured using a microscope. There are different hardness tests that are optimally used with specific materials. These include Knoop, Vickers, Barcol, Brinell, and Rockwell hardness testing (Fig. 13.15). The difference between these testers is the shape and material of the indenter and the force used.

One area where hardness testing is frequently used is in determining the degree of polymerization of composite resin materials. The harder the surface, the greater the degree of conversion. Studies generally compare the hardness of the top surface (closest to the light) and the hardness of the bottom surface. Some of the variables that have been tested include intensity of the light, thickness of the composite resin increment, curing through ceramics, and distance of the light from the resin.



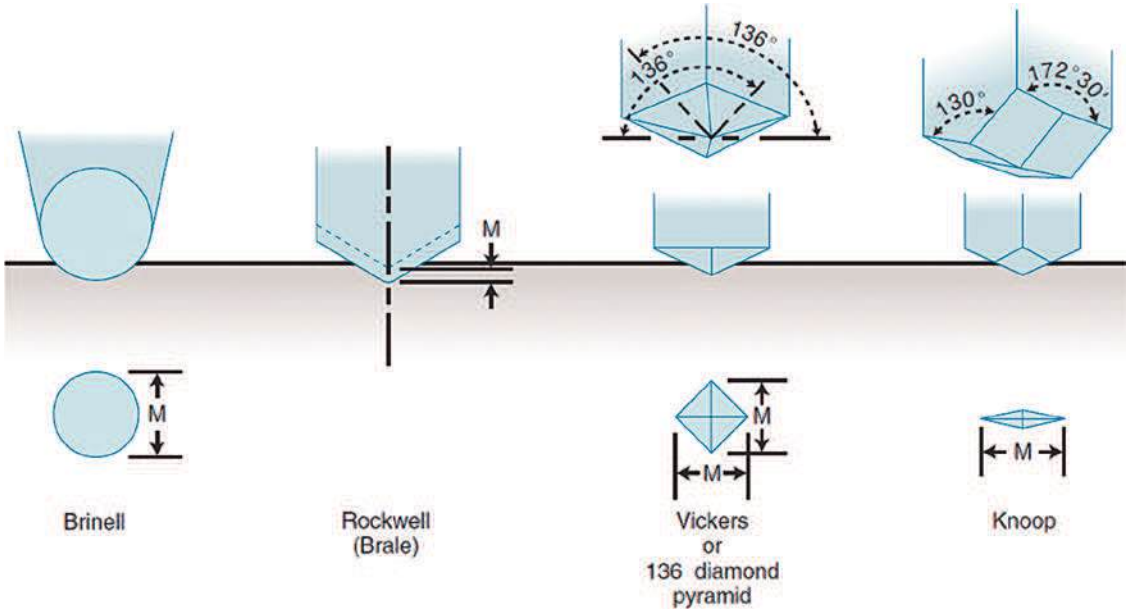
• **Fig. 13.14** Schematic drawing of the contact angle determining wettability. Contact angles less than 90 degrees are considered to be wetting and hydrophilic. Contact angles larger than 90 degrees are considered nonwetting and hydrophobic.

Bond Strength Testing. In the adhesive era there is a need to test the bond of one material to another, and the bond of materials to enamel and dentin. There have been hundreds of studies published on bond strength in recent years. These studies are useful for comparing the relative bond strength of product X to product Y but have relatively low ability to predict clinical performance.[12]

The most common bond strength test is the shear bond strength test. This test is used frequently because it is easy to perform, but the clinical relevance is minimal. Another approach is the microtensile bond strength test. This test became popular because it was initially thought that 8 to 10 samples could be obtained from a single tooth; however, subsequent evaluation determined that data from these samples needed to be combined and in reality only 1 sample was obtained from each tooth. This bond strength test is extremely technique sensitive and produces data with wide standard deviations. It is not clear that the predictive data from microtensile bond strength testing is more predictive of clinical performance than shear bond testing.

It is clear that many bonds are quite strong initially but deteriorate over time. Thus it is imperative that bond strength testing must include initial testing plus testing after storage in water, thermocycling, or fatigue.

Biologic Properties. Biologic properties of biomaterials are concerned with toxicity and sensitivity reactions that occur locally, within the associated tissue, and systemically. It is possible to evaluate local toxic effects on cells by clinical pulp studies or by tissue culture tests. Unset materials may release cytotoxic components; however, in clinical situations this problem is rarely evident. Two



• **Fig. 13.15** Shapes of hardness indenter points *(upper row)* and the indentation depressions left in material surfaces *(lower row).* The measured dimension *M* that is shown for each test is used to calculate hardness. The following tests are shown: Brinell test—a steel ball is used, and the diameter of the indentation is measured after removal of the indenter. Rockwell test—a conical indenter is impressed into the surface under a minor load *(dashed line)* and a major load *(solid line),* and *M* is the difference between the two penetration depths. In the Vickers or 136-degree diamond pyramid test, a pyramidal point is used and the diagonal length of the indentation is measured. In the Knoop test, a rhombohedral pyramid diamond tip is used and the long axis of the indentation is measured. (From Anusavice KJ, Shen C, Rawls RH: *Phillips' science of dental materials,* ed 12, Elsevier, St. Louis, 2013.)

important clinical factors determining toxicity are the exposure time and the concentration of the potentially toxic substance.

Generally, restorative materials harden quickly or are not readily soluble in tissue fluids (or both). Potentially toxic products do not have time to diffuse into tissues. Even more importantly, the concentration makes the poison. Some authorities believe that if the amount of material involved is small, the pulp or other tissues can transport and excrete it without significant biochemical damage. Others believe there is no threshold. A threshold level for toxicity is one below which no effect can be detected.

This section is not meant to be an exhaustive description of all physical and mechanical properties of dental materials. It is intended to highlight some of the more important properties and to describe some clinical examples where these properties might be important. Even though these properties are important, it is imperative to understand that in vitro studies of physical and mechanical properties are poor indicators of clinical performance. As fatigue studies become more sophisticated and clinically relevant, in vitro studies may become more predictive.

THE BOTTOM LINE

1. A large number of physical and mechanical properties of dental materials can be evaluated with in vitro testing.
2. When testing materials, whenever possible they should be tested in the mode in which they are used.
3. Bond strength studies have little relevance to clinical performance.
4. Static in vitro studies of physical and mechanical properties yield minimal data that are predictive of clinical performance.
5. Ceramic materials should be fatigue tested under water.
6. Fatigue testing with cyclic loading is promising but complex.

Direct Restorative Materials

Silver Amalgam

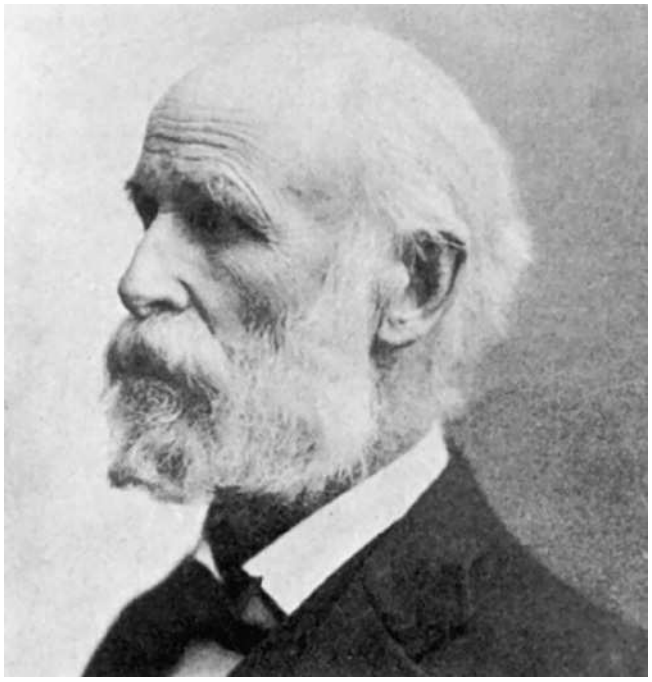
Silver amalgam has been used for the restoration of teeth for well over a century.[6] It has been the most popular restorative material over that period of time, having been used in over 75% of single tooth restorations placed in the past 100 years (see Fig. 13.3).

The use of silver amalgam in North America has diminished somewhat in recent years due to a dramatic reduction in caries rate coupled with a high interest in esthetics and concern for the potential toxicity of mercury.

Nonetheless, silver amalgam will continue to be an important restorative material for many years to come. It will likely be especially valuable in many of the developing countries where dental caries continues to be a major problem and where resources required to deliver alternative (more sophisticated) restorative dentistry options are not available. Additionally, silver amalgam continues to be one of the best materials available for core buildups of extensively damaged teeth to support crowns and fixed partial dentures.

Silver amalgam has had a fascinating and controversial history ever since its introduction to North America by the Crawcour brothers in the 1830s. The actions of these charlatans resulted in the first amalgam war and the infamous "amalgam pledge" that was finally rescinded in 1850. A period of scientific investigation followed, championed by individuals such as Flagg and Hickock, and culminated in G.V. Black's publication of his formula for silver amalgam in 1896 (Fig. 13.16).

In the 1930s the writings of a German physician, Stock, stimulated research into the biocompatibility of silver amalgam, and specifically the effects of mercury. In what could be called the second amalgam war, claims and counterclaims were made, and eventually organized dentistry accepted the belief that amalgam was safe. The first American Dental Association (ADA) Specification (#1) for a dental material was established for silver amalgam, and Stock later went on to repudiate his own writings.



• **Fig. 13.16** G.V. Black. Father of operative dentistry. (From Kock CRD: *History of dental surgery,* vol 1, Chicago, 1909, National Art Publishing.)

The composition of silver amalgam did not change substantially from the publication of Black's formula until the 1960s when high copper amalgam was introduced to the profession by two Canadian researchers (Fig. 13.17).

As subsequent clinical research would clearly demonstrate, the high copper formulations that are almost exclusively used today are clinically superior to the conventional amalgam materials that had been used up to that time. As Greener has explained, the discovery of high copper amalgam was a case of "serendipity of the highest order," yet it remains as one of the most important advances in dental materials in the past 50 years.[6]

Advantages of Silver Amalgam

Silver amalgam has been and will continue to be such a popular restorative material because it possesses a number of well-defined advantages. It is one of the easiest of all restorative materials to manipulate, it can be placed into the prepared cavity in a plastic state, and then it can be carved and shaped to physiologic form before it hardens. Once hardened, it has excellent physical properties that allow it to perform well in the long term as a restorative material in posterior teeth. It is strong enough to provide very predictable long-term service in conservative restorations, and is also valuable as an interim restorative material in very large restorations.

The basic strength of silver amalgam, coupled with the fact that it can be placed relatively rapidly in a single appointment, make it a very cost-effective material. It is less expensive by a wide margin than any of the alternative restorative materials.

Silver amalgam has one additional important advantage: its ability to automatically seal the cavity preparation. This ability,



• **Fig. 13.17** Dispersalloy (dispersed phase alloy) high copper amalgam. (Courtesy Dentsply.)

which is a result of the buildup of corrosion products at the interface between the amalgam and the cavity margins, makes silver amalgam unique among restorative materials. A seal can be produced with excellent technique with some of the alternative materials, but none are as user friendly as amalgam, which creates a seal independent of the operator. Several months are required before the seal becomes complete, especially with high copper amalgam formulations; but once attained, it is a very effective barrier against recurrent caries.

Because it requires some time for the seal resulting from deposition of corrosion products to materialize, a method of obtaining a temporary seal at the time of placement is recommended. This used to be accomplished by placing two or three coats of copal varnish on the enamel and dentin surfaces of the prepared cavity. This copal varnish created an initial seal that would eventually be replaced by the percolation process as corrosion products gradually formed at the interface. In contemporary practice copal varnish is obsolete, and sealing the cavity with a desensitizing agent such as Gluma Desensitizer or G5 prior to amalgam placement is indicated.

Composition

Silver amalgam is supplied to the dentist as a powder or metallic pellets that are then mixed together with liquid mercury in a high-speed triturator to form a plastic mass that can be placed into a prepared cavity. The plastic material allows sufficient working time for successful placement and then sets to form a strong and durable restoration that can provide long-term clinical service.

Amalgam formulations differ from one another both chemically in terms of chemical composition and physically in terms of the size and shape of the alloy particles that are mixed with mercury. These physical and chemical variations result in differences in handling characteristics of the material and in the clinical results that are ultimately attained.

From a chemical standpoint, amalgam can be divided into two groups, *conventional amalgam*, which is similar to Black's formulation, and *high copper amalgam*, which represents the state of the art. Table 13.3 lists the components of both types of amalgam alloy prior to mixing with mercury and also indicates the function of each component.

TABLE 13.3 Amalgam Alloy Composition

Element	Alloy %	Principal Function
Ag	70%	Strength
Sn	16%	Expansion
Cu	13%	Strength
Zn	1%	Deoxidizer

It is useful to understand the chemical reactions that occur with both types of amalgam, as such an understanding will explain the differences in their clinical performance.

A set amalgam is a true eutectic, and micrographic evaluation reveals three distinct phases: gamma, gamma-1, and gamma-2. The reaction for conventional amalgam may be written as follows:

$$\underset{(AgSn)}{\text{gamma}} + \underset{(Hg)}{\text{mercury}} \rightarrow \underset{(AgSn)}{\text{gamma}} + \underset{(AgHg)}{\text{gamma-1}} + \underset{(SnHg)}{\text{gamma-2}}$$

The gamma phase, which consists of unreacted silver and tin, is the major component of the unreacted alloy and is also present in the set amalgam. It is suspended in a matrix of gamma-1, which is the reaction product of silver and mercury. The gamma phase is very strong and corrosion resistant, whereas the gamma-1 phase is less strong and is more corrosion susceptible. The gamma-2 phase, the reaction product of tin-mercury, is also suspended in the gamma-1 matrix. It is the weakest and most corrosion susceptible of the three phases. The major difference between conventional and high copper amalgams is the near elimination of the gamma-2 phase in the high copper amalgams. Elimination of this phase produces a much stronger and clinically superior restoration.

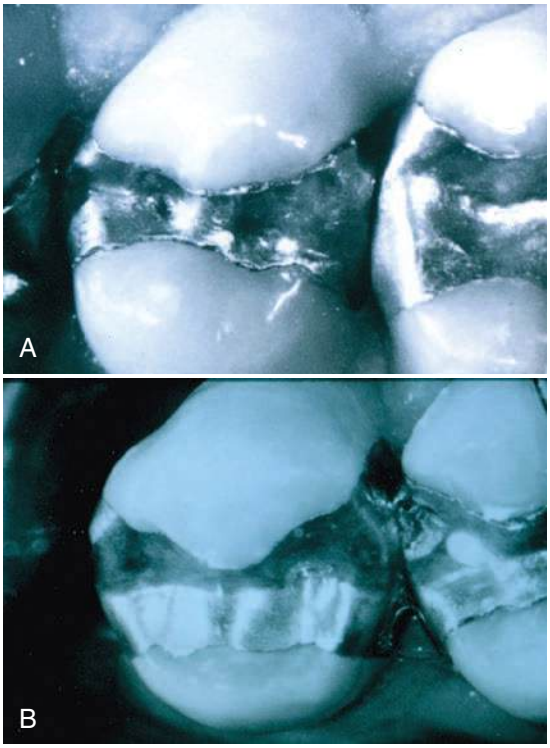
Laboratory studies show that high copper amalgams display less creep than conventional amalgams. The main clinical manifestation of this superiority is improved edge strength, which results in a lack of marginal ditching over the long term (Fig. 13.18).

The chemical reaction for high copper amalgams may be written as follows:

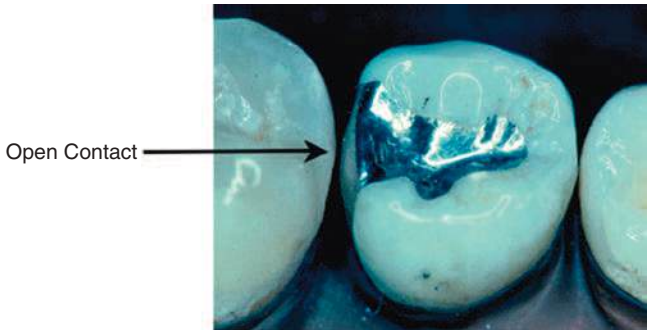
$$\underset{(AgSn)}{\text{gamma}} + \underset{(AgCu)}{\text{copper}} + \underset{(Hg)}{\text{mercury}} \rightarrow \underset{(AgSn)}{\text{gamma}} + \underset{(AgHg)}{\text{gamma-1}} + \underset{(CuSn)}{\text{CuSn}}$$

With such amalgam formulations, where the alloy contains 9% to 30% copper, the copper will preferentially react with the tin and thus prevent formation of the gamma-2 phase. Again, the elimination of this gamma-2 phase strengthens the resultant amalgam and is clinically manifest in improved marginal integrity and minimal marginal ditching over time.

The discovery of high copper amalgams was basically an accident. The metallurgists who developed the first commercial product added small metal spheres to the irregularly shaped alloy particles in an attempt to "dispersion strengthen" the set amalgam. The concept was that the spheres would fit into the interstices between the larger irregular particles and this would make the amalgam stronger. They happened to use copper spheres for this initial product, and it was the elimination of the gamma-2 phase as a result of the added copper that resulted in the improved clinical performance of the alloy, rather than the imagined dispersion strengthening. In fact, it has never been demonstrated that dispersion strengthening is a viable modality.



• **Fig. 13.18** A, Conventional amalgam restoration. B, High copper amalgam restoration.



• **Fig. 13.19** Open proximal contact in a Class II amalgam restoration.

TABLE 13.4 Compressive Strength (MPa) of Different Brands of Amalgam Alloys After 1 Hour and 24 Hours

Alloy	1 hr	24 hrs
Velvalloy	18,000	55,000
Dispersalloy	33,000	64,500
Phasealloy	12,900	65,700
Tytin	34,900	75,000
Valiant	47,800	76,500

Commercial amalgams can also differ from one another in the shape of the alloy particles. The two major types are *admixed* alloys and *spherical* alloys. Both types of alloys produce clinically successful restorations, but the handling characteristics of each are very different.

With traditional amalgam materials, the alloy particles are formed by cutting a block of alloy with a lathe. This method forms relatively large, irregular particles. With admixed alloys, small spheres formed by atomizing the liquid alloy are added to the mix in the hope that a denser final amalgam will result. Such an alloy is termed an admixed alloy, and is the most common material in use today.

Such alloys require considerable force during condensation of the amalgam to ensure optimum packing of the material and intimate adaptation of the amalgam to the cavity walls. Strength of these alloys is time dependent, and 24-hour compressive strength is considerably greater than 1-hour strength (Table 13.4). These alloys should be carved and burnished, and final polishing should be delayed for at least 24 hours.

Spherical alloys handle much differently than admixed alloys. Generally speaking they require slightly less mercury in the initial mix due to the small size of the spherical particles. The consistency of the mix is often described as "buttery," and the amalgam is easily placed into the cavity and adapted to the cavity walls with a minimum of pressure. However, because the fresh amalgam mix will not resist a moderate amount of condensation pressure, it is somewhat difficult to attain an adequate proximal contact in Class II restorations (Fig. 13.19). With experience, this deficiency can be overcome with excellent clinical technique.

Spherical amalgams attain much higher early (1 hour) strengths than do admixed amalgams, and the final (24 hours) compressive strengths are also generally higher. The gain in early strength may possibly reduce the incidence of isthmus fractures due to occlusal force exerted shortly after placement of the amalgam; however, there is no clinical data that would support that. It may also allow preparation of the freshly placed amalgam 15 minutes after insertion in those situations where it is being used as a build-up prior to crown fabrication. The gains in final compressive strength are likely of minor importance.

Evaluation of Silver Amalgam Materials

Dental materials are generally initially evaluated in the laboratory and subsequently in patients in controlled clinical trials. Laboratory tests are of minimal value in differentiating clinical performance between different alloys. The only laboratory test that has proven to be of predictive value in the clinical setting is that of "creep." Alloys with low creep values typically demonstrate excellent marginal integrity over time in clinical trials.[13] Generally high copper amalgams display low creep values and are clinically successful alloys.

Thus an amalgam alloy should be selected on the basis of proven clinical performance. Other variables, such as quality control, and handling characteristics also may play a role in the decision to select a given alloy. Thus today it would be a reasonable choice to select a high copper formulation that has a consistency and rate of set that is desired by the individual practitioner. In fact, the intelligent practitioner may well choose to keep a couple of different formulations on inventory and choose the appropriate formulation to match the specific clinical situation.

For example, many prefer an admixed zinc containing alloy for the majority of conventional restorations where a dry field can be maintained and where achieving good interproximal contact is desirable. They would choose a spherical zinc containing alloy in those situations where a dry field can be maintained but using strong condensation force is difficult, for example when restoring a Class V lesion on the distobuccal of a maxillary second molar. Such a formulation is also desirable when building up an extensively

broken down tooth prior to preparing that tooth for a cast crown. Here the high early strength is useful, as is the ease of condensation especially when small slots, potholes, and small undercut areas are utilized as the primary retentive features. Finally, a nonzinc spherical alloy is useful in those situations where adequate moisture control is not possible, such as certain Class V situations and surgical endodontics where amalgam is used to seal the small prepared cavity in the root apex.

Manipulation of Silver Amalgam

The quality of a given mix of silver amalgam is controlled by the quality of the product as supplied to the dentist, and can be modified by the manipulation of that product by the dentist. The manipulative variables include the mercury/alloy ratio, trituration, condensation, and carving and finishing of the restoration.

In general terms, the less mercury that remains in the final restoration, the better it will be both in terms of strength and corrosion resistance. Thus the initial mix should be made with only as much mercury as is required to make an adequate mix. With most admixed alloys this will be about 50% mercury, whereas with spherical alloys it will be slightly less. Proper condensation and carving technique will remove a significant amount of mercury from the material, thus enhancing final strength. In the past care had to be taken with mercury dispensers to maintain control of the mercury/alloy ratio, but today the majority of practitioners utilize precapsulated products. This is beneficial in that it allows precise control of the mercury/alloy ratio, minimizes handling of the mercury, and minimizes the mercury vapor aerosol created during trituration.

Mixing the amalgam is called trituration. Adequate trituration of the amalgam is essential to produce a workable mass and to ensure optimum physical properties. Each brand of amalgam has specific instructions regarding optimum time and speed of trituration. The instructions are specific for each brand of amalgamator. The longer and the faster an amalgam is mixed, the more rapidly it will set. The manufacturers' guidelines for mixing should be followed and can be modified slightly to obtain a faster or slower setting material.

Condensation of the amalgam is probably the most critical manipulative variable. Adequate condensation results in intimate adaptation of the amalgam to the cavity walls, as well as a denser material to maximize strength. The prepared cavity is always overfilled and carved back to proper anatomic contour. This is because condensation results in a mercury-rich superficial layer, which is removed during the carving procedure. The resultant amalgam surface has considerably less mercury and subsequently superior physical properties.

The most common error made by dentists is under condensation. This results in poor adaptation to the cavity walls, increased microleakage, and recurrent caries. The final amalgam is weaker and more fracture prone, and cannot be polished nearly as well as a properly condensed amalgam.

With the high copper formulations currently in use, both precarve and postcarve burnishing is recommended. This is generally accomplished with a large, egg-shaped burnisher. The amalgam is carved after achieving a firm set, with careful attention being paid to eliminating all marginal flash and carving the amalgam flush with the margins of the cavity preparation. Any marginal flash will eventually fracture, and it is rare that it fractures right at the margin. Inevitably, marginal ditching will occur.

After carving back to anatomic form, the restoration is lightly burnished to a smooth sheen. The restoration may be polished after 24 hours. There are no solid clinical data that support the routine polishing of amalgam restorations, but certainly highly polished restorations are more comfortable for the patient and will be less likely to retain bacterial plaque. Polishing amalgams also indicates a certain amount of pride in workmanship and is a source of great personal satisfaction.

Bonded Amalgams

With adhesive dentistry becoming so popular in recent years, many innovative techniques in all areas of restorative dentistry have been proposed. One of the most intriguing is the concept of *"bonded"* amalgams. This technique has the advantages of forming an instant seal of the prepared cavity, a more conservative preparation design, and the *potential* to strengthen the prepared tooth and reduce the long-term incidence of cuspal fracture.[14] The word *potential* is stressed because the concept of tooth strengthening has not been proven clinically at the time of writing.

Bonded amalgams utilize any one of the current generation of enamel/dentin bonding agents to seal the dentin and "bond" the amalgam to the prepared tooth. The amalgam is condensed into polymerizing dual cure resin that is placed on conditioned dentin and etched enamel. The amalgam or the oxide of amalgam will chemically bond to the resin, and the resin is bonded to the tooth structure primarily by micromechanical retention. Thus the resin will partially retain the amalgam in place, reducing the need to create retentive undercuts in the tooth. Elimination of these undercuts, which have been shown to weaken cusps, coupled with the potential strengthening effect may improve the long-term prognosis of these restorations. Clinical research thus far has shown that bonded and nonbonded amalgams perform equally well. There seems to be little advantage to bonding amalgams, and the technique is more complex and time consuming.

THE BOTTOM LINE

1. Silver amalgam has been an important restorative material for many years and will continue to be an important material for years to come.
2. Advantages of silver amalgam are excellent physical properties, cost, long-term service, and the ability to self-seal the cavity.
3. Amalgam is the most user-friendly material in restorative dentistry.
4. High copper amalgams are superior to conventional amalgams demonstrating excellent marginal integrity over time. This is due to elimination of the gamma-2 phase (tin-mercury).
5. It is easier to create a good proximal contact with admixed versus spherical alloys.
6. Spherical alloys can be prepared 15 minutes after insertion and are indicated for amalgam core buildups.
7. Proper condensation of the amalgam is the most important manipulative variable.
8. Bonding amalgam to tooth structure is possible, but there is minimal clinical benefit.

Mercury and Silver Amalgam

Silver amalgam contains between 43% and 52% mercury when it is freshly mixed, prior to condensation into the prepared cavity. The procedures of condensation, precarve burnishing, and carving remove a portion of this mercury. The remainder of the mercury remains in the finished restoration. Because high doses of mercury vapor can cause serious systemic illness, this mercury remaining in silver

amalgam restorations has been a source of continued controversy for over a century.[6,15] This controversy continues unabated today in spite of a total lack of scientific evidence demonstrating a link between silver amalgam and systemic disease. In fact, quite to the contrary, ample scientific data support the continued use of silver amalgam and document its enviable record of safety.

Historical Review

Amalgam has been the source of controversy ever since it was first introduced to North America by the Crawcour brothers in 1836. These charlatans from France placed their version of amalgam, called "Royal Mineral Succedaneum," into the teeth of patients at carnivals and medicine shows. The material, which was derived from filings from silver coins mixed with large amounts of mercury, tended to expand significantly on setting. This resulted in an incredibly high incidence of postoperative pain, tooth fracture, and elevated occlusal contacts. These postoperative sequelae, coupled with the use of a potentially toxic material such as mercury, precipitated the first amalgam war.

The debate raged furiously for many years, dividing organized dentistry into two main camps. One group—naturally intrigued with the possibilities of a truly plastic material that could be placed into the mouth in a softened state, conformed to the cavity preparation in the tooth, and then hardened—sought to continue to use the material and to experiment with modified formulations. The other group—which comprised the main body of organized dentistry at the time—chose to outlaw amalgam and in fact drafted the now infamous "Amalgam Pledge." The essence of the pledge was that the signer agreed to never consider using silver amalgam. Anyone wishing to belong to the organization was required to sign this pledge.

In 1850 the Amalgam Pledge was rescinded and a period of scientific experiment began, led by individuals such as Hickock, Flagg, and others. This period culminated in 1897 when Black published his formula for amalgam alloy. This alloy, and minor variations thereof, was the type essentially used by all North American dentists until the serendipitous discovery of modern high copper alloy by Innes and Youdelis in 1963.[16] Because silver amalgam is relatively strong, inexpensive, and essentially user friendly, it became the most popular restorative material for posterior teeth in the latter half of the 20th century and continues to be used extensively today. In spite of this popularity and tremendous utility, silver amalgam has had to survive periodic attacks relative to its safety.

A second amalgam war resulted from the writings of a German medical doctor named Stock. His writings appeared in the European medical literature and consisted of anecdotal reports related to mercury toxicity in patients with silver amalgam restorations. No controlled scientific data was reported, and in fact Stock later repudiated that information in his own articles in the 1940s. However, at the time, his statements were sufficiently provocative that scientists initiated a number of studies that would eventually help to establish the safety of this fine material.

In the 1990s, renewed controversy surrounding amalgam resulted in what might be called the third amalgam war. This particular episode in the turbulent history of amalgam has been precipitated by a combination of renewed scientific investigation coupled with an unprecedented amount of charlatanism and media hype.[17] To the uninformed, the safety of silver amalgam is once again in question. This doubt concerning its safety, coupled with an ever-increasing demand by the public for tooth-colored posterior restorations and a concomitant decrease in the incidence of caries in developed countries, has resulted in a drastic reduction in the use of silver amalgam. The reduction in the caries rate is a remarkable achievement, and the desire for more esthetic restorative materials is understandable. However, the widespread practice of removing functional amalgam restorations in the name of patient safety is to be deplored. This results in unwarranted loss of tooth structure, unnecessary expense, and, especially if the replacement restoration is tooth colored, limited longevity.

Perhaps a brief review of the classical and contemporary literature on the topic of mercury in silver amalgam will clarify the issue. For those who would prefer to simply receive the bottom line on this issue, it is this: There is absolutely no scientific evidence to indicate that silver amalgam restorations pose a health risk of any kind to patients. There is a large body of well-conducted and controlled scientific literature to support this statement. Unwarranted removal of amalgam restorations in order to prevent or cure systemic health conditions is wrong and in fact is considered unethical in many jurisdictions.

Classical Research

There are a number of classic studies that had established the safety of silver amalgam to the satisfaction of the scientific community up until the findings of the 1980s. The broad body of this research has been the subject of an excellent review by Bauer and First in 1982, and only a few classic papers will be reported here.[18] The first of these was conducted primarily to repudiate the writings of Stock, published in 1931.[19] A selenium sulfide mercury detector was used to determine if mercury vapor was released from the surface of silver amalgam. The solubility of amalgam in a number of acidic solutions was also tested. No mercury vapor was detected, nor was there any tendency of the amalgam to dissolve in any of the solutes tested. The results of this study corroborated the commonly held belief that the mercury in silver amalgam is chemically reacted and tied up within the body of the amalgam restoration.

A second study that was part of a multifaceted evaluation of industrial mercury measured urinary mercury in patients with multiple amalgam restorations versus patients with no amalgam restorations.[20] No differences in urinary mercury were found, lending credence to the belief that mercury from silver amalgam did not contribute to the overall body burden of mercury.

As part of an in-depth study of mercury in amalgam, Frykholm radioactively labeled mercury and then placed multiple amalgam restorations in a number of patients.[21] He then monitored the amount of radioactivity in the urine over time. The level of radioactivity rose after the restorations were placed, and then dropped back to zero after 8 days. This indicated that there is a small but measurable exposure to mercury vapor when amalgams are placed, but the mercury is then excreted a few days after placement.

Thus, based on these studies, years of use of the material with few reported side effects, and numerous other studies, most authorities were confident of the safety of silver amalgam. However, new studies reported in the 1980s and 1990s have questioned that conclusion.

Research Questioning the Safety of Silver Amalgam

In 1981 researchers at the University of Iowa reported that small amounts of mercury vapor were released from the surface of amalgam restorations when patients chewed gum.[22] This finding has since been confirmed by other researchers.[23] Presumably, sufficient heat is generated during mastication that a miniscule amount of mercury vapor is released intraorally. A portion of this mercury vapor will

undoubtedly be absorbed into the bloodstream. This fact has been accepted by the scientific community, but considerable debate has occurred concerning the actual amount of mercury that is released and whether or not it represents a risk for the patient. This will be discussed in some depth in the subsequent section.

Additional research has established that patients with multiple amalgam restorations have slightly higher blood mercury levels than do patients with no amalgam restorations.[24] Autopsy studies have also shown that patients with multiple amalgam restorations have slightly elevated levels of mercury in the kidneys and in certain areas of the brain.[25,26] Again, the debate hinges on the question of whether these relatively minor elevations of mercury have any demonstrable clinical consequence. At this time, the evidence supporting the safety of amalgam is overwhelming, and there is not a shred of scientific data to support the negative viewpoint.

Much of the emotional outcry against silver amalgam has resulted not from scientific data reported in peer reviewed journals, but from anecdotal stories and sensationalized media coverage of such anecdotal information. Such anecdotal data are simply not admissible in scientific circles. In these case histories, patients have experienced a vague series of symptoms, and an approximate diagnosis is made but not necessarily confirmed. Solid baseline data are never recorded, and some treatment, in this case usually removal of some or all of the existing amalgam restorations, is performed. Rarely is this done in a controlled manner; and often the treatment is carried out concomitantly with a number of additional therapies, including a regime of various medications, alteration in diet, etc. The patient may report some short-term amelioration in symptomatology, again in a very subjective manner. This improvement may or may not be real; if real, it could be a result of any one of the therapies that have been utilized or it could be the result of a placebo effect. No attempt seems to have been made to gather long-term data or to present that data in the scientific literature.

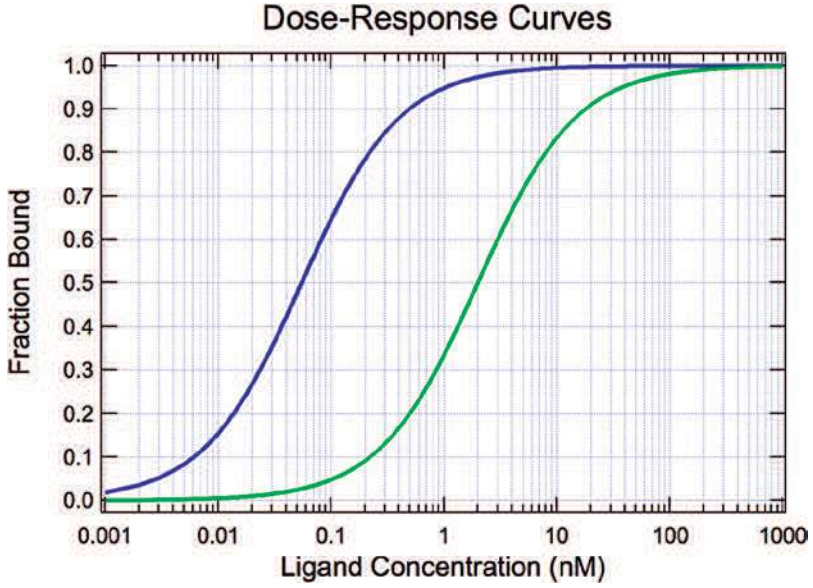
Critique of Anti–Amalgam Research

Most of the hysteria regarding mercury and amalgam has resulted from the anecdotal accounts mentioned previously. These reports have virtually no scientific validity. Medicine and dentistry have made tremendous strides in the past century, by insisting that all proscribed therapies be based on data from controlled scientific studies. Adhering to the scientific method has been essential in developing sound techniques of diagnosis and predictable therapeutic modalities. To be guided by anecdotal reports would be tantamount to returning to the days of witchcraft and medicine shows.

It is important to understand that what is of consequence in this discussion is mercury vapor. Liquid mercury and particulate amalgam are of no importance except to the extent that mercury vapor may be emitted from their surface. High levels of mercury vapor can produce definite neurologic symptoms and be toxic to certain target organs such as the kidney. These effects are essentially toxic in nature. True allergy or hypersensitivity to mercury is extremely rare. It is often claimed that certain individuals are "hypersensitive" to mercury; however, this is an inappropriate use of the term. The implication is that these individuals and perhaps their neurologic and/or immune systems are negatively affected by extremely low doses of mercury. There is virtually no scientific evidence to support this claim.

The critical factor in evaluating the scientific literature related to mercury in amalgam is to determine the actual amount of mercury exposure a patient might receive from his or her amalgam restorations. Mercury, like most elements or therapeutic agents, has a classic dose-response curve (Fig. 13.20). With these therapeutic agents at very low doses, no clinical manifestations are observed; at slightly higher doses a therapeutic effect may be seen, with untoward side effects increasing as the dose is increased. At even higher doses, toxic effects become apparent. In the worst-case scenario the mercury exposure to a patient with amalgams in all of their posterior teeth would represent a dose in the first category, and virtually no clinical manifestations would be evident.

This statement may be made with absolute assurance when comparing the exposure of mercury that would occur from silver amalgam to that allowed a worker who has occupational exposure to mercury vapor. As with other potentially toxic materials, a safe level of mercury vapor that can be tolerated without untoward effects has been established by industrial safety boards. That level is known as the *threshold limit value (TLV)*. This level has been adjusted downward over the years as more and more is known



• **Fig. 13.20** Dose-response curves.

about mercury vapor and its effects. That level is currently either 50 μg/m^3 (U.S. Occupational Safety and Health Administration [OSHA]) or 25 μg/m^3 (World Health Organization [WHO]) and represents the amount of mercury vapor that can be safely allowed in the air for workers who spend 8 hours/day, 40 hours/week in that environment. The worst dose a patient could receive from 10 or 12 amalgam restorations would be about 1/100 of the current TLV.[27]

The actual dose that a patient might receive from amalgams has been the focus of a spirited debate in the literature. Vimy and colleagues calculated that the dose could be as high as 10 μg/day, but their results have been assaulted by other researchers; and it is generally accepted that this number is at least 10 times higher than reality.[28-30] It must be understood that the machines utilized to measure intraoral mercury are not designed to do that, but are used to measure mercury vapor in the air of a well-ventilated room. Thus certain calculations are required to convert the number retrieved from the mercury detector to the actual dose to the patient. Flow rates, respiration rates, and other factors must also be accounted for. When these factors are considered, Mackert concluded that the maximum potential dose to a patient would be in the area of 1.2 μg/day.[29] This number is in close agreement with the adjusted calculations of Berglund, an independent researcher who estimated the dose at 1.7 μg/day.[30]

Thus, the dose of mercury vapor to a patient from 10 or 12 amalgam restorations is extremely low and has virtually no clinical consequences. To put it into perspective, the amount of mercury vapor a patient could absorb per day from his or her amalgams would be 1/10 to 1/20 of that which they receive from a normal healthy diet.[31] Recently, calculations related to dose have estimated that for the most sensitive of patients to demonstrate even minor symptoms of mercury poisoning, patients would have to have 450 to 500 amalgam restorations in place.[32] Of course, this is patently impossible and illustrates that the doses of mercury vapor received by patients from their amalgam restorations are indeed miniscule.

A great deal has been made in the media of studies in sheep and in primates, which purport to demonstrate extremely high concentrations of mercury in the kidneys and other organs as a result of amalgam placement. Reduced organ function in these animals has also been reported. Rather than criticizing these extremely flawed studies, it is more pertinent to review existing human data on mercury concentration in these end organs and the function of these organs. From the autopsy studies mentioned previously, the concentration of mercury in the kidneys of humans with 10 to 12 amalgam restorations has been shown to be 0.38 μg/mg, which is 12 to 20 times less than that reported in the animal studies.[25] More importantly, an evaluation of enzyme function, a critical index of overall organ function, has demonstrated that patients with multiple amalgams have no diminution of normal kidney function.[33] In fact, this study demonstrated that there was no difference between patients with multiple amalgams and patients with no amalgam restorations in terms of kidney function.

Research Supporting the Continued Use of Silver Amalgam

It is clear that there is no scientific data to indicate that silver amalgam poses any significant health risk for patients. It is equally clear that there is an impressive body of research that demonstrates that amalgam is completely safe. The articles by Langan and Fan, as well as the more recent paper by Mackert, review the contemporary literature published subsequent to Bauer's initial review article.[27,34]

Brodsky published a classic epidemiologic article regarding the pregnancy outcome of dental personnel.[35] Using a large patient sample of women either employed in a dental office or married to a dentist, the outcome of pregnancy was compared to that of a nondental group. The underlying assumption of the study was that dental personnel are exposed to much higher concentrations of mercury vapor than are the general population. Thus, if mercury vapor exposure in the concentrations experienced in dental offices was a problem, a difference in pregnancy outcome between the two groups would be expected to emerge. The results of the study indicated that there was absolutely no difference between the two groups in any of the indices evaluated (spontaneous abortion, stillbirth, birth defects, etc.). It must be emphasized that the levels of mercury vapor to which dental personnel are routinely exposed, while usually within the established TLV, are far greater than those that any patient would ever be exposed to as a consequence of his or her amalgam restorations.

Directly related to this finding is the general health of American dentists. The American Dental Association has conducted surveys on the health of its member dentists for a number of years. The data are gathered at the annual meetings of the ADA. These surveys have unequivocally demonstrated that, in spite of their chronic exposure to higher than normal levels of mercury vapor, American dentists are slightly healthier than the general population. If indeed low levels of mercury vapor exposure were etiologic of various autoimmune and other disorders, one would expect an epidemic of such maladies to occur in the dental population. Such is clearly not the case.

Concern has been expressed regarding the potential for mercury in amalgam to suppress the level of circulating lymphocytes in the bloodstream. Most of this concern is based on a report that purported to demonstrate that patients with amalgams had suppressed numbers of T lymphocytes.[36] The scientific validity of this report has been questioned particularly in terms of sample size (three patients) and lack of controls. A recent study with ample numbers of patients has reported absolutely no difference in white blood cell populations (including T lymphocytes) in patients with and without amalgam restorations.[37] As mentioned, additional research has noted that the function of several organ systems is completely normal in patients with multiple amalgams and identical to that of patients with no amalgams.

Mercury is a neurotoxin that has been speculated to play a role in the pathogenesis of Alzheimer disease. An evaluation of 129 Roman Catholic nuns from 75 to 102 years of age indicated that existing amalgams are not associated with lower performance scores on eight different cognitive function tests.[38] Data-related amalgam status and dental history were correlated with mercury levels in the brain at autopsy of 68 patients with Alzheimer disease and a control group of 33 subjects without the disease.[39] There was no significant association with the number, surface area, or history of having amalgam restorations with differences in brain levels of mercury or Alzheimer disease.

It has also been speculated that long-term exposure to low levels of mercury vapor from amalgams can cause or exacerbate neurodegenerative diseases such as amyotrophic lateral sclerosis, multiple sclerosis, and Parkinson disease. An extensive review evaluated existing epidemiologic investigations and concluded that these studies had failed to provide evidence of a role of amalgam in these degenerative diseases and there is no clear evidence supporting the removal of amalgams.[40]

Several countries have suggested that women should not receive dental amalgams during pregnancy because it could possibly result

in children with low birth weight. A population-based case control study of over 1000 women with low-birth-weight infants and over 4400 women who had normal-birth-weight infants found no evidence that silver amalgam fillings placed during pregnancy increased low-birth-weight risk.[41] A recent Norwegian study of over 108,000 pregnancies that occurred between 1999 and 2008 evaluated the number of amalgams in the mother's teeth and pregnancy outcome.[42] Investigators found no relation between the number of amalgam restorations and the incidence of preterm birth, low birth weight, malformation, or stillbirth.

A number of independent health agencies have extensively reviewed the issues of safety and efficacy of silver amalgam in recent years and have all concluded that available data do not justify either the discontinuance of use of amalgam or the removal of existing amalgam restorations. These agencies include the Food and Drug Administration (FDA) in 1991 and 2002, WHO along with the FDI World Dental Federation in 1997, the National Institutes of Health (NIH) and the National Institute of Dental Research in 1991, and the U.S. Public Health Service in 1993.[43,44] The National Council Against Health Fraud warns consumers about dentists recommending unnecessary removal of serviceable amalgam restorations and states, "Promoting a dental practice as mercury-free is unethical because it falsely implies that amalgam fillings are dangerous and that mercury-free methods are superior."[45]

The Life Sciences Research Office, a nonprofit scientific organization in Bethesda, Maryland, released results of its independent review of all articles published on this topic between 1996 and 2004. Among 950 published articles, 300 were accepted based on the scientific methodology. The report concluded that "there is little evidence to support a causal relationship between silver fillings and human health problems."[46]

Two very important randomized controlled clinical trials were published in the *Journal of the American Medical Association* in 2006.[47,48] One was called the New England Children's Amalgam Trial and the other, the Casa Pia Children's Trial. Both studies included over 500 children who received multiple restorations of either composite resin or amalgam. One of the trials was 5 years in duration, the other 7 years. They measured subtle neurobehavioral effects to determine if there were any negative effects due to amalgam. Both studies basically found that there were no neurobehavioral effects that could be attributed to amalgam. In fact the very scant data suggested that composite resin may have more effects than amalgam, but the data were of very low scientific validity.[49-51] The studies also showed that composite resin restorations needed to be repaired or replaced 7 times more than amalgam restorations and that composite resin restorations were 2.5 times more likely to have recurrent caries.

Environmental Impact

Concerns have been raised regarding the impact mercury in amalgam has on the environment. Major environmental sources of mercury include volcanoes (50%), coal combustion (33%), and gold production (6%). Amalgam's only contribution in this area is in waste disposal, largely through mercury resulting from crematories, and is less than 1% of the total. Many jurisdictions require dental offices to utilize sophisticated amalgam separators, and a 2016 FDA regulation requires all offices in the United States to install such separators within 3 years.

In January 2013, a number of nations signed the Minamata Convention in Geneva. This meeting was convened to discuss the influence of mercury on health and the environment. Discussions on amalgam were a small part of that meeting, and there were no curbs imposed on the use of silver amalgam. The group declared that the use of amalgam should be reduced in coming years through improved preventive measures and that research should be encouraged to develop materials that will work as good or better than amalgam.

An analysis of the data presented in this section leads to the conclusion that mercury in silver amalgam restorations poses absolutely no problem for dental patients. This conclusion has been reached by experts in the field, by consumer interest groups, and by thorough reviews by governmental agencies.

Impact of Unwarranted Amalgam Removal

A small minority of the profession currently recommends to their patients that their existing serviceable amalgam restorations should be removed as a health precaution. This is not only incorrect, but also unscrupulous, dishonest, and unethical. Removal of an amalgam restoration requires loss of additional tooth structure—sometimes considerable amounts of tooth structure if an indirect restoration that requires a certain amount of "draw" is to replace the amalgam. This additional loss of tooth structure, coupled with the inevitable inherent operative trauma, can result in compromise of the pulpal tissues and eventually in the necessity for endodontic therapy.

The restorative alternatives to silver amalgam all carry with them certain disadvantages. Cast gold, while undoubtedly the finest possible posterior restorative service, is rather expensive (currently 10 times the cost of amalgam) and time consuming.[52] The cost of well-done cast gold restorations is simply prohibitive for many patients. The tooth colored alternatives are also more expensive, with direct composite resin restorations costing about 3 times the cost of an amalgam, while indirect resin or porcelain inlays are closer to 10 times as expensive. None of these tooth colored alternatives has a clinical track record of success equal to that of amalgam, and some of the procedures must be considered to be essentially experimental at this time. Suffice it to record here that some of the problems with them are fracture, excessive wear, microleakage at the cervical margin, and extreme technique sensitivity.

One other tragic situation that has occurred frequently in recent times is the false hope given to patients suffering from incurable diseases such as multiple sclerosis and other autoimmune conditions that by removing their amalgam restorations they would be cured. Amalgam is not the cause of such ailments; nor is amalgam removal the cure. The authors have personally met many individuals inflicted with such dreaded diseases who have undergone great suffering and expense in having their amalgams removed only to have the symptoms continue.

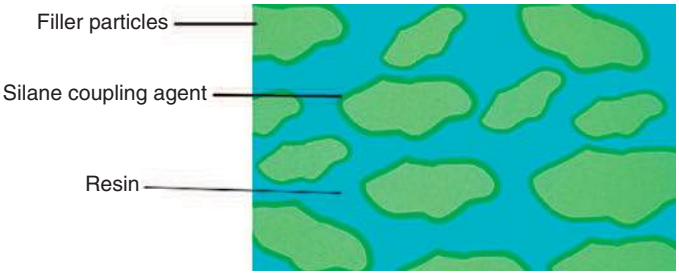
THE BOTTOM LINE

1. Small amounts of mercury vapor are released from the surface of silver amalgam restorations during mastication.
2. The amount of mercury vapor released is very small and poses no risk to the integrity of the restoration or to the systemic health of the patient.
3. There are virtually no published scientific papers validating the belief that mercury in silver amalgam is potentially dangerous to patients.
4. There is an ample and growing body of research that demonstrates the safety of silver amalgam.
5. This research plus the over 100-year history of use of the material supports its continued use.

6. While mercury in silver amalgam poses no health risk to dental patients, even in dental offices where excellent mercury hygiene is practiced, dental personnel are occupationally exposed to higher levels of mercury vapor than the general population. This is due to vapor that is released when amalgams are placed and, even more importantly, when they are removed, or when teeth with amalgam core buildups are prepared for crowns. While the levels of mercury vapor to which dental personnel are routinely exposed are well above those of the general population, they are generally below the current TLV. This occupational exposure to mercury vapor poses no risk as long as proper procedures for mercury hygiene are utilized. All dental personnel should be familiar with the procedures essential for the proper practice of mercury hygiene.



• **Fig. 13.21** Class III cavity restored with poly(methyl methacrylate) (PMMA).



• **Fig. 13.22** Main components of composite resins.

Composite Resins

History

Prior to the introduction of composite resin materials, dentists had two types of tooth colored restorative materials that could be utilized to restore carious defects in anterior teeth: silicate cement and poly(methyl methacrylate) (PMMA).

Silicate cements were supplied as a powder and liquid, with the main constituent of the powder being ground alumino-silicate glass and the liquid phosphoric acid. These materials demonstrated decent initial esthetics but had poor physical properties and color stability and most importantly dissolved readily in oral fluids.

Silicate restorations were expected to serve 3 to 5 years, at which time they would have to be replaced. The main advantage of this material was that as it dissolved, it released substantial amounts of fluoride, with the result that recurrent caries almost never occurred with silicate restorations (see Fig. 13.9).

The second available tooth colored direct material was PMMA (Fig. 13.21). These materials were also supplied as a powder and liquid. The powder was essentially ground prepolymerized methyl methacrylate and initiator, and the liquid was monomeric methyl methacrylate. PMMA materials provided good initial esthetics but also had poor physical properties and lacked color stability; however, their main disadvantage was considerable polymerization shrinkage (three times the shrinkage of modern composites) resulting in large contraction gaps between the restoration and the cavity margins. These gaps permitted significant microleakage that resulted in a very high incidence of recurrent caries.

Thus, while PMMA materials were chronologically developed later than silicate materials, they were actually inferior in terms of clinical performance.

Glass ionomer restorative materials were developed in the 1970s and later evolved into resin-modified glass ionomer materials.[53] These materials are currently indicated for the restoration of root caries lesions.

Composite Resin Materials

These materials were introduced to the profession in 1955 by Dr. Raphael Bowen.[54] It is interesting to note that the discovery of the concept of acid-etching enamel by Dr. Michael Buonocore occurred almost at the same time.[55]

The initial composite resin materials were superior to their predecessors (silicate and PMMA) in terms of physical properties, color stability, solubility, and clinical performance. However, these initial materials were not suitable for use in posterior teeth and had problems resulting from polymerization shrinkage even though the amount of that shrinkage was substantially less than the shrinkage of PMMA materials. Composite resin materials have evolved constantly over the past 50 years and contemporary composite resin materials are vastly superior to the original material in both clinical performance and esthetic potential.

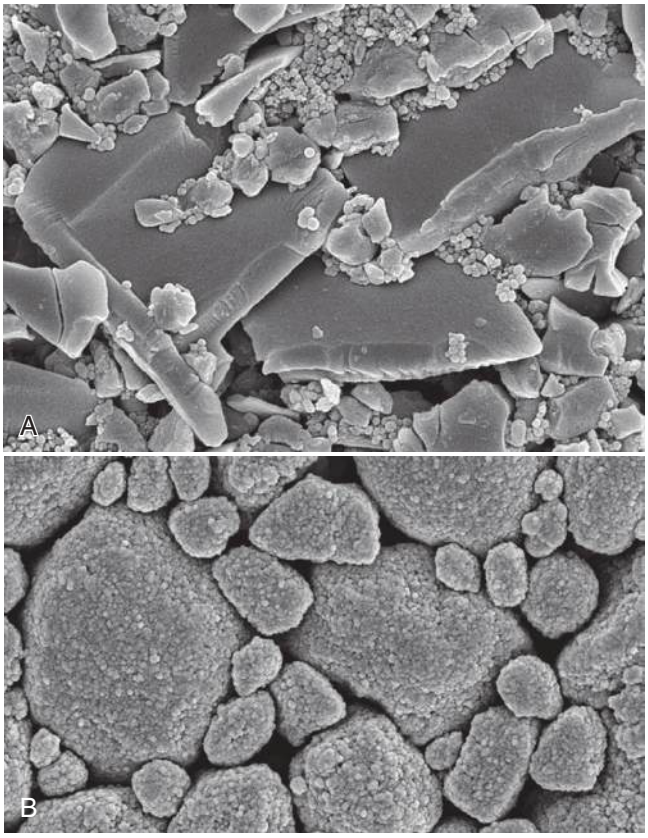
Composition of Composite Resin Materials

There are three main components of all composite resin materials (Fig. 13.22). These are the resin matrix; filler particles, which vary in composition, size, and shape; and silane coupling agents, which chemically bind the filler particles to the matrix.

The resin matrix of most North American composite resin materials is Bis-GMA. Some composite resins utilize urethane dimethacrylate (UDM) as the matrix and both resins are substantially equivalent in terms of clinical performance. Bis-GMA was synthesized from a type of epoxy by Bowen. It sets by means of a polymerization reaction and its major advantage over methyl methacrylate is that the monomeric molecule is substantially larger than monomeric methyl methacrylate. This results clinically in significantly less polymerization shrinkage.

The filler particles in composite resin are composed of quartz or silica, and vary considerably in terms of size and shape (Fig. 13.23). The incorporation of fillers into the resin matrix increases the strength of the material and further reduces the amount of polymerization shrinkage.

There are two rules regarding filler particles in composite resins. The first is the more filler, the better. Increasing the relative amount of filler in a material increases the strength and reduces the amount of polymerization shrinkage. Most modern composites experience a 2.4% to 2.8% polymerization shrinkage. A contemporary composite resin material for restorations should contain at least 75% filler by weight. Most modern materials meet or exceed this specification.



• **Fig. 13.23** A and B, SEM of composite resin displaying different sizes and shapes of filler particles. (Images courtesy Dr. Jorge Perdigao.)

The second rule regarding filler particles is the smaller the average filler particle size, the better. Materials with small filler particles are inherently more polishable, retain their polish for longer periods, and have improved wear resistance. One difficulty that manufacturers have faced is that it is difficult to maintain a high filler content with very small filler particles. This is because small filler particles possess a high ratio of surface area to weight, and this resulted in materials that were not usable with high filler content. Modern technology has resulted in a constant evolution of materials, the crux of which has been the ability to create useful materials with a high filler content using smaller and smaller particles.

The filler particles are chemically bound to the resin matrix by silane coupling agents. Silanes are complex, bifunctional molecules that have two different end groups, one of which bonds to the filler particle and the other bonds to the matrix. The use of silane allows for stronger bonding of the filler to the matrix, which improves wear resistance, and also permits the incorporation of more filler into a given amount of resin matrix. The silanization process is complex and involves coating the particles with silane and then utilizing acetone washes to thin the thickness of the silane to a monomolecular layer.

Composite resin materials also contain numerous other components including pigments, viscosity diluents, cross-linking agents, and initiators.

Classification of Composite Resin Materials

Several different classifications for composite resin restorative materials have been proposed over the years—for example, according to polymerization reaction initiation or according to size of filler particles.



• **Fig. 13.24** Chemical cure composite resin. (Courtesy BISCO.)

According to Polymerization Reaction Initiation

The polymerization reaction for composite resin materials can be initiated chemically or with visible light.

Chemical Cure. Initial composite resin materials were chemically initiated and were called chemical-cure, auto-cure, or self-cure composite resins. These materials had a limited working time and suffered from poor long-term color stability as they contained tertiary amines that turned yellow or orange after several years of service. Chemical cure materials are used today primarily as core materials for the buildup of extensively damaged teeth (Fig. 13.24).

Light Cure. The original light-cure materials were cured using ultraviolet light, but contemporary composite materials all utilize visible light, which will activate the initiator (camphorquinone). Light-cure materials have the substantial advantage of being able to cure on command and thus provide essentially infinite working time so that the restoration can be sculpted and shaped into final form prior to polymerization. They are also color stable due to the elimination of the tertiary amines (Fig. 13.25).

Dual Cure. Dual-cure composite resin materials, which utilize both chemical cure and light cure technologies, are primarily used as cements or core materials. With these materials, the setting reaction is initiated by exposure to visible light but the reaction will slowly continue over time in the absence of light (Fig. 13.26).

According to Filler Particle Size

Early composite resin materials utilized rather large filler particles and are called "macrofilled" composite resins. As esthetic composite bonding became popular, manufacturers developed composite resin materials with very small fillers that are called "microfilled" composite resins. Modern composite resin materials contain a variety of sizes of filler and are called "hybrid" composite resins. The evolutionary trend has been to develop hybrid composite resin materials with smaller and smaller fillers. Thus, contemporary hybrid composite resin materials may be subdivided into subgroups of "microhybrid" and "nanohybrid" composite resin materials. Other categories of contemporary composite resin materials include "flowable" composite resins and "packable" composite resins.

Macrofilled Composite Resin Materials. The original composite resin materials were macrofilled composite resins. These materials had a relatively high level of loading (approximately 75% by weight), with particles that ranged in size between 4 and 40 microns.

Commercial examples of macrofilled composite resins include Adaptic (Johnson & Johnson, New Brunswick, NJ) and Concise (3M Oral Care, St. Paul, MN). Because of the relatively high loading these materials were quite strong, but the rather large size of the filler particles resulted in two significant disadvantages.

First, these materials were not polishable. The filler particles were harder than the abrasives used to polish them. As these materials were polished, the relatively soft matrix was worn away, thus exposing the large, hard fillers; and the more these materials were polished, the rougher they got. This is obviously a problem for an "esthetic" restorative material (Fig. 13.27).

Second, these large particles could be "plucked" from the surface of a posterior restoration by opposing cusp tips and result instantly in a quantum loss of restorative material (40 microns). Thus the materials had very poor wear resistance, a major deficiency clearly demonstrated in early clinical trials (Fig. 13.28).



• **Fig. 13.25** Light cure composite resin. (Courtesy 3M.)



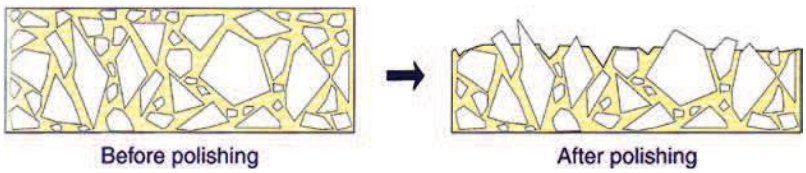
• **Fig. 13.26** Dual cure composite resin. (Courtesy BISCO.)

Microfilled Composite Resin Materials. The poor polishability of macrofilled composite resins directly led to the introduction of microfilled composite resin materials. These materials had a relatively low level of filler loading (45%–55% by weight), with a uniform filler particle size of 0.04 microns. This resulted in restorations that were very polishable and thus provided an excellent esthetic result. The low level of filler loading resulted in a material with a low modulus of elasticity (MOE), which made it unsuitable for stress bearing restorations. These materials were contraindicated for Class IV restorations and for the restoration of posterior teeth. This low modulus of elasticity made these materials the material of choice for the restoration of NCCLs, which many consider to have a flexural etiology.

The low MOE allows the material to flex along with the tooth. Commercial examples of microfilled composite resin materials include Silux Plus (3M Oral Care, St. Paul, MN), Renamel (Cosmedent, Chicago, IL), Durafil VS (Kulzer, South Bend, IN), and A110 (3M Oral Care, St. Paul, MN). Current evidence indicates that microfilled composite resins may not be needed because the filled bonding agents act as stress breakers.

Hybrid Composite Resin Materials. Hybrid composite resin materials have a combination of small and large filler particles, to combine high filler loading with a sufficient quantity of small particles to ensure improved wear resistance and also permit adequate polishability. These materials have been in use since the early 1980s and have proven to be excellent materials for both anterior and posterior restorations. Most contemporary hybrids have a filler content of 75% to 80% by weight, and the trend has been to produce materials with smaller and smaller average filler particle size. With most contemporary materials the largest filler particle would be in the range between 1 and 2 microns. These materials are quite strong, polishable, and have wear resistance equivalent to amalgam (6–15 microns/year). The esthetic potential of these materials is excellent and many products offer a wide range of shades and translucencies. Typical commercial products in the hybrid category include Herculite XRV (Kerr Corporation, Orange, CA), Z100 (3M Oral Care, St. Paul, MN), and TPH (DENTSPLY Sirona, York, PA).

The most contemporary hybrid composite resin materials have been described as microhybrids and nanohybrids. These materials contain a mixture of small and smaller particles and have excellent handling characteristics and esthetic potential. High filler loading is made possible with advanced technology that permits the formation of clusters of very small particles that reduce surface area yet function as individual nanoparticles to ensure adequate wear resistance.



• **Fig. 13.27** Schematic illustration of macrofilled composite resin before and after polishing.

Commercial products in this category include Esthet-X (DENTSPLY Sirona, York, PA), Point 4 (Kerr Corporation, Orange, CA), and Filtek Supreme (3M Oral Care, St. Paul, MN) (Fig. 13.29). The major advantage of nanocomposites is that they polish to a very high luster and they maintain that luster over time.

Packable (Condensable) Composite Resin Materials. Several products have been introduced to the market in recent years that have been described as "packable" or "condensable" composite resins. These materials have been designed by manufacturers to possess handling characteristics similar to amalgam.

These materials are basically hybrid composite resins to which large fillers have been added that can be condensed or packed in a manner similar to amalgam.

The addition of these large fillers has resulted in documented poor wear resistance of some of these materials. While some dentists prefer the handling characteristics of these materials, it is doubtful that the benefits of improved handling offset the reduction in clinical performance that has been demonstrated by some of these products.

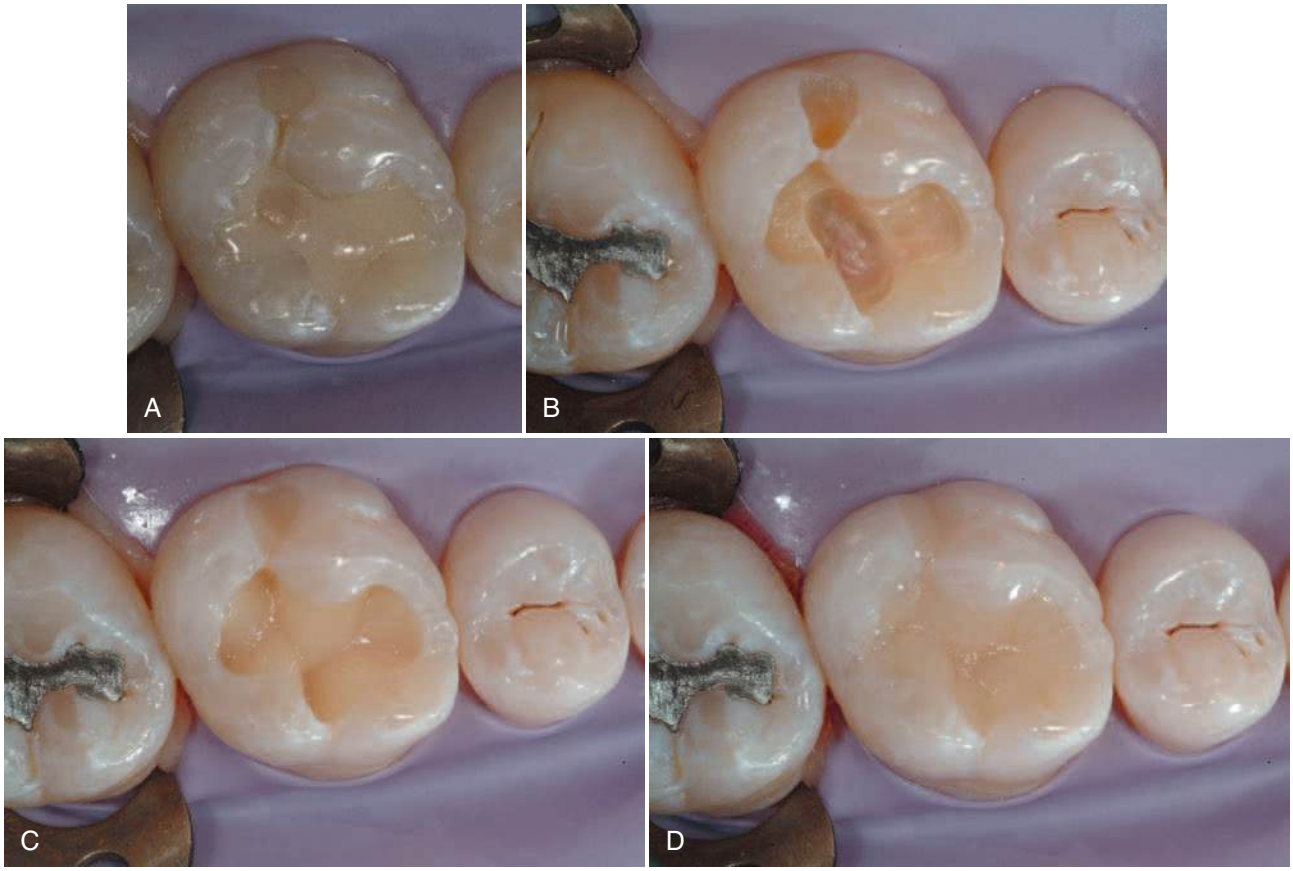


• **Fig. 13.28** Teeth numbers 12, 13, 14 restored with macrofilled composite resin.

Flowable Composite Resin Materials. In recent years manufacturers have introduced a group of materials designated as "flowable" composite resin materials. These materials represent a wide range of products, with filler content ranging between 35% and 65% by weight. These materials are very convenient to use because their viscosity allows them to be injected into the cavity preparation with a syringe. However, as would be expected, the reduction in filler content is accompanied by reduced physical properties and increased polymerization shrinkage. The use of these materials should be restricted to "niche" functions such as a 0.5-mm-thick liner under large posterior composite restorations.

Low Shrinkage Composite Resin Materials

Most modern hybrid composite resin materials exhibit linear polymerization shrinkage of between 2.2% and 2.4%. This amount of shrinkage, if not compensated for by clinical technique, can



• **Fig. 13.29** A–D, Tooth number 15 restored with a hybrid composite resin. (Photos courtesy Dr. Phillip Timmins.)

result in contraction gap formation, microleakage, postoperative sensitivity, and a potential to develop recurrent caries.

Some manufacturers have introduced products to the market that are described as "low shrinkage" composite resin materials. The linear shrinkage of these materials ranges between 1.4% and 1.7%. The reduction in shrinkage has been accomplished by technology that permits higher filler loading. These materials have not undergone extensive clinical testing at the time of writing. More clinical evidence of their efficacy is essential before they can be recommended for routine use.

Recently, composite resin materials based on silorane technology have been introduced on an experimental basis. These materials use a resin polymer that exhibits very low shrinkage and thus does not depend on increased filler loading.

As with other low shrinkage materials, more clinical evidence is required before these materials can be considered for routine use.

Bulk-Fill Composite Resin Materials

Bulk-fill resin materials were recently introduced to allow clinicians to divert from using an incremental layering technique when placing composite restorations, thus simplifying the procedure and reducing chair time. This technique was developed to minimize negative effects of polymerization-induced shrinkage stress and curing light penetration. The undesired side effects—a time-consuming and complex clinical procedure—contribute significantly to make composite restorations technique sensitive.[56]

Different from traditional composites, which are typically restricted to increments of 2 mm or less, these materials are designed to be used in increments of 4 to 5 mm.[57,58] Common manufacturers' claims involve greater depth of cure and lower polymerization-induced shrinkage stress. Although these composites are usually referred to as if they were clearly identifiable as a class of materials—bulk-fill composites—scientific evidence contradicts the assumptions that they would behave similarly.[59,60] The variety in handling characteristics, as well as physical and mechanical properties, is a direct consequence of the wide range of different compositions used by manufacturers, largely related to their filler content.[61]

The different types of bulk-fill composites currently available can be categorized as *flowable base bulk-fill composites* and *full-body bulk-fill composites*. Flowable base bulk-fill composites require a conventional composite as the occlusal increment and therefore are used for dentin replacement only; full-body bulk-fill composites can replace dentin and enamel in a single increment.[60] Overall, bulk-fill composites are more translucent than other composite resin materials. To a certain extent, this is attained by decreasing the filler content and increasing the filler size. Increased translucency facilitates light penetration, which is closely associated with depth of cure and increased polymerization rates toward the deepest areas of the restoration.[62] Evidence appears to support light-curing times of at least 20 seconds.[60] Despite the large array of materials being marketed, bulk-fill composites appear to deliver an increased depth of cure and reduced polymerization-induced shrinkage stress.[63,64]

Manipulative Variables With Composite Resin Restorative Materials

The contemporary clinician has a wide variety of quality composite resin restorative materials from which to choose. *How* these chosen materials are manipulated in the clinical setting is more important to long-term results than *which* material is chosen. The following narrative will briefly discuss some of the important variables when using composite resin restorative materials.

I. Most contemporary composite resin materials are cured or set by means of exposure to visible light using a curing light. In general, most composite resin materials should not be cured in thick sections, and the recommended thickness of an increment of composite resin is about 2 mm. Curing sections greater than 2 mm may result in insufficient curing of the undersurface of the composite resin material. The depth of cure is determined by a number of variables:
   - The shade of the composite resin
   - The thickness of the material
   - The irradiance of the curing light, and the time of cure

II. Curing lights should be checked periodically with a radiometer (most curing lights have one built in) to ensure that the light emitted is of sufficient output irradiance to efficiently cure the material. The light should be held in close proximity to the material (the closer the light, the greater the irradiance) and the composite resin should be exposed to the light for a sufficient time frame. In this regard, the manufacturers' recommended curing times should be closely followed.

III. The major deficiency of composite resins is their inherent polymerization shrinkage (2.2%–2.4%). This can result in the formation of substantial contraction gaps between the restorative material and the cavity margin, which can result in postoperative sensitivity and recurrent caries. The shrinkage of composite resin materials cannot be avoided, but the stress created by the shrinkage can be reduced by the placement technique. Reducing polymerization shrinkage stress can result in intact restoration margins and the prevention of microleakage.

There are five primary strategies that can be used to reduce polymerization shrinkage stress:

- The first is to create a relatively thick primed layer with the chosen dentin bonding agent. OptiBond FL (Kerr Corporation, Orange, CA) has a relatively high amount of filler content and, when properly used, will result in a thick primed layer. This primed layer has a relatively low modulus of elasticity and can thus deform slightly when the composite resin is curing and shrinking and thus absorb some of the stress and maintain the integrity of the tooth/restoration bond.
- The second strategy is to use a thin liner (0.5 mm) of resin-modified glass ionomer (Vitrebond Plus, 3M Oral Care, St. Paul, MN) under the composite resin restoration. These materials bond predictably to tooth structure and also have a low modulus of elasticity to absorb the shrinkage stress.
- The third option is to place a thin layer (0.5 mm) of flowable composite resin as a liner. This again will theoretically effectively absorb the shrinkage stress.
- A fourth concept that may be used to reduce shrinkage stress is the use of "soft-start" polymerization.[65] This approach utilizes a number of procedures to prolong the gel phase of the setting reaction of the composite resin material. The theory is that the longer the setting composite can maintain a gel or flowable condition, the better it can distribute the stresses resulting from shrinking. Procedures to do this include the use of ramped lights, pulse curing, or simply holding the curing light some distance from the material for the initial exposure. Several clinical and laboratory studies have demonstrated the effectiveness of soft-start polymerization, while other studies have demonstrated limited benefit.
- The final approach to reduce stress is to place the composite resin in increments, always keeping in mind the concept of



• **Fig. 13.30** Glass ionomer restorative material. (Courtesy GC.)

"C-factor."[66] The C-factor relates to the configuration of the cavity walls, and is the relative ratio between the bonded surface area of the composite resin to the unbonded surface area.

A class IV composite restoration has a wealth of unbonded surface area and a relatively small bonded area and thus has a low C-factor. The unbonded composite resin material will shrink toward the bond, thus reducing the shrinkage stress.

On the other hand, small Class I cavities have a high C-factor because they have a huge bonded area and relatively small unbonded area. When these restorations are cured, there is a high amount of competition for the bond, and the result is that the weakest bond (pulpal floor) gives away under the shrinkage stress creating a fluid-filled gap below the restoration and a high likelihood of postoperative sensitivity. These restorations should be placed in anatomical increments so that the material can shrink freely in one direction and thus eliminate or reduce the shrinkage stress.

Summary and Conclusions About Composite Resin Materials

Contemporary composite resin materials have improved significantly over the years. Today's materials have good wear resistance and a wide variety of shades and translucencies. They are quite polishable and provide excellent esthetics.

However, the major disadvantage of composite resin materials is the inherent polymerization shrinkage. This results in a significant level of technique sensitivity. Composite resin restorations must be placed with excellent isolation (rubber dam), in conjunction with meticulous bonding procedures. The placement technique itself is critical, as is meticulous finishing and polishing. Properly placed, composite resin materials can provide a very esthetic and durable restoration, and the results are gratifying to both patients and clinicians.

THE BOTTOM LINE

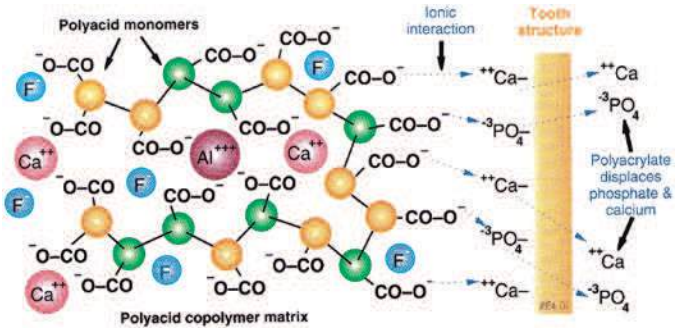
1. There is a significant interest by the public in tooth colored restorative materials.
2. Composite resin restorative materials have improved significantly over the years.
3. High filler content (75% weight) of small filler particles (0.04–1 micron) is the key to that improvement.
4. Polymerization shrinkage (2.4%–2.8%) is the main disadvantage of composite resin materials. This can be accounted for with proper technique.
5. Composite resins must be used in conjunction with contemporary bonding systems.
6. Excellent isolation is critical to success with composite resin materials.

Glass Ionomer Restorative Materials

Glass ionomers (GI) were introduced to the profession by Wilson and Kent in the early 1970s, and since then have evolved into a number of unique dental materials.[53] These materials tend to be misunderstood by many in the profession, and as a result many clinicians underutilize glass ionomer materials, whereas other clinicians tend to overutilize glass ionomers and use them in situations where they are clearly not indicated. This section describes available glass ionomer products and also provides specific indications and contraindications for the clinical use of glass ionomer materials (Fig. 13.30).

Glass ionomer materials are used as dental cements/luting agents, cavity liners, pit-and-fissure sealants, restorative materials (primarily Class V), bases, and core buildup materials. Based on clinical trials and physical properties testing, the efficacy of GI for pit-and-fissure sealants and for large core buildups is questionable.

All GI materials have a number of features in common. While some materials have multiple setting mechanisms, the basic GI setting reaction is an acid-base reaction with polyacrylic acid acting on and partially dissolving alumino-silicate glass. This results in a "burst" of fluoride release at a very high level, but this burst of fluoride release is completed in 24 hours. After that, there is a continual low-level release of fluoride that is likely well below the threshold necessary to protect against secondary caries. Most publications on GI materials state that GI is a fluoride releasing material, which implies protection from secondary caries. However, GI materials must be "recharged" with fluoride, which can then be re-released in order to provide protection. This ability for GI materials to act as "reservoirs" for fluoride will be discussed later.



• **Fig. 13.31** Schematic illustration of ionic interaction between glass ionomer ions and tooth structure. (From Albers HF: *Tooth-colored restoratives: principles and techniques*. 2002, PMPH-USA.)



• **Fig. 13.32** Three different dispensing methods for RMGI cements: powder liquid, clicker, and automix. (Courtesy 3M.)

Glass ionomer materials form very predictable bonds to both enamel and dentin. The bonds are weak chemical (chelation) bonds, but are predictable, effective, and durable (Fig. 13.31). In general, GI materials have inferior physical properties compared to amalgam and resin composite.

Types of Glass Ionomer Materials

Dental Cements/Luting Agents

Conventional Glass Ionomer Cements

Conventional glass ionomer cements are supplied as powder-liquid materials, with the primary component of the powder being ground aluminosilicate glass, and the liquid consisting of polyacrylic acid with small amounts of itaconic and tartaric acid. Examples include Fuji Type I and Ketac-Cem. The setting reaction is a chemical acid-base reaction.

These cements bond to tooth structure and have good physical properties compared to the "gold standard" cement, zinc phosphate. They release fluoride although it is doubtful that this is of any clinical significance.

The main disadvantage of these cements is that they have been associated anecdotally with an increased incidence of postoperative sensitivity. Clinical trials have demonstrated that this sensitivity is likely related to technique (e.g., overdrying the prepared tooth) or contamination of the setting cement by saliva. With proper technique, the level of postcementation sensitivity should be very low.

Conventional GI cements have decreased in popularity in recent years because of the development of resin-modified GI cements.

Resin-Modified Glass Ionomer Cements

Resin-modified glass ionomer (RMGI) cements are the most popular cements available today. Examples include Rely-X Luting and Rely-X Luting Plus cements, Fuji Plus and Fuji-Cem cements. Resin monomers have been incorporated into these materials, and thus these cements have two setting reactions: the classic acid-base reaction and a chemical-cure resin reaction.

Resin-modified GI cements bond to tooth structure and possess the same fluoride release characteristics as conventional GI cements. As a result of the addition of resin, they also possess superior physical properties, have improved handling characteristics, and are less technique sensitive.

Resin-modified GI cements have almost totally replaced zinc phosphate cement and are indicated for cementation of gold inlays, onlays, partial veneer and full veneer crowns and fixed partial dentures, metal-ceramic crowns and fixed partial dentures, and monolithic and layered zirconia crowns and fixed partial dentures. The manipulation of RMGI cements is relatively simple, making them relatively free of technique sensitivity. They are supplied with three different dispensing techniques (Fig. 13.32).

The original cement was supplied as a powder and liquid. It is important with powder/liquid cements to closely follow the manufacturer's directions regarding the ratio of powder to liquid. A later version of RMGI was the development of a "clicker" dispenser. With this product, it was simply a matter of dispensing the correct amount of cement and mixing. This simplified approach was partially responsible for RMGI cements gaining the lion's share of the cement market. The latest innovation has been the development of "automix" dispensers. With these devices the operator simply dispenses the mixed cement into the restoration and proceeds with cementation.

A short note on dispensing devices: It is clear from research conducted at the University of Washington and at the University of North Carolina that the same brand of cement manipulated by different dispensing systems has significantly different physical properties.[67] Whether or not this has clinical significance remains to be determined.

The prepared tooth should be cleaned with pumice or an intraoral air particle abrasion device if available. The use of a desensitizer such as Gluma Desensitizer or G5 is optional, but has been shown

to not only *not* interfere with bonding, but actually slightly enhances it with RMGI cements. A small amount of cement should be placed in the restoration and the restoration should be seated with firm finger pressure and then completely seated with the "dynamic seating" technique. The cement should be protected from saliva contamination while setting and should be allowed to set completely before removing the excess cement.

Many manufacturers of RMGI cements and some brands of resin cements are suggesting the operator should "preset" the excess setting cement with 5 seconds exposure to a curing light to plasticize the excess cement. The excess cement can then be removed readily with one or two deft moves with an explorer. This is not recommended as it is likely that cement just under the margin will also be removed along with the excess cement.

Glass Ionomer Restorative Materials

Resin-Modified Glass Ionomers

Glass ionomer materials have been formulated for use as restorative materials, with their primary function being restoration of noncarious cervical lesions and root caries lesions. The original materials were chemically similar to GI cements with higher glass content and increased particle size. Examples of these materials would be Fuji II and Ketac Cem. These materials have essentially been replaced with RMGI restorative materials, which are less technique sensitive, have better physical properties, show improved esthetics, and display vastly improved handling characteristics. Examples of RMGI restorative materials are Fuji II LC, Fuji Filling LC, Vitremer Restorative Material, and Ketac Nano.

The setting reactions of these materials are complex. All undergo the classic GI acid-base reaction as well as a light-activated resin polymerization reaction. Some also have an additional chemical-cure reaction and can set in the absence of light.

Resin-modified glass ionomer restorative materials are indicated for root caries lesions due to their excellent adhesion to tooth structure and fluoride release. They are also recommended for repair of recurrent caries around existing indirect restorations. (Removal of the existing restoration is generally preferred in these situations, but is not always practical.) RMGI restorative materials are the materials of choice for the restoration of NCCLs where esthetics is NOT of primary importance. RMGI materials have an excellent clinical history of retention with NCCLs, due to excellent adhesion and low elastic modulus, but the long-term esthetic result is compromised. Resin-modified glass ionomer restorative materials are contraindicated in stress bearing situations.

When placing RMGI restorations, the following technique is suggested. First, obtain the appropriate shade match. Then obtain the appropriate isolation with a retraction cord/cotton roll or preferably rubber dam. Complete the cavity preparation and then clean the preparation with the dedicated conditioner. Conditioners are formulations of 10% to 20% polyacrylic acid, which clean the smeared layer of dentin without opening the dentinal tubules and removing the smear plug. Follow the manufacturer's directions carefully. Some conditioners are designed to be washed off prior to placing the restorative material, while others are not. Mix the material. Some materials are mixed in a triturator (automixed), some are powder/liquid, and some are dispensed with a clicker. With Ketc-Nano, do not attempt to place the material until 30 to 40 seconds after mixing as it is initially very sticky and difficult to handle. Place and shape the material and light-cure for the recommended time, typically 20 to 40 seconds. Finish and polish. Placement of a surface glaze material is optional.

Reinforced Glass Ionomers

A number of reinforced glass ionomer materials have been introduced over the years in an attempt to expand the indications for GI materials. The following is a partial listing of such materials:

- Miracle Mix (Fuji II plus amalgam alloy particles)
- Ketac Silver (a glass cermet where the silver particles are chemically attached to the glass)
- Ketac Molar and Fuji IX (self-cure GI with increased aluminosilicate glass concentrations, polyacrylic acid, and polycarboxylic acid)
- Fuji IX
- Equia and Equia Forte
- Chem-fil Rock

Manufacturers claim that these materials are indicated as core buildup materials for extensively damaged or endodontically treated teeth. The physical properties of these materials are inferior to silver amalgam and resin composite, and thus they should *not* be used for core buildups. They can be used as bases and blockout materials.

Fuji IX is one of the most popular of these materials. It is indicated as a base under large composite resin restorations, as the intermediary material in an indirect pulp cap procedure, and as a temporary restorative material to replace missing restorations, broken cusps, etc. Fuji IX is mixed in an automix device. A conditioner of 10% polyacrylic acid is placed on the tooth for 20 seconds and washed off prior to placing Fuji IX.

Cavity Liners

There are several GI materials that have been specifically formulated to serve as liners. Among the most popular are Vitrebond and Vitrebond Plus. Both of these materials are light cured, and the difference between them is that Vitrebond is a powder/liquid material and Vitrebond Plus is dispensed with a "clicker." Liners are materials that are used at a thickness of no more than 0.5 mm under either amalgam or composite restorative materials. Their function is to seal deep dentin exposed during caries removal with the goal of having about 2 mm of thickness of cavity liner and remaining dentin between the restorative material and the pulp.

Glass ionomer liners are recommended when excavation of caries results in a remaining thickness of dentin that is likely less than 2 mm. They are optional any time the depth of a cavity preparation exceeds the depth of a "minimal" preparation.

Manipulation of glass ionomer cavity liners is straightforward. Complete the cavity preparation and clean and dry the cavity. Do *not* desiccate. Mix the lining material and place into the preparation in a thin layer (no more than 0.5 mm) using a Dycal applicator or similar instrument. Light-cure for 20 seconds. Now regular bonding procedures are completed when placing composite resin restorations, or Gluma Desensitizer or G5 is applied if placing an amalgam restoration.

A Note on Fluoride Release and Secondary Caries Prevention. Dentists understand that fluoride is an effective caries reducing agent and manufacturers understand that dentists will buy any material that contains fluoride. Dentists would likely purchase fluoride releasing tires if someone would manufacture them. However, it is critical to understand that the effectiveness of fluoride as a caries preventive agent is dose related. Fluoride releasing amalgam or composite resin has no chance of reducing caries because the dose of released fluoride is so low.

Glass ionomer–based materials are unique in that the set material consists of an aqueous polygel that permits ion exchange in the

oral cavity. This allows glass ionomer–based restorative materials to absorb topically applied fluoride from toothpastes, gels, etc., and slowly re-release that fluoride over 24 hours.

An almost universal misconception is that all GI materials release fluoride and that this released fluoride is effective in preventing secondary caries. As noted in the introduction, all GI materials do release fluoride. There are two phases to this fluoride release. The first occurs primarily in the first 24 hours after mixing and is described as a "burst" of fluoride release resulting from dissolution of the glass particles in the material by polyacrylic acid. After this 24-hour period, there is a sustained fluoride release over time, but this level of fluoride release is likely far below the therapeutic level essential to prevent secondary caries.

To reach a potentially therapeutic level, it is essential that GI materials are regularly (3×/day) exposed to sources of fluoride (e.g., toothpastes, gels, mouth rinses) that will allow the GI material to uptake fluoride and slowly re-release it into the surrounding environment.[68] Research has demonstrated that this does occur on a predictable long-term basis.

Repeated exposure is essential if the GI material is to act as a reservoir for fluoride. It is unlikely that a GI-based luting cement, in a 25-μm film thickness under a properly fabricated indirect restoration, would be able to absorb and re-release fluoride at a therapeutic level. A GI base or liner under a sealed restoration would not be exposed to fluoride to be absorbed and re-released. Similarly, under a leaking restoration, these materials would not be exposed to sufficient fluoride to be effective against recurrent caries. The reason these materials are preferred as bases and cavity liners is because of their predictable bond and seal of the underlying dentin, not because of their ability to prevent secondary caries.

Thus, the only GI materials that have legitimate potential for prevention of secondary caries are GI restorative materials such as Fuji II LC, Ketac Nano, and Fuji IX. It is important to understand that these materials will only be effective if they receive regular exposures to fluoride (multiple exposures/day).

Polyacid Modified Composites (Compomers)

Polyacid modified composites or compomers were released in the 1990s. They are modestly filled composite resin materials to which has been added some prereacted glass ionomer material. The important distinction between a glass ionomer product and a compomer is that there is *no acid-base glass ionomer reaction* with compomers. Examples of compomer products are Dyract, Compoglass, and F2000. Compomers are intended to be used in non–stress-bearing situations; and they have adequate esthetics, ease of handling, adhesion to tooth structure, but poor physical properties. They do release fluoride, but at a very low level, less than 10% of the release of glass ionomers. Thus it is unlikely that compomers have any meaningful effect on the incidence of secondary caries. There are likely no indications for use of compomers in contemporary dentistry.

THE BOTTOM LINE

1. Glass ionomer materials are useful for several clinical applications in restorative dentistry.
2. Glass ionomer materials provide predictable long-term bonds to enamel and dentin.
3. Resin-modified glass ionomer cement is the most popular cement in restorative dentistry and is indicated for cementation of metal castings and zirconia and porcelain-fused-to-metal based restorations.
4. Resin-modified glass ionomer liners have an excellent track record as lining materials used in films no thicker than 0.5 mm.
5. Resin-modified glass ionomer lining cements should be used to provide a liner/dentin thickness of 2 mm from the pulp.
6. Resin-modified glass ionomer restorative materials provide satisfactory service for the restoration of root caries and noncarious cervical lesions.
7. Reinforced glass ionomer materials should not be used for core buildups.
8. Glass ionomer materials do not provide protection from secondary caries with their inherent fluoride release.
9. Glass ionomer and RMGI materials may provide protection from secondary caries as a result of their ability to act as fluoride reservoirs. In order to accomplish this, they must be regularly exposed to topical fluoride from toothpastes, gels, mouthwashes, etc.

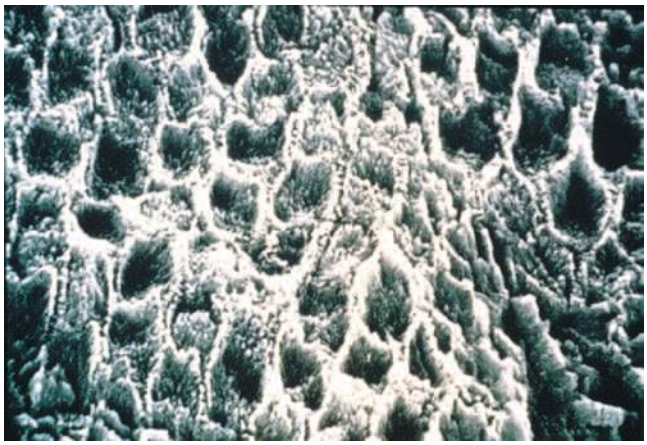
Adhesive Dentistry

Bonding to Enamel

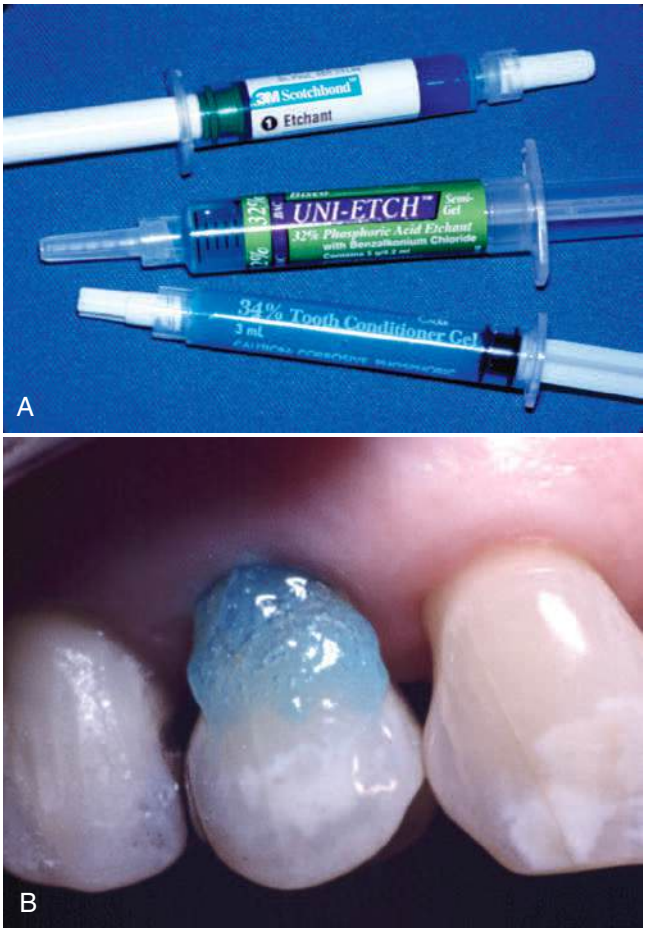
Of the many important scientific advances in restorative dentistry in the past century, perhaps none is as important as the discovery by Buonocore in 1955 of acid etching enamel.[51] Buonocore was investigating the use of phosphoric acid to clean fissures in molar teeth and recognized the retention potential of the etched enamel surface when he examined it microscopically (Fig. 13.33).

The first practical application of his discovery was pit-and-fissure sealants, although routine use of sealants was not to become popular for many years. The concomitant introduction of the first composite resin material by Raphael Bowen in 1957 made possible current concepts of adhesive dentistry.[50] It is worth noting that the concept of bonding composite resin materials to tooth structure did not become popular until the mid-1970s. It is also worth noting that at the present time (2017) the resin bond to acid-etched enamel is the most important bond in adhesive dentistry.

The original acid etchant was the liquid from zinc-phosphate cement, which was basically phosphoric acid. This has been replaced by brightly colored gels (Fig. 13.34A), which aid in ensuring the phosphoric acid is completely washed off the tooth. Phosphoric acid concentrations vary from 30% to 40% with the most common one being 37%. While extended application times may be okay



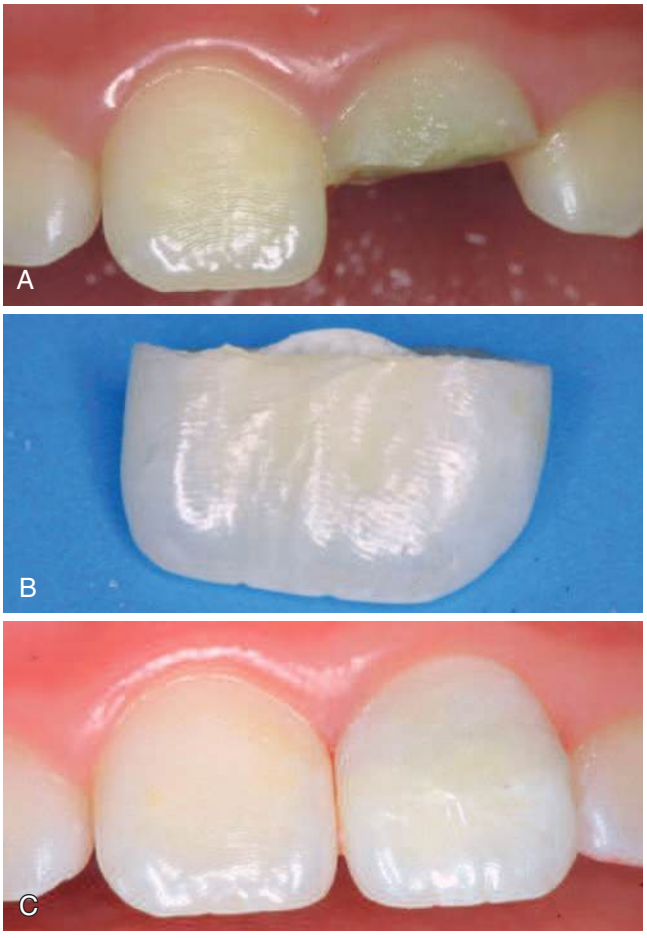
• **Fig. 13.33** SEM image of enamel surface etched with phosphoric acid.



• **Fig. 13.34** A, Phosphoric acid gel. B, Etching of the enamel surface with phosphoric acid gel.



• **Fig. 13.35** Enamel surface appearance after applying the acid etchant. Notice the chalkiness of the enamel surface.



• **Fig. 13.36** A, Fractured tooth #9. B and C, Fragment piece was saved and bonded back in place.

on enamel, there is no benefit in applying the gel for more than 15 seconds (Fig. 13.34B). The operator should always verify that an etch pattern has been achieved before proceeding with bonding (Fig. 13.35).

The acid-etched enamel/resin bond has been studied extensively and is a very predictable long-term bond. This bond is a purely *mechanical bond*, with shear bond strength usually exceeding 20 megapascals (MPa). Properly done, it is a clinically successful bond and should result in sealed margins with composite resin restorations. This is important to reduce microleakage and prevent secondary caries, which is a common cause of failure with composite resin restorations. It's worth repeating that the acid-etched enamel/resin bond is the most important bond in restorative dentistry.

The acid-etched enamel/resin bond is useful for providing retention for Class IV restorations, and for sealing the margins of Class I, II, III, and V restorations. It is also useful for bonding fragments of teeth fractured due to trauma (Fig. 13.36).

The process of etching enamel with phosphoric acid is an example of a "differential acid attack," a process that has been used in industry for many years. A specific acid will preferentially attack one phase of a multiphasic substrate and leave behind a microscopically rough, retentive surface. Phosphoric acid is used to etch enamel, nitric acid is used to etch base metal, and hydrofluoric acid is used to etch feldspathic porcelain. The surfaces created on each of these substrates are similar in appearance and provide mechanical bonds to resin that are equal to or greater than the bonds to etched enamel.

These concepts have resulted in the use of a number of very conservative resin-retained modalities for restoring teeth. One of these procedures is the Maryland bridge (Fig. 13.37), which is a conservative method of replacing missing teeth. Clinical trials have demonstrated that these resin-retained restorations have clinical longevity comparable to conventional fixed partial dentures. A



• **Fig. 13.37** Maryland bridge.

similar concept is the use of bonded metal rests to provide support for removable partial dentures.

Perhaps the most popular of these modalities has been bonded porcelain laminate veneers. Simonsen and Calamia are credited with the first study on etching porcelain with hydrofluoric acid and Harold Horn wrote the first article describing porcelain veneers.[69,70] With conservative intraenamel preparations, these restorations can provide many years of successful service and have a survival rate of 95% at 15 years.[71]

The survival rate of all-ceramic crowns is improved if they are etched and bonded in place. Posterior bonded ceramic restorations in the form of inlays, onlays, and crowns are popular today and must be bonded in place. Etched porcelain pontics can be bonded to etched enamel of the abutment teeth to fabricate a "Carolina" bridge. These are primarily meant to be long-term provisionals for patients with congenitally missing lateral incisors during their teenage years prior to placing an implant (Fig. 13.38).

In summary, the acid-etched enamel/resin bond is a strong, reliable, mechanical bond that has proven to be useful with many different clinical procedures. It is the most important bond in adhesive restorative dentistry.

THE BOTTOM LINE

1. The acid-etched enamel/resin bond is the most important bond in adhesive restorative dentistry.
2. Enamel etchants are 30% to 40% phosphoric acid.
3. Normal etch time is 15 seconds.
4. If the etched surface is not achieved after 15 seconds, etch again until it is obvious.
5. Isolation is imperative. Any contamination of the etched surface prior to bonding will negatively affect the bond.
6. The etched-enamel/resin bond is a durable mechanical bond. It can provide adequate retention for restorations such as porcelain laminate veneers and can also result in sealed margins with direct composite resin restorations.

Bonding to Dentin

Bonding to acid-etched enamel is a rather straightforward procedure that has been used effectively for many years. Bonding to dentin, however, has proven to be a much more difficult process. Research into dentin bonding began in the late 1970s and it continues to be an active area of both clinical and laboratory research. Dentin



• **Fig. 13.38** A and B, Missing teeth 7 and 10 restored with a Carolina bridge.

bonding has grown through many generations, and the process has been one of evolution and simplification. The following narration will describe the evolution of dentin bonding agents in some detail and will identify preferred adhesives at the time of writing (2017).

The first three generations of dentin bonding agents (DBAs) were generally ineffective, and the fourth generation DBAs introduced in the 1990s were one of the most effective bonding agents ever available until recently. However, the number of steps in the fourth generation DBAs makes their use technique sensitive, and many dentists reported poor results when using them. Most of the later generations of DBAs were designed by the manufacturers to be simpler to use, but that simplification has generally resulted in diminished clinical performance.

There are three main groups of DBAs: initial bonding agents that were generally ineffective, etch-and-rinse bonding agents that include the fourth and fifth generations, and self-etching primers. Basically the contemporary dentist must decide on whether to use an etch-and-rinse product or a self-etch product. Recent products have been marketed that can be used either as a self-etch, selective etch, or etch-and-rinse materials.

There are a number of factors that have made bonding to dentin more difficult than bonding to enamel.

I. The compositions of enamel and dentin are different (Table 13.5). The major differences are that dentin is much less mineralized and has much more water content than enamel.
II. Dentin is a heterogeneous substrate. One of the major factors in this regard is the dentinal tubules. Superficial dentin, close to the dentinoenamel junction (DEJ), has a relatively small number of tubules and the orifices of the tubules are small. Thus there is a large amount of intertubular dentin to bond to and successful dentin bonding is relatively predictable and easy. With deep dentin, there are more tubules and the orifices of the tubules are wider, so there is little intertubular dentin to bond to (Fig. 13.39). Bonding to deep dentin is difficult and unpredictable.
III. Dentin is a dynamic substrate. It changes after bonding occurs. It also contains enzymes called matrix-metalloproteinases (MMPs), which can be activated by the bonding procedure.

These activated MMPs gradually break down the collagen matrix and hybrid layer. The result of this is that dentin bonds tend to decrease significantly with time.

Initial Bonding Agents

For practical purposes, the first three generations of DBAs can be clumped together and called "initial" bonding agents. These materials were ineffective because at the time they were formulated dental scientists believed resin-based materials were deleterious to the pulp. Thus these materials were designed to bond to minerals in the smeared layer of dentin, which scientists believed would protect the pulp from the resinous bonding agents.

The smeared layer of dentin is 3 to 5 microns thick and consists of ground dentin, fragments from the rotary instruments, denatured protein, and both live and dead bacteria (Fig. 13.40). It does have a significant protective function, but most importantly for dentin bonding, it is not firmly attached to the underlying dentin. Thus when shrinking polymers such as DBAs are bonded to it, the smeared layer becomes detached from the underlying dentin.

These initial DBAs formed excellent bonds to etched enamel, and weak bonds to dentin. Most importantly, however, they had significant microleakage that lead to both postoperative sensitivity and recurrent caries.

Etch-and-Rinse Bonding Agents (Previously Called "Total-Etch")

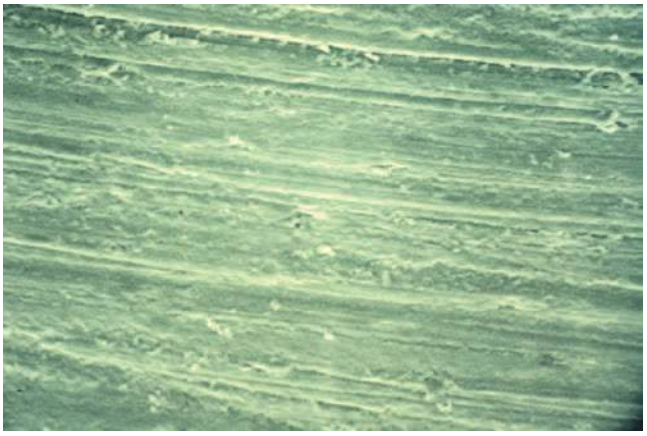
The ineffectiveness of the initial DBAs led to the development of total-etch systems in the late 1980s. Credit must be given to Fusayama and Nakabayashi from Japan for developing this concept and to Bertolotti for recognizing the value of their work and introducing it to North America. John Kanca also deserves credit for creating the first effective total-etch product developed in North America. Kanca has suggested that total-etch DBAs be called "etch-and-rinse" DBAs to clearly differentiate them from the self-etch products developed later on.

These DBAs constitute the fourth and fifth generations of DBAs. The fourth-generation DBAs were multiple bottle systems that required three distinct procedures for bonding. These steps are as follows: (1) The tooth (enamel and dentin) is etched and then rinsed. (2) A primer is placed. (3) A bonding agent is placed and cured. The primer bonded to enamel and dentin, and sealed the dentin. The bonding agent bonded to the primer and to the restorative material. Inherent to this process was the necessity of bonding to "moist" dentin, and the difficulty of precisely defining exactly what "moist" meant led to many dentists having difficulty using these products successfully. Commercial products of fourth generation DBAs include All-Bond 2 (Bisco, Inc.), Scotchbond Multi-Purpose (3M Oral Care, St. Paul, MN), OptiBond FL (Kerr Corporation, Orange, CA), and Tenure Multi-Purpose (Den-Mat).

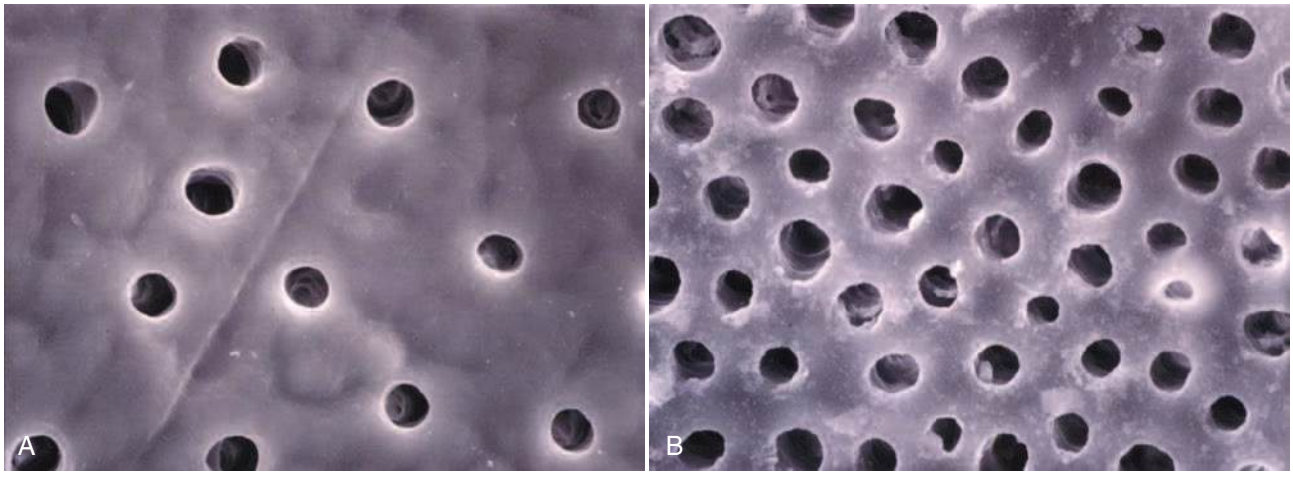
The technique sensitivity of the fourth-generation DBAs led directly to the development of the fifth-generation DBAs, which were basically single bottle systems and were easier to use. With these systems only two steps are required: (1) The tooth is etched and rinsed, and (2) priming and bonding are carried out as a single

TABLE 13.5 Composition of Enamel and Dentin

	Enamel	Dentin
Mineral	80%	45%
Organic	2%	30%
Water	12%	25%



• **Fig. 13.40** SEM image of smear layer of dentin.



• **Fig. 13.39** A, Superficial intertubular dentin. B, Deep intertubular dentin. (Photos courtesy Dr. Ed Swift.)

step, because both the primer and the bonding agent are in the same bottle. While this procedure is a significant simplification, it still requires bonding to "moist" dentin, which remains a dilemma for many dentists. Commercial examples of fifth-generation DBAs include One-Step Plus (Bisco, Inc.), Adper Single Bond Plus (3M Oral Care, St. Paul, MN), OptiBond SOLO Plus (Kerr Corporation, Orange, CA), and Tenure Quick (Den-Mat).

Many of the initial fourth-generation products were not particularly successful because the manufacturers were still not convinced it was safe to use phosphoric acid on dentin. Thus they used two different etching agents, one for enamel and one for dentin. For example, an early version of Scotchbond Multi Purpose used only one acid to etch both enamel and dentin; however, it was a weak (10% maleic) acid. Maleic acid works well on dentin, but again does not adequately etch enamel. As stated several times in the section on acid-etching enamel, the most important bond in adhesive restorative dentistry is the resin bond to acid-etched enamel.

Today's fourth-generation DBAs all use 30% to 40% phosphoric acid to etch both enamel and dentin, as research has shown conclusively that it is both safe and effective. The components of the primers include a hydrophilic monomer such as HEMA, PENTA, BPDM, and now MDP in a volatile carrier such as acetone or ethyl alcohol and water. One of the keys to successfully using fourth-generation DBAs is to completely volatilize this carrier. This can be done by air thinning or by use of high volume vacuum. The components of bonding agents are Bis-GMA and a hydrophilic monomer such as HEMA.

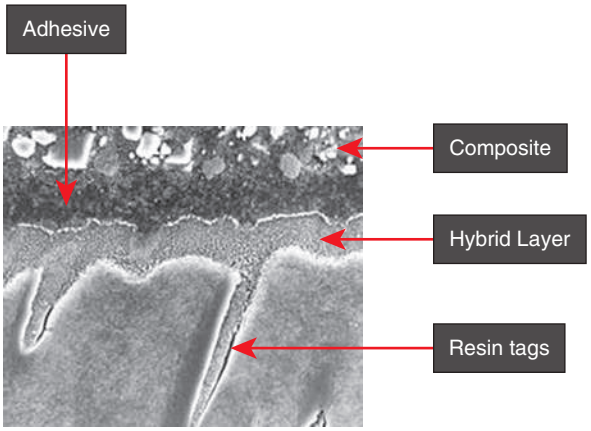
The fifth-generation DBAs are similar in composition, but components of both the primer and bonding agent are in the same bottle.

Almost all contemporary fourth-generation and fifth-generation DBAs contain quartz filler particles to improve strength, reduce shrinkage, and provide a thicker primed layer. Research has shown that DBAs with a thicker primed layer are generally more efficacious than those with a thin primed layer.

Etch-and-rinse DBAs basically remove the smeared layer of dentin, and etch both enamel and dentin. Etching dentin is often called "conditioning." Etching dentin involves removing a small amount of mineral from the superficial layer of dentin. This exposes bundles of collagen, which the primer can infiltrate and form the "hybrid" layer, composed of infiltrated collagen, mineralized dentin, and the primer. The bond is primarily to etched intertubular dentin. Bonding to the peritubular dentin around the dentinal tubules is difficult and ineffective. Resin tags look spectacular in SEMs but provide no real benefit in dentin bonding (Fig. 13.41).

Superficial dentin has an abundance of intertubular dentin and allows predictable successful bonding. With deep dentin there is relatively little intertubular dentin available due to the presence of multiple large orifice tubules. Predictable bonding is difficult with deep dentin, and a resin-modified glass ionomer may be placed over deep dentin prior to etching and bonding.

Comparing the fourth and fifth generations of etch-and-rinse DBAs, it is clear the fourth generation has the best clinical performance. The fourth-generation DBAs have three separate steps: (1) The enamel and dentin are etched and rinsed, (2) the primer is applied, and (3) the bonding agent is applied and cured. With the fifth-generation DBAs, the enamel and dentin are etched and rinsed, and the priming and bonding steps are done simultaneously. Clinical research has shown clearly that the fourth-generation (three-step) DBAs provide superior clinical performance when compared to the fifth-generation (two-step) DBAs.



• **Fig. 13.41** SEM image of the hybrid layer and resin tags.

The sequencing of the bonding procedures and their rational will be given in "The Bottom Line" at the end of this chapter.

Self-Etching Primers

Self-etching primers are basically acidic monomers that etch through the smeared layer of dentin. The primers are simply applied and light cured. They are not rinsed off. Since the smeared layer is usually not completely removed, the dentinal tubules remain sealed and postoperative sensitivity is almost completely eliminated.

There are two types of self-etching primers. One type has two bottles. The first bottle is the self-etching primer that is placed on the prepared tooth and dried. The second bottle is the bonding agent that is placed over the primer and cured prior to placing the restorative material. Commercial products in this category are Clearfil SE Bond (Kuraray America, Inc., New York, NY), Simplicity (Apex Dental Materials, Inc., Lake Zurich, IL), Opti-Bond XTR (Kerr Corporation, Orange, CA), and All-Bond SE (Bisco, Inc.).

The second type of self-etching primer is the single bottle systems that have the primer and bonding agent in the same bottle. Current research has shown that the single bottle materials are not reliable and should be avoided at this time. Commercial products in this category include Xeno IV (DENTSPLY Sirona, York, PA) and Adper Prompt-L-Pop (3M Oral Care, St. Paul, MN).

Self-etching primers create very predictable bonds to dentin and eliminate postoperative sensitivity, but they have one major problem: They form weak bonds with uncut enamel. If the proposition that the acid-etch/resin bond to enamel is the most important bond in adhesive restorative dentistry is accepted, this is a significant deficiency. Clinical trials using self-etching primers frequently report marginal staining after 3 or 4 years indicating microleakage at the enamel/restoration margin. Clinicians using self-etching primers should consider using the selective etch procedure to obtain a better bond to enamel. That technique will be described in the "Bottom Line" section of this chapter.

In summary, the etch-and-rinse DBAs have two groups. One group has three distinct steps including etching, priming, and bonding. The other group has two steps, etching then priming and bonding simultaneously. The self-etch materials also have two groups. The best self-etch materials have two steps, etching and priming simultaneously and then bonding. The other group etches, primes, and bonds all at once.

New Developments

While today's DBAs are clearly superior to early generations of DBAs, research has demonstrated that the bonds to dentin deteriorate over time. This is partially due to hydrolysis of the DBAs and also to breakdown of the collagen matrix in the hybrid layer due to the action of MMPs activated during the bonding procedure. Part of the problem seems to be that many DBAs are too hydrophilic and act as semipermeable membranes allowing water-tree formation and osmotic blistering. Newer products are considerably less hydrophilic than their predecessors.

The use of 2% chlorhexidine (Consepsis, Ultradent Products; Cavity Cleanser, Bisco, Inc.) after etching seems to be somewhat effective in inhibiting the ability of MMPs to break down the hybrid layer, although this has not yet become a common practice. Kerr Corporation has introduced a product called OptiBond XTR, which is the first self-etch material to demonstrate enamel bonding equivalent to that of the fourth-generation DBAs. The long-term clinical efficacy of such bonds has yet to be determined, but laboratory testing has been promising.

A major development in dentin bonding has been the expiration of the patent held by Kuraray on the monomer MDP. This is the chief ingredient in Clearfil SE Bond, which has been shown in long-term clinical trials to be as effective as the fourth-generation etch-and-rinse materials. This is a mild two-step self-etching primer, which has proven to be a very effective DBA and also does not activate MMPs. Many of the major manufacturers have now incorporated MDP into their DBAs, and it is anticipated that they will be effective. For example, 3M Oral Care has introduced a new product called Scotchbond Universal that purportedly can be used in the self-etch, selective-etch, or etch-and-rinse modes. This adhesive utilizes the monomer MDP. Laboratory tests indicate that superior bond to enamel is achieved using the etch-and-rinse procedure. For the different bonding protocols, please refer to Chapter 5.

THE BOTTOM LINE

1. Dentin bonding agents have evolved rapidly over the past few decades.
2. Contemporary DBAs are considerably more effective than early generations.
3. Modern DBAs contain fillers, which result in a thicker primer layer and seemingly improved efficacy.
4. Contemporary DBAs are either etch-and-rinse materials or self-etching primers.
5. Etch-and-rinse DBAs are either three-step (fourth-generation) or two-step (fifth-generation) materials. Both groups are effective, but clinical trials have clearly demonstrated the superiority of the three-step systems.
6. Both etch-and-rinse materials involve bonding to "moist" dentin. An effective technique for bonding to "moist" dentin will be described later in this section.
7. Self-etching primers are supplied as either two-step or one-step products. The one-step products are not recommended at this time.
8. Most self-etching primers do not completely remove the smeared layer of dentin, and postoperative sensitivity is rarely reported with these products.
9. Enamel bonds are compromised with most self-etching primers. This deficiency may be overcome using the "selective etch" technique that will be described later on in this section.
10. Bonding to superficial dentin is more predictable than bonding to deep dentin. With deep cavities a thin resin-modified glass ionomer liner should be used.
11. Successful dentin bonding requires the formation of a hybrid layer resulting from resin penetration into a collagen matrix that is partially demineralized by etching with either phosphoric acid or acidic monomers.
12. Dentin bonds tend to deteriorate over time.

Bases and Cavity Liners in the Adhesive Era

Bases and cavity liners have been used for many years in restorative dentistry. With the development of more effective adhesives, the use of materials as bases and liners has diminished. The topic of bases and cavity liners is quite controversial today and there is a general lack of consensus among authorities, practitioners, and educational institutions as to indications for such materials.[72] The following is a brief description of materials used as bases, liners, varnishes, and sealing agents available today.

Base Materials

To begin, a base is any substance placed under a restoration that blocks out undercuts in the preparation, acts as a thermal or chemical barrier to the pulp, and/or controls the thickness of the overlying restoration (Fig. 13.42).

Any dental cement mixed to a very thick consistency can be used as a base. Zinc-phosphate, polycarboxylate, zinc oxide-eugenol (IRM), and glass ionomer cements have all been historically used as bases for direct and indirect restorations.

The primary contemporary function of a base is to block out undercuts for indirect restorations, to seal the dentin against leakage in deep cavities, and to act as replacement dentin with large composite resin restorations to reduce the volume of composite resin. The contemporary material most often utilized for these functions is Fuji IX (GC America, Alsip, IL), which is a reinforced, self-cure glass ionomer.

Liner Materials

A cavity liner is a fluid paste applied in a thin layer as a protective barrier between dentin and the restorative material. Liners should *not* be used in layers thicker than 0.5 mm.

A number of materials have been used historically, including calcium hydroxide, zinc oxide-eugenol, glass ionomer, and resin-modified glass ionomer. Calcium hydroxide (e.g., Dycal, DENTSPLY



• **Fig. 13.42** Base material used to block out undercuts in the preparation of teeth 9 and 10.

Sirona, York, PA) and resin-modified glass ionomer (e.g., Vitrebond, 3M Oral Care, St. Paul, MN; Fuji Lining Cement, GC America, Alsip, IL) lining materials are the most commonly used. Resin-modified glass ionomers are used most frequently, as few clinical situations today call for the use of calcium hydroxide.

Resin-modified glass ionomer liners (Vitrebond and Vitrebond Plus) are supplied as powder/liquid or paste/paste systems, respectively. They are light activated and have improved handling characteristics and physical properties compared to regular glass ionomer lining materials. They bond very predictably to dentin, provide an excellent seal, and are very compatible with the pulp. Typically they are used as a thin liner (0.5 mm) over dentin in deep cavity preparations (Fig. 13.43).

Calcium-hydroxide materials have a very high pH (9–14), are antibacterial, and promote sclerosis, hence reducing permeability. They must not be used as a base because they have very poor physical properties and should only be placed as a very thin (0.5 mm) liner over the deepest portion of the cavity preparation. Calcium hydroxide materials have traditionally been used when performing a direct pulp cap, as the high pH irritates the pulp tissues and stimulates the production of secondary dentin. Calcium hydroxide has largely been replaced by bioactive cements such as mineral trioxide aggregate (MTA) and biosilicate cement (Biodentine, Lancaster, PA) for this function. Direct pulp capping should only be considered with teeth with no record of spontaneous pain and no radiographic evidence of periapical pathology.

Cavity varnish is a solution of a natural gum or synthetic resin in an organic solvent. Copal varnish (Copalite, Temrex, Freeport, NY) has been traditionally used under amalgam restorations to reduce initial levels of marginal leakage. Cavity varnishes are no longer popular and have been replaced by cavity sealers.

A cavity sealer is a liquid primerlike material that seals the dentin surface prior to placing definitive restorative materials. Current products used as cavity sealers under silver amalgam restorations are Gluma Desensitizer (Kulzer, South Bend, IN) and G5 desensitizer (Clinician's Choice, New Milford, CT). These sealers contain both HEMA and glutaraldehyde and are very effective sealing agents for dentin.

Clinical Recommendations

When using bases and liners, it should be remembered that the ultimate goal is to have a minimum of 2 mm of tooth structure or a combination of tooth structure base and liner between the restorative material and the pulp.[73]

Amalgam

It is no longer considered necessary to use bases to build up deep cavity preparations to "ideal" form. With deep preparations close to the pulp, a very small liner of calcium hydroxide may be placed, followed by a thin liner of resin-modified glass ionomer (Vitrebond Plus). As an alternative, the calcium hydroxide liner may be omitted. Deep preparations that are not close to the pulp should receive a thin RMGI liner. Gluma Desensitizer or G5 should be applied to all preparations, including those in which liners have been placed.

Composite Resin

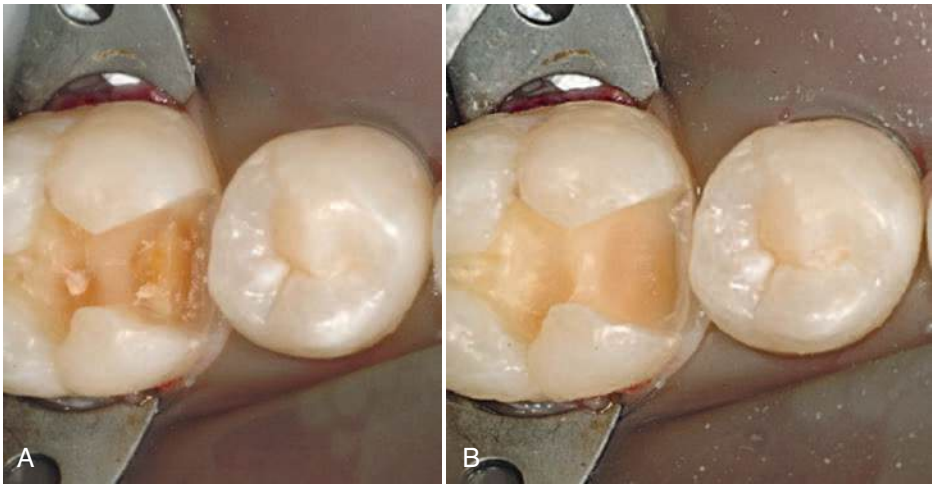
For any preparation deeper than a minimal preparation, a thin resin-modified glass ionomer liner should be placed in the deepest portion of the preparation. In this situation (deep dentin), dentin adhesives tend to be less effective because there is less intertubular dentin to interact with and form an effective hybrid layer. With posterior composite resin preparations where the proximal box ends close to or beyond the CEJ, a resin-modified glass ionomer liner should be placed to reduce postoperative sensitivity. Adhesive bonding systems *must* be used with all composite resin restorations regardless of whether a lining material has been placed.

Cast-Metal Indirect Restorations

Deep restorations should receive a RMGI liner prior to impression making. With very deep lesions, vitality testing must be done before administration of local anesthetic to determine the need for endodontic therapy. Resin-modified glass ionomer, Fuji IX, or other base materials can be used to block out undercuts. Gluma Desensitizer or G5 should be applied immediately after tooth preparation prior to making the provisional restoration.

THE BOTTOM LINE

1. There is still a need for bases and liners in the adhesive era.
2. Bases are primarily used to block out undercuts.
3. Liners should be used in a thin film, no greater than 0.5 mm thick.
4. Resin-modified glass ionomers are used with deep caries prior to bonding because bonding to deep dentin with available dentin



• **Fig. 13.43** A, Cavity preparation of tooth 30 before placing cavity liner. B, Preparation after placement of cavity liner.

bonding agents is not predictable due to the lack of intertubular dentin.
5. Cavity sealers should be used routinely under amalgam restorations.

Indirect Restorative Materials

Impression Materials

When fabricating indirect tooth-supported or implant-supported restorations, accuracy of the impression is critical. It is clear that impression materials have improved steadily over the past 50 years, and the contemporary clinician has a great number of excellent impression materials to choose from. While it is clear that contemporary impression materials are vastly superior to their predecessors in the 1960s and 1970s, the quality of impressions sent to commercial dental laboratories has failed to show a concomitant improvement (Fig. 13.44).[74]

The major deficiency seen with these impressions is failure to record the entire prepared cervical margin in the impression. A second deficiency is failure to follow the fundamental principles in the manipulation of impression materials. Use of flexible plastic stock trays ignores the need to control bulk and risk the possibility of hydraulic distortion, especially with use of heavy body and/or putty materials.[75] Putty-wash materials are also used extensively, often in an inappropriate manner resulting in impressions with less than optimal accuracy.

The primary cause of these problems is a lack of appreciation of the principles of manipulation of impression materials and poor decisions related to margin placement and gingival displacement techniques. Many dentists are excited about the ability to make optical impressions, and the technology related to optical impressions is impressive; however, it is clear that dentists who cannot predictably make a traditional impression will not be able to predictably make an optical impression.

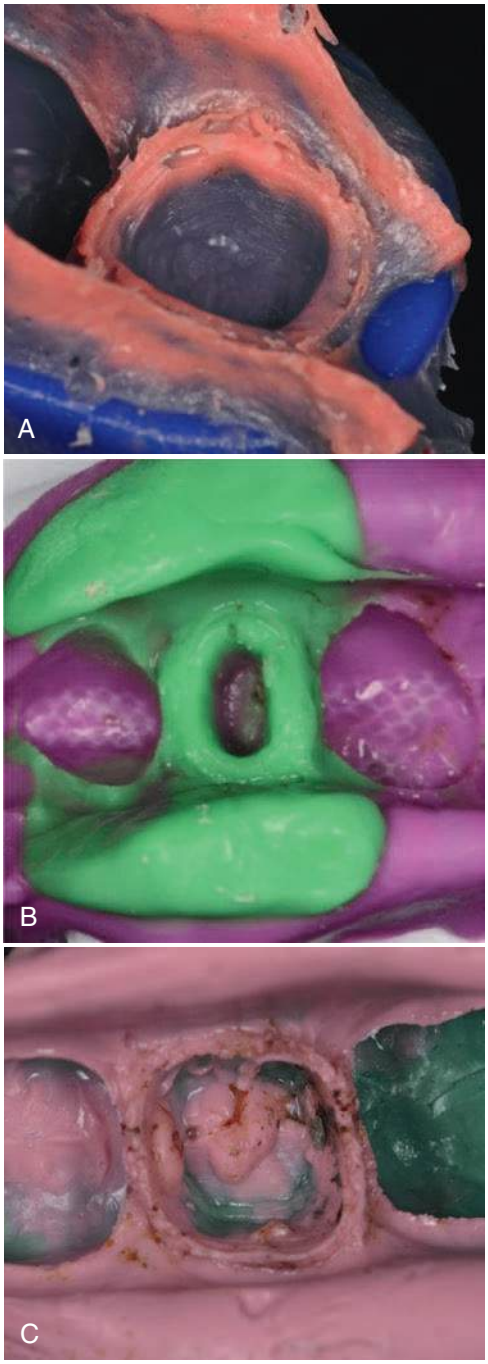
Classification of Impression Materials

There are two main types of impression materials: nonelastic impression materials, which are used primarily in removable prosthodontics, and elastic impression materials (Fig. 13.45). There are two types of elastic impression materials: water-based and elastomeric (rubberlike) impression materials. The two types of water-based impression materials are reversible and irreversible hydrocolloid (alginate). There are four types of elastomeric impression materials: polysulfide rubber (also known as rubber base, thiokol rubber, and mercaptan rubber); polyether; condensation silicone; and addition reaction silicone (also known as polyvinyl siloxane (PVS) impression material).

This section will outline the ideal properties of impression materials. Critical manipulative variables will be highlighted in the subsequent section. Properties to be discussed include accuracy, elastic recovery, dimensional stability, flow, flexibility, workability, hydrophilicity, shelf life, patient comfort, and economics.

Accuracy

There are three main factors related to accuracy of impression materials: fine detail reproduction, dimensional accuracy, and elastic recovery, which will be discussed separately. According to ADA #19, elastomeric impression materials used to fabricate precision castings must be able to reproduce fine detail 25 microns or less. All currently available impression materials meet this specification.



• **Fig. 13.44** A–C, Examples of poor intraoral impressions.

Classification of Impression Materials

1. Nonelastic Impression Materials
2. Elastic Impression Materials
   - A. Water-Based Impression Materials
     1. Irreversible Hydrocolloid
     2. Reversible Hydrocolloid
   - B. Elastomeric Impression Materials
     1. Polysulfide Rubber
     2. Polyether
     3. Condensation Silicone
     4. Addition Reaction Silicone (PVS)

• **Fig. 13.45** Classification of impression materials.

Polyvinyl siloxane is the best in this regard and reversible hydrocolloid is the worst. It is not likely that improved ability to record fine detail results in improved accuracy of the final restoration, because the limiting factor in this area is the ability of the gypsum die materials to record fine detail. The corresponding specification for gypsum die materials related to fine detail is 50 microns. Some die materials can provide better fine detail reproduction than the 50-micron specification but fall well short of the impression materials in this regard.

There are differences in the ability of different viscosities of elastomeric impression materials to record fine detail. Typically the lower the viscosity, the better the ability to record fine detail. Putty viscosities in general are very poor in this regard and are only required to record fine detail of 75 microns.[76] Some putty-wash techniques result in putty recording some or the entire gingival margin, which has a deleterious effect on the accuracy of the gypsum die.

A second aspect is dimensional accuracy, which is evaluated by measuring tooth-to-tooth distances in the same quadrant and also cross-arch distances. There is some evidence that reversible hydrocolloid is slightly superior to the elastomers in this respect, but it is likely of limited clinical significance.[77]

All of the available contemporary impression materials provide superb accuracy if they are manipulated correctly. Although PVS materials are likely to be slightly more accurate than other impression materials, differences in accuracy are not likely to be clinically significant.

Elastic Recovery

Elastic recovery is the ability to flow into an undercut, set in that position, and then spring around the undercut on removal of the impression and rebound to its original shape. No impression material has 100% elastic recovery; and for all impression materials, the greater the depth of the undercut, the greater the permanent distortion of the impression material is.[78,79] Polyvinyl siloxane impression materials have the best elastic recovery at over 99% with a specific test undercut. This elastic recovery coupled with excellent dimensional stability make it the material of choice for clinicians who prefer to double pour impressions to obtain a backup die.

At one time, dentists were cautioned to wait 30 minutes prior to pouring impressions to allow maximum elastic recovery to occur. With contemporary impression materials, this delay is no longer necessary as elastic recovery occurs almost instantly. Keeping in mind the greater the depth of the undercut, the greater the distortion, an important procedure is to always block out undercuts in tooth preparations prior to impression making. This often occurs when preparing teeth with existing restorations. Many dentists fail to do this, assuming the laboratory can block out the undercuts on the die. While this is possible, blocking out the undercuts with resin-modified glass ionomer materials will reduce the overall distortion due to improved elastic recovery.

Dimensional Stability

One of the most important properties when selecting an impression material is dimensional stability. An ideal impression material would be dimensionally stable over time, allowing the impression to be poured at the convenience of the operator. This is extremely important in contemporary dentistry, as most dentists have opted to not pour up their own impressions and instead send them to dental laboratories to be poured. The time between removal of an impression from the mouth and the time when it is poured by the laboratory varies considerably, and if the impressions are mailed to the lab, it will be well over 24 hours.

Dimensional stability of an impression material is determined by the composition of the material, its susceptibility to imbibition and syneresis, and the volatility of the by-product of the setting reaction.[80] Impressions made from water-based impression materials, such as reversible and irreversible hydrocolloid (alginate), are approximately 80% water, which evaporates at room temperature. This evaporation can result in clinically significant distortion of the impression. Because of this evaporation (syneresis), water-based impression materials should be poured within 10 minutes after removal from the mouth. Water-based impression materials are also subject to imbibition or the absorption of water. The common practice of wrapping alginate impressions with wet paper towels should be avoided, as the impression material will absorb water from the towel (imbibition), which results in distortion of the impression. These impressions are properly stored in sealed plastic bags prior to pouring.

The by-product of the setting reaction of condensation silicone impression materials is ethyl-alcohol, which evaporates rapidly at room temperature. Impressions made with condensation silicone should be poured within 30 minutes of removal from the mouth.

The by-product of the setting reaction of polysulfide rubber (rubber base) impression materials is water (Fig. 13.46). This evaporates more slowly than it does from water-based impression materials, but is still significant. Thus impressions made with polysulfide rubber should be poured within 30 minutes.

Polyether impression materials have no volatile by-product of the setting reaction but are sufficiently hydrophilic to absorb water from the atmosphere. This absorption is dependent on relative humidity (RH); and if the RH is over 50%, absorption of water is sufficient to cause clinically significant distortion.[81] Polyether impressions should be poured within 1 hour and should never be mailed to a dental laboratory.

There is no by-product of the setting reaction for PVS impression materials, so PVS impressions possess almost unlimited dimensional stability. If impressions are to be mailed to the dental laboratory, the only reasonable material to use is PVS. Indeed, the case can



• **Fig. 13.46** Rubber base impression material.

be made that if the impression is going to be poured by a local laboratory, it is safer to use PVS because there is no way to be certain it will be poured at the appropriate time. This exceptional dimensional stability, coupled with the excellent elastic recovery of PVS impression materials, make it the best choice for clinicians who prefer to double pour impressions to obtain a backup die. Table 13.6 displays discussed impression materials and recommended pouring times.

Flow and Flexibility

Impression materials need to flow readily into the minute details of cavity preparations and accurately capture grooves, pinholes, and cervical margin detail. Most brands of impression material provide light body or syringe materials with low viscosity to accomplish this. These are used with heavy body materials in a tray to provide more rigidity to the impression and assist in forcing the low viscosity material into the gingival sulcus and accurately and completely capture the prepared cervical margin.

Many early versions of low viscosity PVS impression materials had excellent flow characteristics but tended to flow away from the prepared tooth after injection of the impression material. This created problems when attempting to make an impression of several prepared teeth at one time. The composition of most contemporary PVS and polyether materials has been modified to render them more thixotropic, and stay in place when they are syringed around the prepared teeth, but then flow readily when the heavy body tray materials are placed over top of them.

Set impression materials vary considerably from one another with regard to flexibility. Polyether impression materials tend to be more rigid than other materials. This can present a problem with long, thin preparations of mandibular incisors or periodontally involved teeth. Fracture of the delicate gypsum dies is a common occurrence due to the rigidity of polyether materials. Another problem related to this rigidity is tearing of the impression material in the gingival sulcus. While the tear strength of polyether impression materials is excellent, the amount of force required to remove the rigid impression may exceed the tear strength of the material. More recent formulations of polyether materials are less rigid than early versions but are still more rigid than PVS materials.

Set PVS materials are relatively rigid but seem to fall below the threshold where problems with fracture of dies are common. Although rarely used today, reversible hydrocolloid impression material is the least rigid of all impression materials and may be the material of choice when making impressions of multiple prepared periodontally involved teeth.

With dual-arch impressions, it is advantageous to use a relatively rigid material. Many of the commonly used trays for dual-arch impressions are somewhat flexible, and a rigid impression material can compensate for this flexibility. Polyether impression materials work well in this regard. Many PVS products have heavy body versions that also provide the essential rigidity.

TABLE 13.6 Multiple Impression Materials and Recommended Pouring Time

Impression Material	Pour Time
Water based	10 minutes
Polysulfide rubber	30 minutes
Condensation silicone	30 minutes
Polyether rubber	60 minutes
Polyvinyl siloxane	At operator convenience

Workability

Impression materials were originally supplied as two tubes for each viscosity. One tube was labeled as the base, the other the catalyst (Fig. 13.47). The base and the catalyst were different colors, and equal lengths of each were dispensed on a mixing pad. The materials were mixed until the material became a uniform color with no streaks. When making a clinical impression, one pad with heavy body material was used and one with light body material. The materials were mixed simultaneously, the light body material loaded into an impression syringe and the heavy body material loaded into a tray. This process required excellent coordination between the dentist and assistant(s); thus impression making was a stressful procedure.

Contemporary materials are supplied with "automix" handheld devices. This is a substantial improvement and results in a standardized mix with fewer inherent porosities, increased working time, and economic savings due to less waste of material. Even better are sophisticated electronic mixing devices such as the Pentamix (3M Oral Care, St. Paul, MN).

Working times of impression materials vary by the manufacturer, and most automix devices provide materials with standard-set and quick-set capabilities. When making a single crown impression with a quadrant or dual-arch tray, the operator may opt for a quick-set material with a shorter working time. When making a full-arch impression with multiple prepared teeth, it might be prudent to choose a material with a longer working time. When making an impression with several prepared teeth, another option to extend working time is to refrigerate the low viscosity material, which increases working time without sacrificing accuracy.[82]

Hydrophilicity

Reversible hydrocolloid impression materials are truly hydrophilic and can effectively make accurate impressions in the presence of moisture. *All elastomeric impression materials require a completely dry field to make a successful impression.*

Polyether impression materials are hydrophilic as witnessed by their tendency to absorb water from the atmosphere. However, polyether impression materials absolutely require a dry preparation surface to make an accurate impression.



• **Fig. 13.47** Two-tube impression material: base and catalyst.

The manufacturers of many contemporary PVS impression materials claim their products are "hydrophilic," and incorrectly imply that their materials can effectively make impression in the presence of blood and/or saliva. The hydrophilicity of a material is measured by the contact angle a sessile drop makes with the material. If the contact angle is below a specific value, the material can be called hydrophilic. Many contemporary PVS impression materials meet that criterion, and therefore the manufacturers are technically correct. However, the implication that these materials can make impressions in a wet environment is totally incorrect.[83,84] These materials require a dry environment to make successful impression.

The advantage these "hydrophilic" PVS materials have is they are significantly easier to pour. Early versions of PVS were hydrophobic and very difficult to pour. They also released hydrogen for approximately 3 hours after setting, so delaying pouring for that period was recommended. Contemporary PVS materials have nonionic surfactants grafted onto the polymer chains, which makes them hydrophilic and thus easier to pour. They also contain small quantities of spongy palladium, which absorbs hydrogen and allows pouring of the impression immediately upon removal from the mouth.

Shelf Life

The exact shelf life of impression materials is not known, but it is inadvisable to use impression materials that have passed the expiration date established by the manufacturer. Dentists must become familiar with manufacturers' codes for the materials they are using and ensure that current materials are consistently utilized. It is arbitrarily suggested that no more than a 6-month supply of impression material should be kept on hand at any time.

Patient Comfort

Contemporary impression materials are far more patient friendly than the old polysulfide materials, which were messy, had a bad taste and odor, and permanently stained clothing. Reversible hydrocolloid materials were uncomfortable for patients because they required the use of bulky, water-cooled trays. Contemporary impression materials are essentially odorless and tasteless.

The rigidity of polyether impression materials can be a disadvantage, especially if the patient has existing fixed prostheses or multiple open gingival embrasures due to periodontal disease. In these situations, it is advisable to use a more flexible material, and to block out the undercuts with utility wax prior to impression making. The use of the dual-arch impression technique, where indicated, is also pleasant for patients in that it requires a minimum amount of material and avoids the necessity for opposing arch impressions.

When full-arch impressions are indicated, the use of a custom tray is advocated. Many studies indicate that custom trays are more accurate than stock trays and the level of patient comfort is substantially improved. In addition, significantly less material is used, which offsets the costs of fabricating the custom trays.

The ultimate in patient comfort with impression making is the use of optical impressions. Digital impressions are becoming more and more popular and are very patient friendly. Imaging systems are becoming more economical and it is anticipated that the majority of impressions made in the next decade will be made optically. It is important to point out that while optical impressions for single tooth preparation are as accurate as conventional impressions, the same factors that result in failed contemporary impressions will result in failed optical impressions.

Economic Factors

There can be significant differences between the costs of different impression materials. Reversible hydrocolloid impression material is considerably less expensive than elastomeric impression materials. However, there is a significant cost involved with using reversible hydrocolloid, considering conditioning and tempering baths need to be purchased along with an appropriate selection of water-cooled stock trays. Polyether and PVS materials are comparable in costs at about $12 per impression. Competing elastomeric materials are less expensive and are popular in underdeveloped countries.

It is likely true, however, that the cost of impression materials is of minimal consequence in most contemporary practices. Practitioners can reduce costs by using automix devices, by using the dual-arch technique where indicated, by using custom trays when full-arch impressions are indicated, and by reducing the need for remakes by making successful impressions on the first attempt.

THE BOTTOM LINE

1. Many excellent impression materials are available to the contemporary clinician.
2. The quality of impressions sent to commercial dental laboratories needs improvement.
3. All contemporary impression materials are sufficiently accurate if manipulated properly.
4. The only impression material that should be mailed to commercial laboratories is PVS.
5. Polyvinyl siloxane materials have the best fine detail reproduction, elastic recovery, and dimensional stability of all impression materials.

Principles of Impression Material Manipulation

Proper manipulation of impression materials is probably more important in determining the accuracy of an impression than which type of material is selected. This section will discuss the important manipulative variables that must be considered when making impressions for indirect restorations.

Control of Bulk

Impression materials shrink when setting. Irreversible hydrocolloid and elastomers shrink due to polymerization reactions, and reversible hydrocolloid undergoes thermoplastic shrinkage as it cools from the warm to cool state. The effect of this shrinkage can be controlled by paying close attention to the bulk of impression material used. Water-based impression materials are most accurate when used with a uniform cross-sectional thickness of 4 to 6 mm. This means that successful impressions can be made using stock trays that will provide approximately 4- to 6-mm thickness of material.

Elastomeric impression materials are most accurate when used with a cross-sectional thickness of 1 to 2 mm.[7] Use of a stock tray with elastomers results in excess thicknesses of impression material, which possibly could result in an inaccurate impression. Elastomeric full-arch impressions are best made using a custom tray that will provide the desired cross-sectional thickness of impression material (Fig. 13.48).

Numerous studies have evaluated the accuracy of elastomeric impressions made in stock trays versus impressions made in custom trays. Almost all conclude that impressions made in custom trays are more accurate, yet very few North American dentists routinely use custom trays.[85]

The difference in cross-sectional thickness of impression material in a stock tray versus a custom tray is only about 2 mm, but that difference does reduce the accuracy of the impression.[86] This

mandates that custom trays be fabricated with precision. Custom trays should be fabricated on the diagnostic cast using one layer of base plate wax as a spacer. Trays can be fabricated with PMMA, photo-cure bisacryl (Triad, DENTSPLY Sirona, York, PA), or PVS. PMMA trays should be fabricated at least 24 hours in advance to ensure stability.

Occlusal stops are critical for proper orientation of the tray in the patient's mouth. Three occlusal stops are ideal, with at least one stop posterior to the prepared teeth. Stops should be placed on nonfunctioning cusp tips to minimize distortion in the area of the stops. The stops are prepared by removing the base plate wax from the nonfunctioning cusp tips with a hand instrument (Fig. 13.49). The wax spacer should be covered with tin foil before making the tray to facilitate removal of the wax from the tray and to prevent incorporation of a wax residue on the internal surface of the tray. This can happen due to the inherent exotherm reaction that occurs during the setting of the PMMA tray material. This residue can interfere with the proper functioning of tray adhesives.

Adhesion of the Impression Material to the Tray

It is imperative that the impression material adheres to the tray. With proper adherence, the impression material shrinks *toward* the tray as it polymerizes, creating a slightly larger die. Adhesion is achieved using specific chemical tray adhesives. Adhesives must be specifically matched with the impression material. They should be painted in a thin layer on the internal portion of the tray and the tray borders (Fig. 13.50). The tray should be painted 7 to 15 minutes prior to impression making to permit formation of an adequate bond strength of the impression material to the tray.[87]

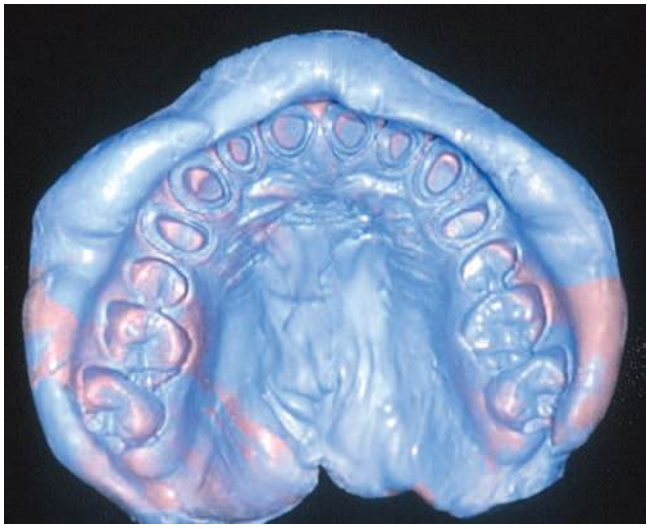
Pouring of Impression Materials

One of the most important manipulative variables with impression materials is the time limit after removal from the mouth to when the impression is poured with the appropriate gypsum product.[88] The factors that determine when an impression should be poured are discussed in depth in the previous section (see Table 13.6).

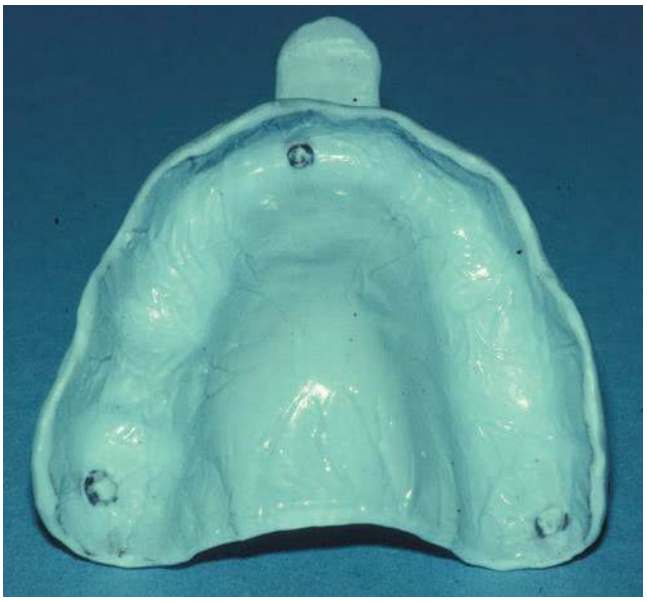
Viscosity Control

Impression materials are supplied in a number of viscosities ranging from very low viscosity to very high viscosity putty materials. The main difference between these materials with different viscosities is in the amount of inert filler material in the impression material. With low viscosity materials, there is relatively little inert filler and thus more chemical reagent. Low viscosity materials are excellent at recording fine detail, but also undergo more polymerization shrinkage during the setting reaction. Thus the optimum method of impression making is to use only as much low viscosity material as is necessary to capture the fine detail of the prepared margin, grooves, and boxforms of the preparation; and the bulk of the impression should be made with high viscosity material. This high viscosity material has correspondingly more inert filler and less reagent and displays less polymerization shrinkage on setting. The heavy body material also assists in pushing the low viscosity, light body material into the gingival sulcus.

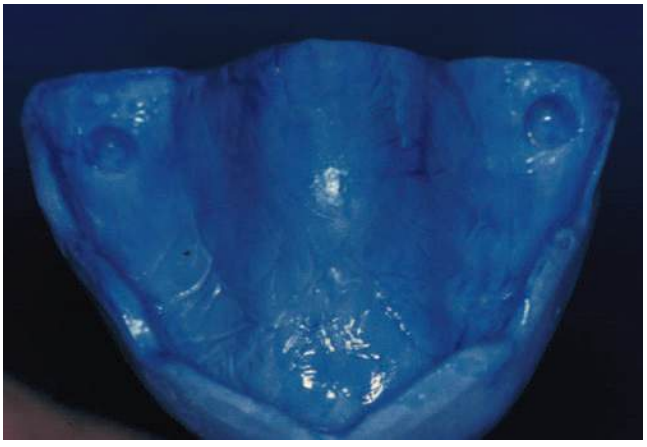
Monophase PVS and polyether materials are supplied by many manufacturers. In theory, such materials do not provide the same level of accuracy as provided by the proper use of a combination of low viscosity/high viscosity materials, but the actual differences in accuracy are likely not clinically significant. The convenience of having to use only one viscosity of material makes monophasic materials very practical.



• **Fig. 13.48** Full-arch impression taken with a custom tray.



• **Fig. 13.49** Custom tray showing three occlusal stops.



• **Fig. 13.50** Custom tray coated with tray adhesive.

Adequate Mixing

As discussed in the previous section, almost all impression materials today come with an automix system. This provides optimum mixing with fewer inherent voids, extends the essential working time of the materials, and reduces waste because the materials are loaded from the dispenser directly into the syringe or tray.

Disinfection

Microorganisms in the oral cavity can be transmitted from impressions to the dental laboratory. Dentists must disinfect impressions prior to pouring the cast or sending the impression to the laboratory.

The first step of any disinfection technique is to rinse the impression thoroughly in cold tap water. This step removes a significant portion of the microorganisms from the impression. Disinfection techniques involve spraying the impression with disinfection agents or immersion of the impression materials in chemical agents such as sodium hypochlorite. Polyvinyl siloxane materials are stable in this regard, but special care must be taken with water-based materials and polyethers to ensure that adequate immersion times are used to eliminate microorganisms but that extended immersion times are avoided to prevent excess imbibition of the disinfecting solution and distortion of the impression.

Polyvinyl Siloxane Impression Materials and Latex

Polyvinyl siloxane impression materials have been available since the mid-1970s and have continued to evolve and improve.[89] Polyvinyl siloxane impression materials are easily the best selling impression materials on the North American market, and they have the best fine detail reproduction and the best elastic recovery of all available impression materials. Polyvinyl siloxane materials have remarkable dimensional stability and are odorless and tasteless and pleasant for patients. They are provided in a wide variety of viscosities, rigidities, and working and setting times so they can be used in a variety of clinical situations.

Polyvinyl siloxane impression materials have one major disadvantage: They have a significant interaction with latex (rubber dam and latex gloves). Any contact of unpolymerized PVS material with latex results in inhibition of polymerization of the impression material. This can occur if the dentist or assistant mixes putty materials while wearing latex gloves or even if latex gloves were worn prior to mixing. Direct inhibition of polymerization also can occur if the impression material is in contact with a rubber dam.

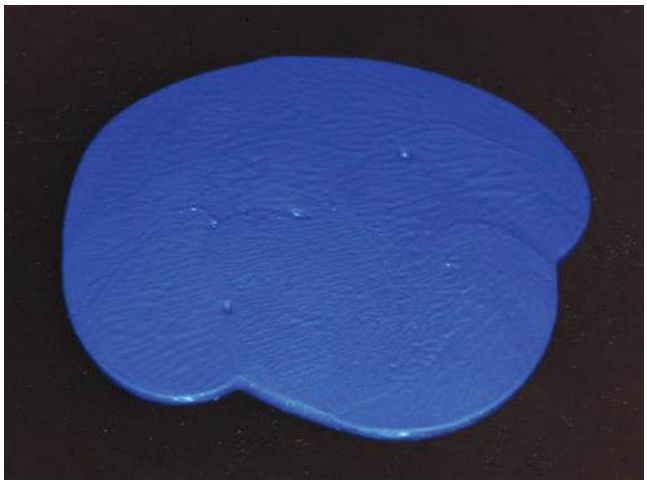
Indirect inhibition of polymerization also can occur intraorally when latex gloves contact tooth preparations and the surrounding periodontal tissues during tooth preparation or gingival displacement procedures.[90-93] Such inhibition of polymerization is often subtle and limited to small isolated areas of the surface of the impression. It is often not detected with the initial inspection of the impression and may be noticed only after pouring and separation of the gypsum cast. The presenting signs of inhibited polymerization are a film of unset material in isolated areas or the presence of a sticky, slippery substance on the surface of the impression (Fig. 13.51). It is similar to the feel and appearance of the oxygen-inhibited layer that is seen with photo-cure composite resin restorative materials.

Although these isolated areas of inhibited polymerization are subtle and not easy to detect (Fig. 13.52), depending upon their location they can render an impression unusable. Clinicians must inspect impressions and recovered casts carefully to ensure contamination of critical areas has not occurred.

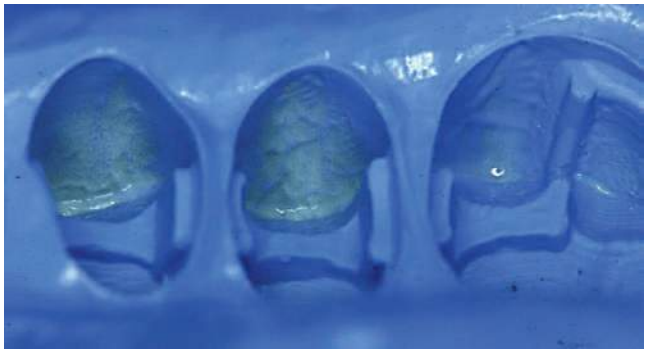
The mechanism of inhibition of polymerization is thought to be a result of contamination of the chloroplatinic acid catalyst of the PVS material with unreacted sulfur, which is present in natural latex gloves. Synthetic latex gloves, vinyl gloves, and the powder commonly found on gloves do not cause this inhibition of polymerization. Sulfur-containing gingival displacement chemicals do *not* cause inhibition of polymerization.[94]

Clinicians should avoid touching tooth preparations and adjacent gingival areas with latex gloves. When this is not avoidable, wearing vinyl gloves over the latex gloves is recommended. Once contamination of the preparations has occurred, cleaning with water and other agents is not effective in removing the contamination. Routine cleansing of tooth preparations with flour of pumice may be indicated prior to impression making.

Contemporary PVS impression materials may not be as sensitive to latex as the products that were used in the 1990s. It does appear that direct contact of contemporary PVS with latex or nitrile gloves will result in inhibited polymerization; however, these materials are not susceptible to indirect contamination.[95] It appears this reduced sensitivity is a result of changes to the impression materials rather than changes in gloves and rubber dam materials. Contemporary PVS materials have increased concentrations of platinum in the chloroplatinic acid catalyst, which makes it less sensitive to sulfur. Also nonionic surfactants have been grafted onto the polymer chains and these surfactants may act as a separating medium and prevent the sulfur from contacting the catalyst.



• **Fig. 13.51** Unpolymerized polyvinyl siloxane material due to handling of the material with latex gloves.



• **Fig. 13.52** Impression displaying areas of inhibited polymerization of the prepared teeth.

Putty-Wash Impression Techniques

While the recommended technique for full-arch impressions is use of heavy body/light body viscosities of impression material in a custom tray, many dentists prefer to make "putty-wash" impressions.[80] There are three approaches to putty-wash impressions. One approach is appropriate and acceptable, another can provide acceptable impressions but has some potential drawbacks, and a third approach is unacceptable.

An excellent technique for putty-wash impressions is to use the putty material to fabricate a custom tray. It is fabricated in the same manner as with PMMA materials or light cure materials. One layer of base plate wax is placed over the diagnostic cast as a spacer, and wax is removed from nonfunctioning cusp tips to provide occlusal stops. A putty impression is made in a stock tray, producing a PVS putty custom tray (Fig. 13.53).

A second approach is to use a relieved putty impression. In this technique, a preoperative putty impression is made intraorally prior to tooth preparation. Plastic sheets may be placed over the teeth to prevent putty material from entering gingival embrasures. In the area where the teeth are to be prepared, putty impression material is removed with a bur or scalpel to provide relief, and the impression is "washed" or relined with low viscosity PVS material.

This approach can be successful, but there are two potential problems. It is difficult to confine the wash materials to the area of the relieved impression, and some wash material enters the unrelieved impression. This results in an inaccurate occlusal pattern for the resultant cast. Thus the entire impression rather than just the relieved area should be "washed." This creates the problem of hydraulic distortion of the putty material as the impression is seated in the mouth. This is impossible to detect on a clinical level but may have a deleterious effect on the accuracy of the impression and resulting restoration.

The third approach to putty-wash impressions is the "simultaneous" technique. With this technique a stock tray is loaded with putty material, and the syringe material is injected around the prepared tooth or teeth. The tray containing the putty material is squashed over the syringe material, and the impression is made with the putty and syringe material setting simultaneously. This approach is unacceptable because it is impossible to control the thickness of the impression material, and excess bulk is used. It is impossible to control what material records the margin detail of the preparation(s). Usually portions of the prepared margin are captured in the putty, and putty materials are essentially deficient in their ability to record marginal detail (Fig. 13.54).

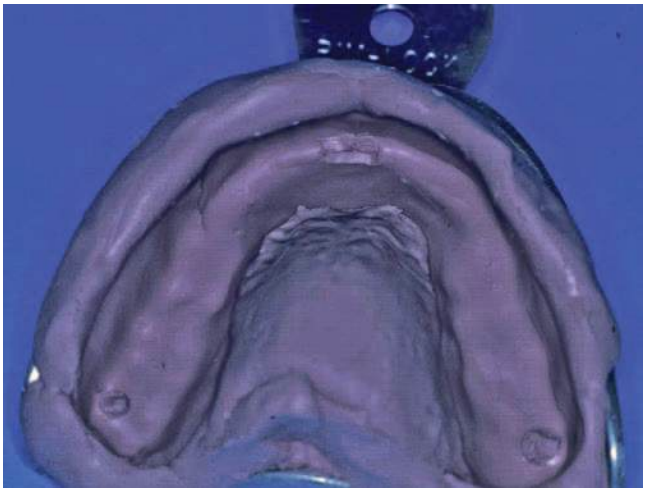
THE BOTTOM LINE

1. Water-based impression materials are most accurate with a uniform cross-sectional thickness of 4 to 6 mm. This allows impressions made with these materials to be made in a stock tray.
2. Elastomeric impression materials are most accurate when made with a uniform thickness of 1 to 2 mm. This implies the routine use of custom trays.
3. The rational for custom trays is to make impressions with a uniform cross-sectional thickness of 1 to 2 mm.
4. Custom trays can be made with PMMA, putty, and photo-cure bisacryl.
5. Occlusal stops should be placed on nonfunctioning cusp tips.
6. Low viscosity materials should be used only around the prepared teeth.
7. Specific impression adhesives should be used and applied to custom trays 7 to 15 minutes prior to making the impression.
8. Impressions should be poured at appropriate times for each type of material.
9. Impressions should be disinfected prior to pouring or sending to the laboratory.
10. Some putty-wash techniques are superior to others.

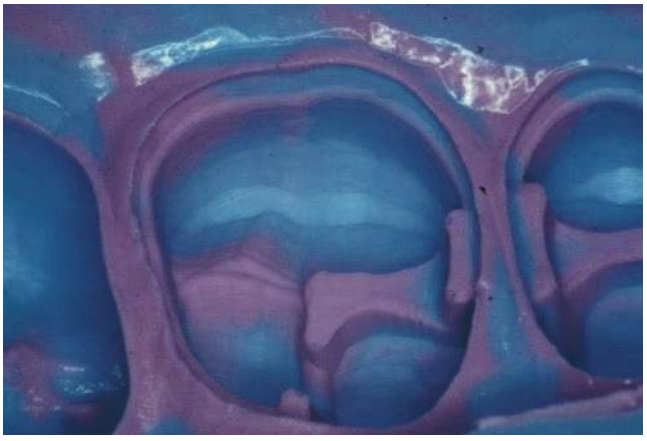
Gypsum Products

Gypsum products are widely used in a variety of applications in restorative dentistry. In many situations, obtaining maximum accuracy and strength with the selected gypsum product is critical to the long-term success of the restoration or prosthesis to be fabricated. To predictably attain such success, the clinician must first *select* the optimum product for the use intended and then must *manipulate* the gypsum product properly.[96,97]

Fortunately there are a relatively small number of gypsum products available, and the critical manipulative variables are rather simple. However, many dentists consistently select inappropriate materials for the task at hand. Additionally, even though proper



• **Fig. 13.53** Polyvinyl siloxane putty custom tray.



• **Fig. 13.54** Impression taken utilizing the "simultaneous technique." Notice portions of the margin captured with putty material, which may result in lack of details.

manipulation of gypsum is not complicated, it is frequently ignored, thus compromising both strength and accuracy.

The Nature of Gypsum Products

Mineral gypsum exists naturally as calcium-sulfate *dihydrate* and is treated by the manufacturer to remove water. It is then supplied to the dentist as a powder, which chemically is calcium-sulfate *hemihydrate*. The dentist then adds water, and a chemical reaction occurs converting the gypsum back to the *dihydrate* form. The chemical reaction for these processes may be written as follows:

$$Ca_2SO_4 \bullet 2\,H_2O + HEAT \rightarrow Ca_2SO_4 \bullet 0.5\,H_2O + 1.5\,H_2O$$

$$Ca_2SO_4 \bullet 0.5\,H_2O + 1.5\,H_2O \rightarrow Ca_2SO_4 \bullet 2\,H_2O + HEAT$$

Differences in the various types of gypsum exist due to the conditions under which the water is removed during the manufacturing process, the size and shape of the particles of the hemihydrate, and the effects of various chemical additives. From a practical standpoint, the properties of a given gypsum product can be predicted from the water/powder ratio suggested by the manufacturer, and an understanding of the concept of gaging water.

A finite quantity of water is required to chemically react with a given amount of gypsum powder supplied as calcium-sulfate hemihydrate. This amount of water is the same, whether the material to be mixed is Type I, Type II, Type III, Type IV, or Type V gypsum. However, depending upon the size and shape of the gypsum crystals, an additional amount of water is required to produce a workable mix of material. This extra water is known as gaging water. This gaging water does not react chemically and remains trapped within the matrix of the setting gypsum. With time (24 hours), most of this water evaporates, leaving air-filled voids in the set gypsum. As a general rule, the greater the porosity of a set gypsum material, the weaker it will be. It will also display diminished abrasion resistance and greater expansion, although this latter factor is somewhat variable depending upon whatever additives are present in the formulation.

For example, the recommended water/powder ratio for a Type II gypsum product is 45 cc/100 gm. For a Type III product it is 30 cc/100 gm, and for a Type IV stone it is 22 cc/100 gm. With 100 gm of each of these gypsum products, approximately 19 cc of water will be used up in the ensuing chemical reaction. The rest of the water will be unreacted and will eventually result in porosity in the set gypsum. The gaging water in the products would be 26 cc, 11 cc, and 3 cc for Types II, III, and IV, respectively. Thus, a cast from a Type II stone would be porous, weak, and have poor abrasion resistance. Casts from the Type IV materials would have the least porosity and the greatest strength and abrasion resistance. Casts from the Type III materials would be intermediate between the Type II and IV products.

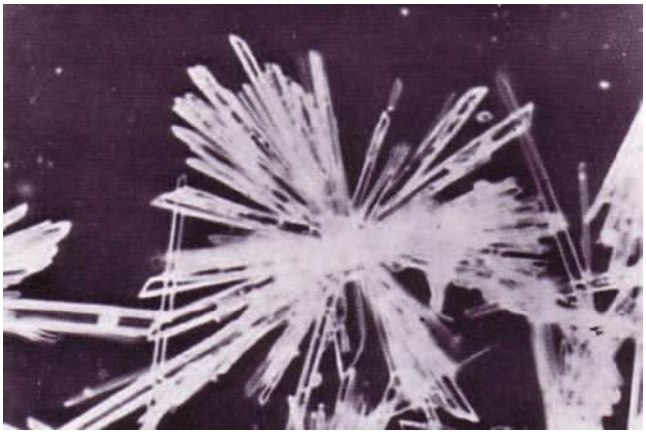
Understanding the nature of the setting reaction for gypsum is essential to understanding the clinical behavior of the materials. When the calcium-sulfate hemihydrate is mixed with water, insoluble crystals of calcium-sulfate dihydrate form and begin to precipitate from the mass. These crystals, which can be visualized three dimensionally as resembling sea urchins, serve as precipitation centers or nuclei of crystallization (Fig. 13.55).

As the reaction proceeds, more and more nuclei precipitate out and all continue to grow in an outward direction. The branches of these growing crystals intermesh and intercept one another, creating a tendency for the crystals to be pushed apart. This results practically in a net volumetric expansion (Fig. 13.56). Continued mixing while the reaction is underway can result in increased expansion because of the breakup of the gypsum crystals creating many more nuclei of crystallization.

Classification of Gypsum Products

Gypsum products are classified as Types I through V, and described by ADA Specification #25 (Table 13.7). Types I and II are plaster products. Type III is commonly known as dental stone or hydrocal. Types IV and V are high strength dental stones, also called improved dental stone or densite. The old terms *alpha* and *beta* calcium-sulfate dihydrate have been deemed obsolete because there is no actual chemical difference between the two types, and all differences are a result of the physical characteristics of the gypsum particles.

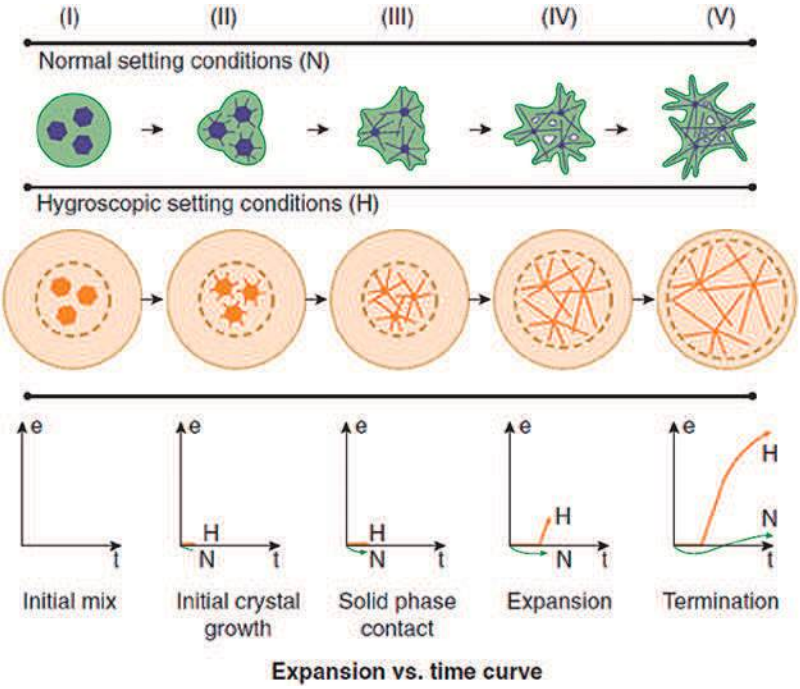
Type I dental stones are referred to as "impression plaster," and are typically weak, quick setting, and possess relatively low expansion values, as additives cause the materials to "set" before the normal setting expansion (due to crystalline growth) can occur. The primary use for Type I stones today is in mounting casts in an articulator. The minimum expansion displayed by these materials coupled with their quick setting capabilities make them ideal for this task. Because virtually zero expansion is desirable in a mounting stone and because expansion is bulk related, the two-stage mounting technique is recommended. With this technique, the cast is attached to the mounting ring with as small an increment of plaster as is deemed practical. After this material has set, a subsequent mix may be made and added to the initial mix for strength and cosmetic purposes. Impression plaster is also useful in certain indexing procedures, such as the soldering of fixed restorations, occlusal registrations, and in implant dentistry.



• **Fig. 13.55** Crystal of calcium-sulfate dihydrate.

TABLE 13.7 Classification of Gypsum Products

Type I	Impression plaster
Type II	Model plaster
Type III	Dental stone (hydrocal)
Type IV	Improved dental stone
Type V	Die stone high expansion



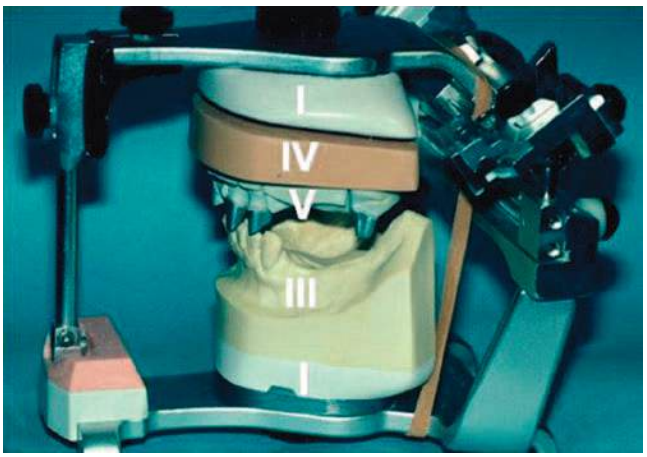
• **Fig. 13.56** Diagrammatic representation of the setting expansion of gypsum products. In the top row, the crystal growth is inhibited by the surface tension of water surrounding the growing crystals. In the middle row, the gypsum mixture *(the area surrounded by the dashed circle)* is immersed in water during setting *(represented by the larger solid circle);* the immersion water provides more room for longer crystal growth. The bottom row shows the expansion *(e)* over time *(t)* for hygroscopic setting expansion *(H)* and normal setting expansion *(N).* (From Anusavice KJ, Shen C, Rawls RH: *Phillips' science of dental materials,* ed 12, Elsevier, St. Louis, 2013; adapted from Mahler DB, Ady AB: An explanation for the hygroscopic setting expansion of dental gypsum products. *J Dent Res* 39:378-379, 1960.)

Type II dental stones are referred to as "model plaster" and have little use in restorative dentistry, although they are often utilized for the fabrication of diagnostic casts. These materials are weak and suffer from excessive expansion, which is desirable only in certain situations.

Type III gypsum is commonly referred to as "dental stone." Dental stone is considerably stronger and more resistant to abrasion than plaster. It also exhibits much less expansion. This product is used extensively in restorative dentistry for the fabrication of diagnostic casts, opposing arch casts, and in removable prosthodontics where a high strength stone may actually be a disadvantage when attempting to recover a processed prosthesis from a denture flask.

Type IV gypsum, often referred to as densite or improved dental stone, is commonly called "die stone." This stone is quite strong and abrasion resistant, and demonstrates low expansion values. It is used primarily for the fabrication of precision casts and dies in the lost wax process.

Type V gypsum is a high strength and high expansion stone, and is utilized today in the fabrication of casts and dies for metal-ceramic restorations, particularly when the casting shrinkage of the metal utilized is high and warrants extra expansion in the dies (Fig. 13.57).



• **Fig. 13.57** Cast model showing four types of gypsum products.

Selection and Manipulation of Gypsum Products

The selection of gypsum products is relatively simple, with the main variables being the strength and expansion desired for any specific procedure. Once the appropriate product is selected, proper manipulation is essential to ensure that the maximum physical properties of the material are attained.

The most important manipulative variable is paying attention to the proper water to powder ratios. Gypsum is best purchased in prepackaged envelopes. This ensures purity and prevents contamination that may occur when using bulk gypsum, and also provides a consistent volume of powder. Distilled water should be used and measured carefully with a pipette or glass cylinder. Often tap water contains minerals and other contaminants that can affect the setting reaction.

Controlling porosity in the set cast is critical to achieving maximum strength. This is accomplished in several ways. Adhering to the correct water/powder ratio helps this initially by minimizing the amount of gaging water. Next, the powder is always added to the water rather than the other way around. This minimizes air inclusions that occur when the water is added to the powder. Vacuum mixing should be utilized whenever dental stones or improved dental stones are mixed. The mixing times recommended by the manufacturer should be closely adhered to. Impressions should also be poured utilizing vibration devices to prevent air entrapment.

Separating the set cast from the impression must be done at the appropriate time. First the gypsum must have attained sufficient strength to avoid fracture during removal. Generally speaking, waiting 45 minutes to 1 hour will accomplish this. Additionally, especially when using water-based impression materials such as reversible or irreversible hydrocolloid, separation should occur within this time frame to avoid interaction between the impression material and the surface of the set gypsum. Such interaction would deleteriously affect the surface of the stone cast.

In most cases, handling of the casts should be avoided for 24 hours. There is a dramatic difference between the wet strength (1 hour) and dry strength (24 hours) of gypsum products. If casts are trimmed or margins on dies delineated before maximum strength is attained, it is likely that such casts and dies could be excessively abraded. Also gypsum casts are very susceptible to erosion by water during trimming. This can be minimized by soaking the aged (24 hours) cast in a solution of slurry water for 1 to 2 minutes prior to trimming. It will also prevent sludge from the trimming process from adhering to the surface of the cast. An alternative approach is to soften a sheet of base plate wax and adapt it to the surface of the cast prior to trimming. This will effectively prevent water contamination and sludge adherence.

Casting Investments

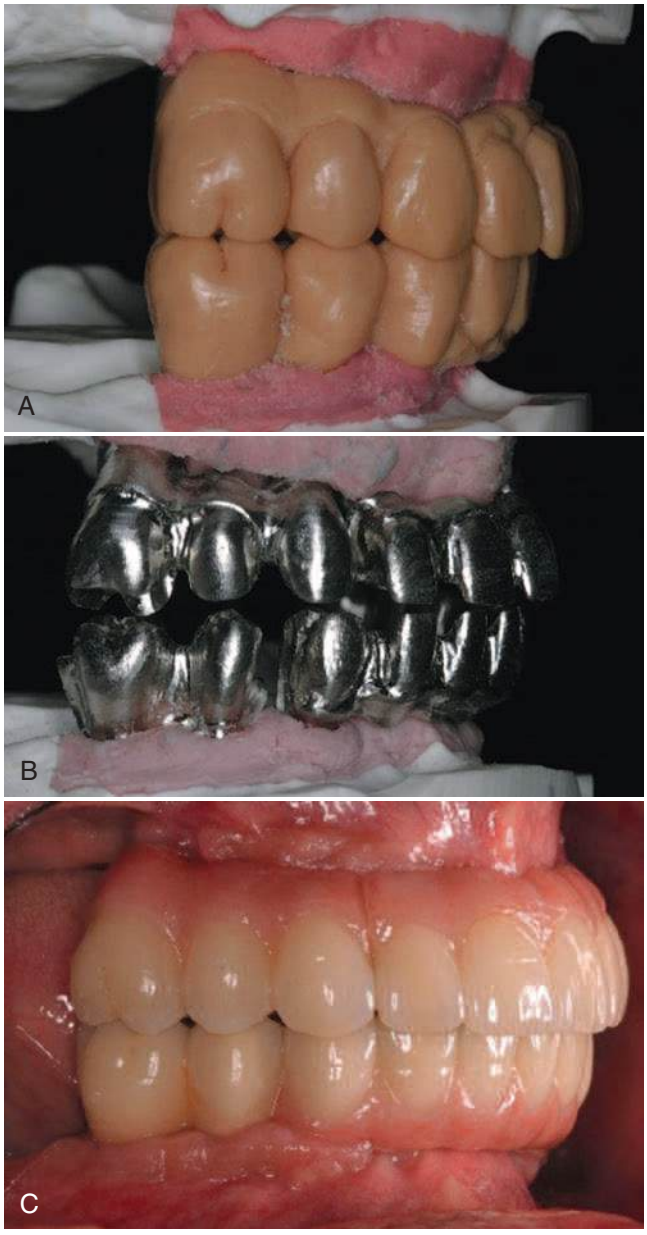
The lost wax process was introduced to the profession in 1907 by Taggart.[98] In order to fabricate a precise gold casting, it is necessary to balance a number of expansions and contractions related to dental materials (Fig. 13.58).

One of the major shrinkage factors in the process is the casting shrinkage of the metal alloys as they cool from the liquidus temperature to bench temperature. This shrinkage is roughly proportional to melting temperature: The higher the melting temperature the greater the shrinkage of the alloy. Type III gold alloys shrink approximately 1.2% while base metal alloys have double the casting shrinkage at 2.4%. This shrinkage must be compensated in the entire process. The major mechanism for accomplishing this is the use of casting investments.

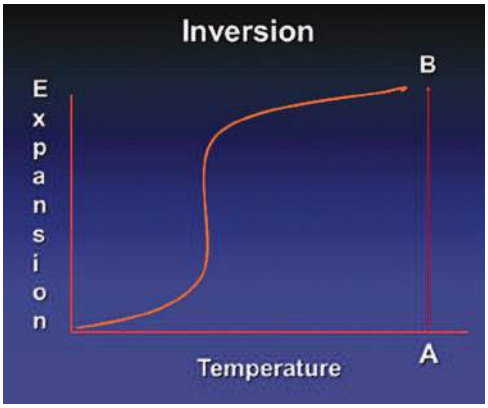
There are three major types of casting investments. Gypsum bonded investments are primarily used with gold casting alloys. Phosphate bonded investments, also known as high-heat investments, are primarily used with porcelain bonding alloys because they can withstand the higher temperatures required to melt porcelain bonding alloys. Ethyl-silicate investments are primarily used with removable partial denture castings.

With gypsum bonded investments there are three major components. The refractory is composed of materials such as cristobalite, tridymite, and quartz. The refractory supplies most of the thermal expansion of the investment through the process of inversion. As the investment is heated in a burnout oven, the investment expands in proportion to the increase in temperature. Then a significant amount of expansion suddenly occurs when a specific temperature, called the inversion temperature, is reached. This expansion occurs as a result of a change in crystalline form of the refractory; it changes from the alpha crystalline form to the beta crystalline form. This form change is associated with specific expansion, called inversion (Fig. 13.59).

The second component of gypsum bonded investments is the binder. The binder is basically gypsum, starting out as calcium sulphate hemihydrate powder converting to calcium-sulphate dihydrate after being mixed with water. The setting reaction of gypsum is associated with expansion, which contributes along with the thermal expansion to compensate the casting shrinkage of the alloy.



• **Fig. 13.58** A, Wax-up of maxillary and mandibular arches. B, Metal casting of maxillary and mandibular arches. C, Final restoration of maxillary and mandibular arches.



• **Fig. 13.59** Inversion diagram.

The third component of gypsum bonded investments is the modifier, which may take the form of oxidizing agent or reducing agent, depending on the commercial product.

There are three types of expansion associated with gypsum bonded investments. The first and most important is thermal expansion as a result of refractory expansion, as described previously. The second is setting expansion provided by the setting reaction of the gypsum binder. The third type is hygroscopic expansion provided by soaking the ceramic liner of the casting ring. At one time, asbestos was used for this function. The liner allows the investment and casting to be easily removed from the casting ring after casting is completed. By soaking it in water before placing it into the ring, the liner supplies a small amount of gaging water to the setting gypsum, which in turn allows a small amount of extra expansion. NOTE: There is a separate type of investment called a hygroscopic investment. These are very accurate investments that rely primarily on hygroscopic expansion and will not be described here because they are rarely used at this time.

Phosphate bonded investments were developed to provide the extra expansion required with metal alloys that melt at high temperatures and also had to be formulated to withstand those higher melting temperatures. Phosphate bonded investments come with a "special liquid" that is basically liquid silica. The liquid replaces some of the water used to mix the investment, and the basic rule is the more liquid silica, the greater the expansion of the investment.

Ethyl-silicate investments are used with removable partial denture framework castings. Basically the master cast is blocked out and duplicated with an ethyl-silicate investment. The wax-up is completed on the ethyl-silicate cast and the cast and framework are invested in ethyl-silicate investment; the wax pattern is burnt out and cast with chrome-cobalt alloy.

Manipulation of casting investments is relatively simple. First the appropriate investment materials must be selected. Then the most important variable is use of precise water to powder ratios. In the case of phosphate bonded investments, liquid to powder ratios are used, with the liquid being part water and part special liquid, depending on the amount of expansion required. Investments are vacuum mixed and poured into different sized molds, depending on the size of the casting.

After waiting a minimum of 1 hour, the casting ring is placed into a cold oven, subjected to a controlled burnout, and the restoration is cast.

THE BOTTOM LINE

1. Casting investments are an integral part of the lost wax process.
2. The three types of casting investments are gypsum bonded, phosphate bonded, and ethyl silicate investments.
3. Gypsum bonded investments are used with cast gold; phosphate bonded investments are used with porcelain bonding alloys, and ethyl-silicate investments with removable partial denture frameworks.
4. Casting investments compensate for casting shrinkage of the alloy.
5. Three types of expansion occur with casting investments: thermal, setting, and hygroscopic. The most important expansion is thermal.
6. Special liquid (liquid silica) is used with phosphate bonded investments to allow for the precise amount of expansion depending on the alloy used.

Gold Alloys

Cast gold restorations have been used in dentistry ever since Taggart introduced the lost wax process in 1907.[98] Properly done cast gold restorations provide the longest term survival of all types of restorations. A recent study of 1314 cast gold restorations in place from 1 to 55 years demonstrated that at 30 years, the failure rate was 4.5% (Fig. 13.60).[52] Pure (100%) gold is too soft for use in the oral cavity so it must be alloyed with other metals to produce the essential physical properties.

Gold foil is 98% pure gold and direct gold foil restorations have been placed for many years and provide long-term service with non–stress-bearing restorations (Fig. 13.61). Metals that are



• **Fig. 13.60** Cast gold restorations after 37 years.



• **Fig. 13.61** Gold foil restoration (98% gold) restoring a noncarious cervical lesion.

alloyed with gold in dental casting alloys include platinum and/or palladium for both strength and passivity, silver and copper for strength, and zinc and indium as deoxidizers. Gold is used because it is a noble metal and does not tarnish in the oral cavity. All noble metals share that property. The major noble metals used in dentistry are gold, platinum, and palladium. Nobility is determined by an element's position in the periodic table. Silver is a precious metal, but is not noble as it easily tarnishes in the oral cavity.

There are four types of gold alloy, Types I, II, III, and IV (Table 13.8). Type I gold alloy has the highest concentration of gold at 98%; it is considered a "soft" gold and is indicated in non–stress-bearing situations such as Class V inlays. Type II gold contains approximately 77% gold and is typically used with gold inlays and restorations with finishable margins, as these alloys can be burnished to attain maximum marginal integrity. Type III gold, with a concentration of 72% gold, is the most commonly used gold alloy in dentistry; it is used to fabricate crowns and fixed partial dentures. Type IV gold has the lowest concentration of gold (69%), and was used for the fabrication of removable partial denture frameworks at one time but rarely for that purpose now. It is still used for the fabrication of custom cast dowel and cores, as it is the strongest of the gold alloys and is less expensive due to the lower concentration of gold.

Type I gold is the softest, with the highest concentration of gold; Type IV is the hardest, with the lowest concentration of gold. Types I and IV gold are rarely used at this time, thus the most used alloys in contemporary dentistry are Types II and III.

One technique that can be used to improve marginal integrity of cast gold restorations is heat treatment. To heat treat a gold alloy, the alloy must have a copper content of 12%. Metal alloys can exist in one of three states. With a eutectic mixture, distinct phases can be identified in the microstructure. A good example of a eutectic would be amalgam where gamma, gamma-1, and gamma-2 phases can easily be identified in a micrograph. Eutectic mixtures are hard and brittle.

The second state seen with metal alloys is as an intermetallic compound. With intermetallic compounds there are no distinct phases, but there is molecular ordering, and these alloys are hard and strong. After heat hardening treatment, gold alloys exist as intermetallic compounds. When gold alloys are heat softened, they exist as a solid solution. With solid solutions there is a total lack of molecular order with randomization, and solid solutions are soft and burnishable.

The heat treatment procedure begins by heat softening. The alloy is melted and cast and immediately quenched in cold water. This transforms the alloy into a solid solution, which again is soft and burnishable. The restoration is then taken to the mouth and the finishable margins are burnished to achieve maximum marginal integrity. Then the alloy is heat hardened to render it hard and strong. To accomplish this, the restoration is heated in an oven and then slowly cooled to bench temperature. This transforms the alloy into an intermetallic compound, which will be hard and strong and provide optimum clinical service.

Cast gold restorations unquestionably provide the longest service of any restoration available in dentistry. The use of cast gold has declined significantly in recent years primarily due to the profession's obsession with esthetics and also due to the cost of gold. Most recent graduates of dental schools have not been taught how to fabricate gold restorations with a high level of precision. This is an unfortunate situation. Dentists are encouraged to contact any one of many study clubs that teach the concepts of cast gold restorations and add this wonderful modality to their armamentarium.

TABLE 13.8 Types of Gold Alloys Used in Dentistry

Type	Au	Hardness	Indication
I	98%	Soft	Class V
II	77%	↑	Inlays
III	72%		Crowns
IV	69%	Hard	RPDs

Porcelain Bonding Alloys

Prior to the introduction of porcelain-fused-to-metal technology, the clinician had only two options for the esthetic restoration of extensively broken down anterior teeth. One was the classic feldspathic porcelain jacket crown, which generally displayed inferior marginal fidelity and had a high incidence of fracture. The second option was heat-processed acrylic resin bonded to metal. The acrylic resin was mechanically bonded to the metal using beads and other mechanical retentive features. Type III or IV gold alloys were generally used for these restorations. The major disadvantage of these crowns and fixed partial dentures was the poor wear resistance of the resin veneering materials. At time intervals ranging from 3 to 5 years, the resin veneer would typically begin to age, demonstrating clinically evident wear and staining.

In late 1962 metal-ceramic restorations were patented and introduced to the profession by Weinstein, Katz, and Weinstein.[99] Original research in this area began with typical gold casting alloys, simply because they had been used in dentistry ever since Taggert had introduced the lost wax process to the profession in 1907. Four major alterations needed to be made to these alloys before clinical success could be expected.

1. The melting temperature needed to be raised because the gold alloys melted at temperatures well below the fusing temperatures of porcelain. This was achieved by the addition of an appropriate amount of platinum to the gold alloy.
2. Strength of the alloy needed to be increased. Any flexure in the metal substructure would result in catastrophic fracture of the porcelain veneer. Conventional gold alloys are far too flexible. Again, the addition of platinum was a major factor in improving the strength of the alloy.
3. The thermal coefficients of expansion and contraction of the metal alloy and the veneering porcelain needed to be more closely matched. The COE of the alloy was reduced to a great extent again by the inclusion of platinum, and the COE of the veneering ceramic was raised with the addition of sodium.
4. Predictable bonding of the porcelain to the metal needed to be achieved. Mechanisms of porcelain to metal bonding will be dealt with in more depth later on in this chapter. With these early alloys, chemical bonding was achieved with the inclusion of trace elements such as indium, zinc, and tin. These elements produced a consistent oxide on the surface of the alloy that would provide a chemical bond with the tetravalent oxides in the opaque porcelains.

These early porcelain bonding alloys contained a high percentage of gold (85%) and were used extensively through the 1960s and early 1970s. Although the physical properties of these high gold alloys were less than ideal, the main impetus for research into alternative alloys was the sudden increase in the price of gold in

the late 1970s that approached $1000/ounce. Since that time a number of alternative alloy systems have been introduced and have been extensively researched. The following analysis is given to assist the clinician in alloy selection for this important clinical modality. It is important to understand that the dentist should determine which alloy should be used for each restoration. This important choice should not be delegated to the laboratory technician.

Classification

Because the physical properties of an alloy can be predicted from their component elements, it is useful to classify porcelain bonding alloys according to their composition. Often alloys are casually referred to as "precious," "semiprecious," or "nonprecious" metals, usually based on the approximate gold content of the alloy. These terms lack precision, and several alloy groups with significantly different physical properties could be legitimately termed "semiprecious" (Fig. 13.62).

A more meaningful classification is to use the *nobility* of an alloy to designate the appropriate group to which a metal alloy should be assigned. The nobility of an element is determined by its position in the periodic table, and is a measure of its passivity in the oral environment. The noble metals most commonly used in restorative dentistry are gold, platinum, and palladium. Iridium, rhodium, ruthenium, and osmium are also designated as noble metals. Silver, which has a tendency to tarnish in oral fluids, is generally not considered as a noble metal. Silver is a precious metal, which signifies only that the metal has some intrinsic value. The word "precious" has nothing to do with tarnish potential.

Thus porcelain bonding alloys can be classified into two major groups: noble alloys and base metal alloys. There are three subgroups of noble alloys and three subgroups of base metal alloys (Table 13.9).

They can be differentiated from one another in terms of composition and in terms of certain attributes and physical properties such as strength, fit of castings, and bond potential with porcelain, biocompatibility, and economic considerations. Table 13.10 compares the alloy systems in terms of these variables.

Composition

High gold alloys typically contain a high percentage of gold, usually in the range of 80% to 85%. The alloy is strengthened with the addition of 7% to 10% platinum, and trace elements such as indium, zinc, and tin are added to provide an oxide layer for predictable porcelain bonding. These alloys predominated until the price of gold skyrocketed in the mid-1970s. Examples of commercial alloys in this category are Jelenko-O (J.F. Jelenko Co., Armonk, NY), Silhouette 850SL (Leach and Dillon, N. Attleboro, MA), and SMG-3 (J.M. Ney Co., Bloomfield, CT).

High noble alloys have a gold content typically in the range of 50% to 55% gold. Palladium is present in the range of 35% to 40%, along with the trace elements essential for bonding. These alloys may or may not contain silver. If more than 12% silver is present, it may cause a tendency for the porcelain to take on a greenish hue. Small amounts of silver may improve the wettability of the metal coping with the opaque porcelain, although this has not been scientifically established. Examples of this type of alloy include Olympia (J.F. Jelenko, Armonk, NY), USC Ceramic Alloy (Leach and Dillon, N. Attleboro, MA), and W2 and W3 (Williams Gold Co., Buffalo, NY).



• **Fig. 13.62** Porcelain-fused-to-metal fixed dental prosthesis.

TABLE 13.9 Classification of Metal-Porcelain Alloys

Noble Metal Alloys	Base Metal Alloys
High gold	Nickel-chrome-beryllium
Low gold	Chrome-cobalt
Palladium-silver	Titanium

TABLE 13.10 Comparison of Multiple Alloys According to Different Variables

	High Au	Low Au	Ag-Pd	Base
Composition	85% Au, 7% Pt	58% Au, 38% Pd	60% Pd, 38% Ag	65% Ni, 20% Cr
Strength	0.5 mm	0.3 mm	0.3 mm	0.1 mm
Fit	-	++	+	–
Bond	++	++	+	–
Biocompatibility	++	++	+	–
Cost	$100	$52	$34	$6

Palladium-silver alloys are composed primarily of high concentrations of palladium (60%) and silver (30%). With such high concentrations of silver, discoloration of the porcelain is obviously a consideration unless special precautions are taken during firing. Use of the alloy W1 (Williams Gold Co., Buffalo, NY) along with the low sodium content porcelain specified by the same manufacturer can eliminate this problem. Another example of a palladium-silver alloy is Silhouette 150 (Leach and Dillon, N. Attleboro, MA).

There are three groups of *base metal alloys*. The first, *nickel-chromium alloy,* is the most common of the three and typically contains about 65% nickel for strength, 20% chromium for passivity, and 2% beryllium for castability and control of oxide formation. Commercial products representing this type of alloy include Rexillium III (Jeneric Pentron) and Lite-Cast B (Williams Gold Co., Buffalo, NY). Nonberyllium alloys are commercially available but should not be considered for clinical use. Note: Beryllium is critical to help control oxide formation with base metal alloys; however, it has been associated with the development of berylliosis, a serious occupational pulmonary condition, and laboratory technicians are

at risk. The American Dental Association will no longer certify beryllium-containing alloys.

Chromium-cobalt alloys are commercially available and primarily marketed as "biocompatible" base metal alloys because they are nickel and beryllium free. Other than the exclusion of these elements, these alloys have little to recommend them and thus should not be used. Because they contain approximately twice as much cobalt as they do chromium, it has been suggested that these metals should be designated cobalt-chromium alloys. It is now possible to very accurately mill chrome-cobalt frameworks, which removes one of the major disadvantages (poor fit) of these alloys. There is little evidence at this time that the problems associated with excess oxide formation with these alloys have been solved.

Titanium alloys have become available recently, and copings are currently formed both by casting and by a combination of copy-milling and spark-gap erosion. Both of these systems must be considered experimental at this time and should be used only under strict protocol in an institutional setting. The other issue with titanium fused-to-metal is that veneering ceramics with very low COEs are required, and the esthetic results achieved with these ceramics are poor. Thus only the nickel-chrome-beryllium alloys should be considered for use by the practitioner.

Strength

Strength of a porcelain bonding alloy is of paramount importance to the long-term longevity of the restoration. Any flexure whatsoever in the metal substructure will result in catastrophic failure (fracture) of the porcelain veneer. The greater the number of units in any fixed partial denture framework, the greater is the potential for flexure. An alloy will ideally possess strength in relatively thin sections to allow for minimal tooth reduction and permit the fabrication of delicate frameworks. These substructures will then permit the creation of embrasure forms and connector designs that will allow for optimum cleansibility and the development of physiologic crown contours.

One of the major disadvantages of the high gold alloys is the relative lack of strength possessed by these metals. This lack of strength can be compensated by bulking up the framework and the occlusogingival dimension of the connector areas. This can compromise embrasure form and produce overcontoured restorations. A minimum cross-sectional thickness of 0.5 mm is suggested for use with these alloys. High gold alloys should only be used with single crown restorations on anterior teeth. In this situation, high gold alloys have an advantage because they do not produce a dark oxide, and intense opaque porcelains do not need to be used to block out the color of the oxide. Because of the reduced bite force with anterior teeth compared to posterior teeth, the strength of the gold framework is adequate.

The high noble and palladium-based alloys are considerably stronger than the high gold alloys, and a minimum cross-sectional thickness of only 0.3 mm is required. This reduction of thickness required is only 0.2 mm, but it is extremely significant clinically. The nickel-chrome-beryllium alloys are very strong and can survive with a minimal thickness of only 0.1 mm, although it is very difficult to work within such a tight tolerance without running the risk of perforation of the metal coping.

Fit of Castings

A most important factor in long-term success with cast restorations is achieving excellent marginal integrity. A number of factors totally independent of the alloy utilized are important in this regard. However, there are also differences in the ability of the various alloy systems to predictably produce excellent margins.

Several studies have evaluated the castability of these alloy systems.[100] Variations were found due to the different specific gravities of the alloys and due to variations in melting temperatures. However, current research has established that there is a "metal-mold equilibrium" for each alloy and investment combination that will produce an optimum casting in terms of completeness and the quality of the surface of the casting. This equilibrium is defined as the difference between the melting temperature of the alloy and the temperature of the mold into which the alloy is cast. Once this optimum temperature differential is established, excellent castings can be achieved with all of the systems under discussion. This finding, plus the development of induction casting machines, has determined castability is no longer a concern with any of the alloy systems.

When metal alloys are cast they solidify from the liquid to the solid state and then subsequently cool to room temperature. Casting shrinkage occurs during this time. This shrinkage occurs in three stages: (1) when the liquid metal cools from the melting temperature to the solidification temperature; (2) when it undergoes the transformation from the liquid to the solid state; and (3) when the solidified metal cools to room temperature. It is likely that this latter stage is where the most significant shrinkage occurs.

There are major differences in the amount of casting shrinkage that occurs with the various alloy systems. Casting shrinkage is a function of the melting temperature: The higher the melting temperature of the alloy, the greater will be the casting shrinkage. This shrinkage of course must be compensated by equivalent expansion of the investment. There is some question that this can be accomplished with adequate precision with the alloys possessing the highest melting temperatures. This is especially a problem with large multiunit castings.

Type III gold alloys shrink approximately 1.2%. Gypsum bonded investments can compensate very accurately for this amount of shrinkage. High gold porcelain bonding alloys shrink slightly more at about 1.3%. The casting shrinkage of the high noble and palladium based alloy systems is slightly higher than that of the high gold alloys. Compensation for this shrinkage is accomplished with high heat phosphate bonded investments, using specific amounts of liquid silica to control the expansion. The more silica used, the greater is the expansion.

The nickel-chrome-beryllium alloys shrink 2.4%, which is double the shrinkage of a Type III gold alloy. Early studies with these alloy systems indicated that the castings were too tight and failed to seat adequately. This was simply a result of lack of adequate expansion provided by the investment materials utilized in the investigation. Today it is possible to provide adequate expansion with modern investments. However, the difficulty of producing an accurate casting increases as the expansions and contractions of the materials involved increase. This is particularly true when dealing with castings with nonuniform amounts of bulk, and large multiunit castings. It is difficult to solder base metal alloy units, and as a result, fixed partial dentures fabricated with this alloy are often cast in one piece. A diminution in the quality in the fit of the castings is likely to occur when this is done.

A third factor related to fit of castings is distortion of the metal coping during firing of the porcelain veneer. Clinicians have long noted subjectively that metal copings that fit very well prior to firing the porcelain do not fit nearly as well after firing. The distortion that occurs is a function of both the composition of the alloy and the margin geometry utilized. Additionally, research has shown

that not only do individual copings distort, but so do entire frameworks for larger fixed partial dentures. Because the majority of the distortion seems to occur during the degassing process, the major factor in this latter type of distortion seems to be release of stresses induced in the casting process.

As noted previously, distortion varies with the composition of the alloy. The higher the melting temperature of an alloy, the less distortion will likely occur. Thus the high gold alloys, which have melting temperatures closest to the sintering temperatures of the porcelain, tend to distort the most. High noble and the palladium-based alloys distort somewhat less, while base metal alloys tend to distort very little. However, this latter fact should not be taken to construe that there is no change in fit of base metal castings during porcelain application. In fact the quality of the fit is diminished considerably with base metal castings as a result of inevitable *oxide* formation on the internal of the castings. This oxide layer, which is substantial with base metal alloys, can interfere with seating of the casting and result in marginal opening.

Of course distortion can be minimized with the intelligent use of cervical marginal geometry. Shoulder and shoulder-bevel margins distort relatively little, even with high gold alloys, and should be utilized routinely in nonesthetic situations. Framework distortion can be minimized by proper alloy selection, use of custom sager trays, and use of the postsoldering procedure where indicated.

When all factors are taken into account, it would appear that the high noble/palladium-silver/high palladium alloys would have an advantage over the other systems in terms of the ability to achieve consistently excellent marginal integrity.

Bonding to Porcelain

Clearly the formation of a predictable bond to porcelain is an essential property of an alloy intended for crown and bridge prosthodontics. There are three main mechanisms of porcelain bonding: mechanical, chemical, and compression bonding. The most important of these by far is chemical bonding. However, it is prudent to take advantage of all three methods.

Mechanical bonding occurs due to the inherent microscopically rough metal surface that exists after the degassing procedure. Van der Waal forces play a role here, as does the improved wetting that occurs when the surface of the alloy is air-abraded with 50-micron aluminum-oxide particles prior to applying the opaque porcelain. It has been estimated that such mechanical bonding should constitute only about 10% of the total bond of the porcelain to the metal.

Compression bonding occurs as a result of the slight thermal coefficient of expansion mismatch that exists between the porcelain and a compatible alloy. When a metal-ceramic restoration is taken from the muffle of a porcelain oven and allowed to cool, the metal coping will cool first and begin to shrink slightly. This will pull the overlying porcelain under compression. This compression stress contributes to the overall bond strength. We take advantage of this by attempting to get a "wrap around" effect with our cutback design, both buccolingually and interproximally.

Chemical bonding is the most important means of bonding porcelain to metal. The opaque porcelains are specially formulated with tetravalent oxides that will bond to oxides formed on the surface of the metal. The metallic oxides either form naturally or are induced from trace elements during the degas cycle, and ideally will form a monomolecular layer on the surface of the alloy (Fig. 13.63).

Chemical bonding with the noble metal alloys has been extensively studied and is very predictable. Trace elements such as indium, zinc, and tin are added to the alloy to provide the oxide required as well as refine the grain structure. The metal coping is run through a degas or oxidation cycle to induce the oxide onto the surface. At least 50% new metal is used each time a coping is cast to prevent depletion of the trace elements from the mix. Light air abrading with aluminum-oxide is recommended with many alloys to prevent excess oxide formation.

Chemical bonding is much less predictable with the base metal alloys. The problem with these alloys is with the oxide layer. There is simply too much of an oxide layer, and while the bond failure may occur at either the interface between the metal and the oxide or between the oxide and the opaque, it most frequently occurs within the thick oxide layer itself.

Most porcelain bond tests are flawed in that they all allow for a considerable amount of compression bonding. Thus base metal alloys can appear to possess good ability to bond to porcelain as measured by these tests. However, a majority of the reported bond will be compression bonding, which can fail catastrophically at any time. As stated previously, the major problem is the thickness of the oxide layer, which propagates each time the porcelain is fired. The main function of beryllium in a base metal alloy is to help control the oxide layer in addition to improving the castability of the alloy. In the former function, the beryllium is only partially successful. Nonberyllium base metal alloys have no control of the oxide layer and hence should be avoided.

The tendency of base metal alloys to form thick oxide layers is also problematic when attempting to solder fixed partial dentures. The problem here is that the alloys form the oxide layer before



• **Fig. 13.63** A, Noble metal alloy copings. B, Opaque porcelain applied to noble metal alloy copings. C, Finalized porcelain-fused-to-metal crowns.

the solder can melt and join with the parent metal. This interposed oxide layer can result in weak or nonexistent solder joints. As a result, many fixed partial denture frameworks using base metal are cast in one piece. While the literature on one-piece castings is somewhat equivocal, many feel that the fit of the castings is definitely compromised. Also, it is often essential to section, index, and solder units because they simply do not fit. This can be difficult with base metal alloys.

Biocompatibility

While patients can be theoretically allergic to almost any element in a casting alloy, even gold, the major elements that are of interest in terms of biocompatibility are nickel and beryllium. It is estimated that approximately 22% of women are allergic to nickel. The incidence of nickel allergy in men is less than 10%. The major reason for the difference in incidence between the sexes is thought to be that many women have pierced ears and wear costume jewelry. Much of this jewelry is nickel based, and it is believed that this will induce the allergy.

The signs and symptoms of nickel allergy can be local or systemic. Often the presenting symptom is a rash or eczema on the arm or legs. Patients often do not associate such peripheral symptoms with an intraoral restoration and hence may suffer for prolonged periods before a diagnosis of nickel allergy is made. When use of a base metal alloy is contemplated, the patient should be specifically asked if he or she has any allergy to nickel or reaction to any metal.

Beryllium is primarily a potential problem for the laboratory technician where grindings containing beryllium can be the etiologic agent for a number of respiratory ailments. Proper ventilation and the routine wearing of a face mask can prevent such untoward events. Beryllium has also been shown to migrate toward the surface of restorations, and in that location can also dissolve in oral fluids. The clinical significance of these findings is unknown at this time but is a concern to some.

A similar situation may occur with the high palladium alloys. In a number of proprietary studies palladium has been shown to dissolve in oral fluids and be cytotoxic. This most frequently occurs when the combined concentration of gold and silver is less than 25% of the total alloy. Again the clinical significance of this is unknown.

Economic Considerations

Obviously the cost of a metal alloy needs to be considered. This can be even more important when the price of gold fluctuates extensively. Indeed the genesis of most of the research into alternative alloys, and many of the all-ceramic alternatives to metal-ceramics, was a gold price approaching $1000/ounce.

A pilot study conducted at the University of Southern California School of Dentistry in 1981 compared the price of a metal coping fabricated with different alloys. The price of gold at the time was close to $400/ounce. Fifty metal copings were collected at random, weighed, and an average weight calculated. The average price/coping/alloy could then be calculated. The results do not take into account the lower specific gravity of base metal alloys, which affords additional minor savings. Nor does it take into account the fact that copings made with one alloy could be thinner than those fabricated from another alloy. The data do indicate a significant differential in cost (Table 13.11).

Since 1981, the cost of noble metals has fluctuated considerably, but the price of gold has ranged around $1000 to $1600 an ounce. That means the metal cost of laboratory work has at least doubled over that time, which has a huge effect on laboratory costs and subsequently raises the cost of restorative services to patients. Indeed, a major impetus for the development of all-ceramic alternatives to PFM is the laboratory costs.

Depending upon the philosophic orientation and sociologic makeup of a practice, the differences in cost of alloys may or may not be significant. Far more important would seem to be the predictability of an alloy in terms of castability, consistent fit, and lack of bond failure.

In summary, a host of metal alloys intended to be used with porcelain are currently available commercially. These have been classified according to both composition and the characteristics and physical properties of the major groups compared.

An analysis of the material presented reveals that one group of alloys, the high noble alloys, seems to have no deficiencies and scores high in all of the categories discussed. High noble alloys also happen to be cheaper than high gold alloys, although that is a secondary consideration. These alloys would seem to be the metals of choice. They are stronger and can be used in thinner section than the high gold alloys. They cast easily and accurately, and show fewer tendencies to distort on firing the porcelain. Porcelain bonding is very predictable with no discoloring of the porcelain. Biocompatibility is not a problem.

If cost was a major consideration, use of a palladium silver alloy with a porcelain low in sodium content would likely be a suitable choice. As is usually the case, proper manipulation of the alloy chosen is probably even more important than which alloy is chosen.

THE BOTTOM LINE

1. Porcelain-fused-to-metal crowns and fixed partial dentures continue to be important restorations and provide the best combination of reasonable esthetics with maximum longevity of available esthetic crown options.
2. Many different alloy systems with different properties and costs are available.
3. The dentist should choose which alloy is most appropriate for each restoration.
4. Some of the factors that must be considered when choosing an alloy include strength, ability to achieve excellent marginal integrity, ability to attain a predictable bond with porcelain, biocompatibility, and cost.
5. Base metal alloys are inexpensive but have significant problems with fit, porcelain bonding, and biocompatibility.
6. High gold alloys do not form dark oxides, and have the potential to provide improved esthetics for single crowns on anterior teeth.
7. High noble alloys are less expensive than high gold alloys and seem to be the alloy of choice for metal-ceramic restorations.

TABLE 13.11 Comparison of Costs for Different Alloys in 1981 and 2016

Alloy	Cost 1981	Cost 2016
1. High gold	$50	$100
2. High noble	$26	$52
3. Palladium-silver	$17	$34
4. Base metal	$3	$6

Implant Materials

Dentists have tried for centuries to find a way to implant devices into the jaw bone to replace missing teeth. Many different approaches were tried but no material or technique proved successful until the 1970s. Up until that time, various forms of metal implants primarily fabricated from chrome-cobalt alloy had been used with some short-term success, but all failed in the long term. Blade implants, ramus-frame implants, and other configurations all failed eventually because of epithelial growth between the implant and surrounding bone and subsequent infection. The major problem with these implants was that the surgical site that was prepared was not sufficiently precise to provide initial intimate bone-implant proximity.

Vitreous carbon implants were introduced in the mid-1970s and showed great promise, but suffered the same fate as the implants described previously, as the shape of the implant was rectangular, which made it impossible to surgically prepare the site with sufficient precision.

The first successful implant system was the mandibular subperiosteal implant designed by Dr. Roy Bodine (Fig. 13.64). This chrome-cobalt implant was used with resorbed mandibles and demonstrated survival rates close to 20 years. With this system, an impression was made of the exposed mandible after raising a flap, and the implant framework was placed directly on the periosteum over the alveolar bone rather than in the mandible. These implant frameworks were restored with mandibular dentures that were secured to the metal framework with clips. While subperiosteal implants had acceptable survival rates, the frameworks eventually became infected and had to be removed. This removal often resulted in considerable morbidity, because bone had grown over the framework and had to be surgically removed before the framework could be removed. Maxillary superiosteal implants did not perform nearly as well as mandibular implants because of resorption of the maxillary alveolar process.

The major breakthrough with implants came as a result of Brånemark's research in Sweden. Brånemark was doing capillary research in animals using titanium cylinders to contain the specimens. When trying to remove the cylinders from the bony extremities of the animals in order to reuse them, the researchers discovered the cylinders were very difficult to remove and had in fact integrated into the bone. After several years of animal and human research, Brånemark introduced his system of root form titanium osseointegrated implants in the early 1980s. It is not hyperbole to state that this was one of the most important advances in the history of dentistry.

Brånemark's original clinical research was with "hybrid" prostheses, which consisted of five titanium implants placed in the anterior mandible between the mental foramina (Fig. 13.65). These implants were placed under a strict surgical protocol, then allowed to heal and integrate some time before restoration. The implants were splinted together and restored with a bar and arch-shortened mandibular denture. These restorations were opposed by a maxillary denture. Clinical results were outstanding with 95% implant survival and 100% prostheses survival at 5 years.

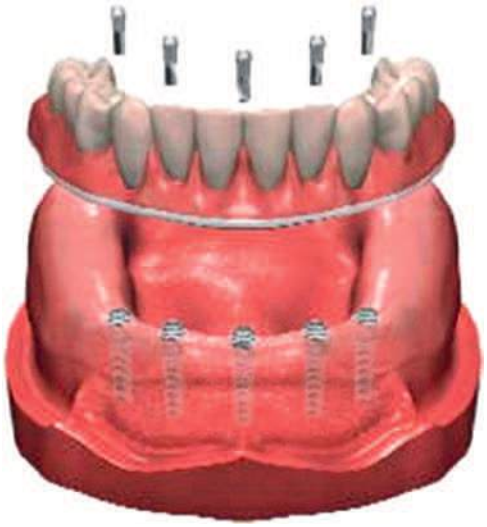
These original implants were composed of 99% commercially pure titanium and had machined surfaces. Many manufacturers developed and marketed a number of different implants with different designs, and data from current research cannot identify a "best" implant design. Several implant systems were introduced with "coatings" of hydroxyapatite and other formulations, but they performed poorly compared to titanium implants, primarily due to breakdown between the "coating" and the implant. Titanium-vanadium alloy implants have been used successfully and are stronger than commercially pure titanium implants, but appear to have no clinical advantage because fracture of implants is a very rare occurrence.

Manufacturers have experimented with different implant surfaces trying to create more rapid osseointegration. Most contemporary implants have microscopically "rough" surfaces, which seem to results in more rapid integration into the bone as indicated by "push-out tests." It is not known whether this is clinically significant.

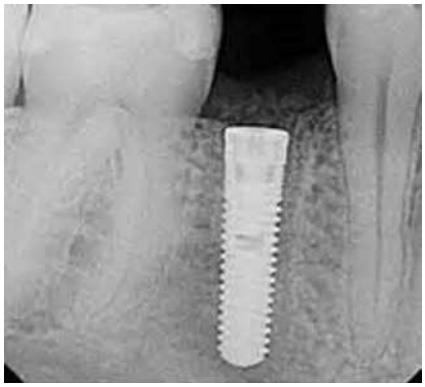
Implants today are used to replace single teeth in all areas of the mouth, to restore multiple lost teeth with fixed partial dentures, and to support full-mouth reconstructions (Fig. 13.66). It is important to understand that Brånemark's original research with 95% survival at 5 years was done with five splinted implants in the anterior mandible. This is the best possible bone for implantation, so these results cannot be extrapolated to implants placed in other areas of the mouth. The least desirable bone for implantation is in the posterior maxilla.



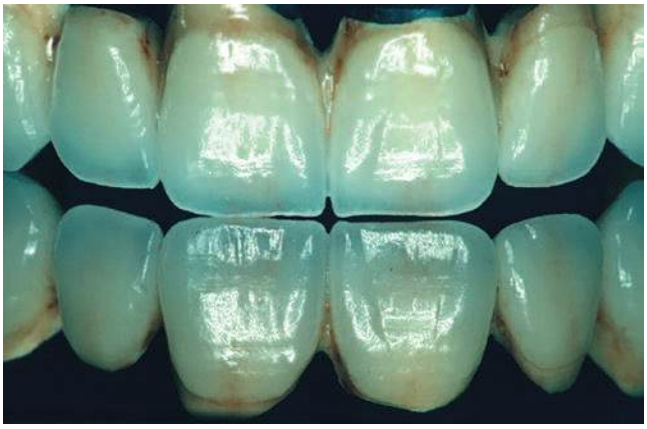
• **Fig. 13.64** Mandibular subperiosteal implant framework.



• **Fig. 13.65** Illustration showing design of hybrid dental prosthesis.



• **Fig. 13.66** Radiograph of single mandibular implant.



• **Fig. 13.67** Porcelain-fused-to-metal fixed prosthesis.

While data from Brånemark's original research cannot be used to predict survival of implants placed elsewhere in the mouth, it is clear that root form titanium implants have adequate survival rates to recommend routine use for the replacement of missing teeth. Restorations can be screw-retained to the implants or cement-retained. Many authorities prefer screw-retained restorations as this allows 100% retrieval, but this approach requires precise implant placement.

One issue that has come to light in recent years is the condition called periimplantitis, defined as periimplant bone loss accompanied by suppuration. The incidence of periimplantitis seems to be significantly higher than originally reported and seems to primarily occur in patients who originally lost their teeth to periodontitis and who smoke. In these patients, the decision to extract teeth and replace them with implants should be undertaken with an abundance of caution.

THE BOTTOM LINE

1. The advent of predictable osseointegration with titanium root-form implants is one of the most important advances in restorative dentistry.
2. Contemporary implants have acceptable survival rates and should be considered as the primary approach for the replacement of missing teeth.
3. No "best" implant design has been determined by clinical research.
4. Coated implants have not survived as well as noncoated implants.
5. Periimplantitis is a serious complication of implant therapy.
6. Smokers who lost their teeth due to periodontal disease are especially at risk for periimplantitis.

Evolution of All-Ceramic Crowns

New ceramic systems continue to be introduced to the profession at exponential rates. It is critical that clinicians understand how these systems have evolved since the mid-1980s and that they understand the deficiencies of many of these systems so they can more effectively evaluate the potential of new systems as they are introduced to the market. The three major concerns with new systems are as follows:

1. What is the real esthetic potential of the system compared to available tried and true systems?
2. What is a reasonable potential life span for the new system? This can be difficult to determine because improvements in physical properties do not necessarily translate into improved longevity. Keep in mind that ceramics do not fail because they are not strong enough; they fail because it is not known how to manufacture flaw-free restorations. Those flaws, called Griffith flaws, tend to propagate over time and result in fatigue failure.
3. What clinical situations are recommended for the new system? Can fixed partial dentures made with this system reasonably be expected to survive? Should its use be limited to anterior teeth or can it be used in molars?

This section has been written to describe the main features of the different ceramic systems as they were introduced to the profession. It is not our intent to examine each system in great detail, but to highlight the important features that led to success or failure. It is also not intended to be all-inclusive but will focus on the major players.

The first esthetic crown was the porcelain jacket crown (PJC) introduced to the profession circa 1910 by Land. To fabricate this crown, a platinum foil was adapted to the die using a tinner's joint. This foil was used as a scaffold to carry the porcelain into the porcelain oven where it was fired and the foil later removed. The PJC had good esthetics, but poor marginal integrity and a short life span. They were cemented in place using different shades of silicophosphate cement. The major deficiency of the PJC was the fact they frequently fractured 1 or 2 years after placement.

In 1962, Weinstein-Katz and Weinstein filed the patent for the first metal-ceramic crown system.[99] At the time, a porcelain-fused-to-metal (PFM) was known as a "Ceramco." It became popular in the late 1960s, but dentists knew little about the important preparation details required to produce a truly esthetic restoration (Fig. 13.67). Dentists did not understand the space requirements needed for layers of metal, opaque porcelain, body, and incisal porcelains and thus most preparations were underprepared. They knew nothing about cervical margin design. Laboratory technicians were also unaware of many of the important details of fabrication. The dental profession and laboratory industry learned about PFMs by trial and error. This resulted in many restorations that lacked the expected esthetic potential and frequent fracturing of the feldspathic porcelain veneer from the metal coping.

In the mid-1970s, individuals such as John McLean, Jack Preston, Lloyd Miller, and Jack Nicholls began to do research that guided the profession in producing PFMs that had improved esthetics and great clinical longevity.

One excellent systematic review published in 1989 indicated the failure rate with PFMs is 5% at 11 years.[101] Several other

studies have shown similar results.[102] Such data must be considered when assessing the utility of new ceramic systems. Numerous ceramic systems have been introduced to the profession (Fig. 13.68). Almost all of these systems have been brought to market with almost no clinical testing. It is also important to repeat the fact that laboratory testing of physical and mechanical properties and load-to-failure testing yields little useful information about expected clinical performance. Thus it is clear that clinical dentists are carrying out the essential clinical trials for the manufacturers.

In 1978 McLean introduced the aluminous core PJC.[103] This restoration used twin platinum foils for fabrication. One foil was used as a die spacer and the other was tin plated and used to form the core with aluminous core porcelain. This was veneered with compatible low-alkali porcelain. The laboratory technique was very demanding, which limited its commercial adaptation. In 1984 Donovan introduced a simplified single foil technique for fabricating aluminous core PJCs, but this simplified technique also failed to catch on.[104]

The twin-foil crown by McLean was the first in a series of foil-supported restorations that attempted to improve clinical longevity by reducing the number of microscopic defects on the internal portion of the crown. Shorer introduced the Renaissance foil system, which used a pleated noble metal foil and a gold colored bonding agent. Asami Tanaka had a similar system with his Sunrise crown. These foil-supported restorations had poor survival data, primarily because the foil took the form of the tooth preparation and frequently there were excesses of unsupported porcelain. The industry standard is that there should never be more than 2 mm of unsupported porcelain, regardless of the core material. With foil-supported crowns this is difficult to achieve in the majority of cases.

In 1993, Shorer and Whiteman introduced the Captek system. Captek is an interesting technology that forms a noble metal coping without casting or machining. A wax sheet containing a powdered metal alloy of 88% gold is adapted to a refractory die and heated in an oven until the wax melts and the metal coalesces into a porous coping. A second wax sheet containing platinum is adapted over the coping and heated. The wax melts and the platinum enters the gold coping by capillary action (hence "Captek"). The gold color of the coping results in excellent esthetics and the platinum provides strength to resist metal flexure. The major deficiency of Captek is the inability to provide adequate support for the veneering ceramic, again because the coping takes on the form of the tooth preparation and few preparations are ideal.

In 1983, Sozio and Riley introduced the Cerestore crown, also called the "shrink-free" ceramic crown.[105] This product used an extremely accurate epoxy-resin die system, on which was injected a high strength alumina-ceramic core. This core did not shrink on sintering and was quite opaque, which resulted in unesthetic cervical opacity. The core could be colored intrinsically and was overlayed with a relatively translucent veneering ceramic. The Cerestore crown system had a relatively short time on the market, partially due to the inherent cervical opacity, which resulted in limiting the esthetic outcome to that attained with metal-ceramic restorations with porcelain labial margins, and partially, due to early clinical failures with fracture.

Experiences with Cerestore taught us one important principle: If an all-ceramic crown system uses an opaque core to improve fracture strength, the esthetic result achieved by that system will be no better than the esthetics achieved with metal-ceramics!

About the same time as Cerestore entered the marketplace, Adair and Grossman from the Dow Corning Company introduced Dicor, which was marketed as the "castable" ceramic crown.[106] Dicor crowns were waxed, invested, and cast using a glasslike material that was very translucent. The restoration was then given a specific heat treatment called "ceraming," which induced a change in crystalline structure and made it slightly less translucent. Then a thin film of extrinsic stain was painted on the crown to attain the desired shade.

Dicor was the first example of a "monolithic" crown system where the final esthetic result was attained by applying a thin layer of extrinsic coloration (Fig. 13.69). In general, such monolithic restorations lack depth and have a high level of surface or specular reflection. Such restorations often experience a high incidence of metamerism where the restoration appears one color with certain light sources and a very different color in different light sources. Thus the major deficiency of Dicor was the inability to achieve intrinsic coloration. Fracture rates for Dicor crowns were also relatively high.

Originally Dicor crowns were intended to be cemented with different shades of zinc-phosphate cement. However, Malament discovered that fracture rates could be improved if Dicor crowns were etched internally and bonded in place with resin cements. The combination of high fracture rates and poor esthetics led to the early demise of Dicor, however.

Experiences with Dicor taught us two important principles: First, monolithic crowns do not allow intrinsic coloration and provide depth of color in the same manner as layered crowns; second, the clinical performance of ceramic crowns can be improved by etching and bonding.

To overcome the esthetic disadvantages of Dicor, Willi Geller introduced the Willi Glass crown in 1987.[107] Willi Glass used an internal translucent core of Dicor, about 0.75 mm thick. This was veneered with Vitadur N ceramic. Willi Glass was undoubtedly the most esthetic restoration ever made available to the profession. However, it had a relatively short clinical life span and suffered

1. Feldspathic PJC
2. Aluminous Core PJC
3. Foil Reinforced
4. Captek
5. Cerestore
6. Dicor
7. High-Ceram
8. Willi's Glass
9. IPS Empress
10. Optec
11. Empress II
12. IPS Eris
13. IPS e.Max
14. Inceram (spinell, alumina, zirconia)
15. Procera All-Ceram
16. Zirconia cored materials

• **Fig. 13.68** Major ceramic systems used in restorative dentistry.



• **Fig. 13.69** Ceramic restoration by Dicor.

from catastrophic fracture. This was partially because the Dicor core was not strong enough at that thickness, and partially due to thermal incompatibilities of the Dicor core and the veneering porcelain.

In the late 1980s, Wholwend and Scharer introduced the original IPS Empress leucite reinforced ceramic system (Fig. 13.70).[108] This system introduced the concept of hot pressing the ceramic, which has the potential to reduce the number of microscopic defects in the ceramic crown.

IPS Empress crowns can be fabricated in two different ways. Monolithic IPS Empress crowns can be waxed, pressed, and stained. These monolithic restorations suffer from issues discussed previously and do not provide optimum esthetics on anterior teeth. Layered IPS Empress, with a leucite reinforced core and matched veneering porcelain, provides the most esthetic and durable ceramic option currently available for restoring anterior teeth. Clinical trials have indicated 95% survival of bonded, layered IPS Empress crowns on anterior teeth. Survival rates drop considerably with crowns on posterior teeth.



• **Fig. 13.70** Leucite-reinforced glass ceramic crowns (IPS Empress).

It is critical to understand that this high survival for IPS Empress crowns on anterior teeth is only achieved if they are internally etched and bonded in place. There are three approaches to accomplishing this. The first is the traditional approach of bonding the restoration in place using a fourth-generation or fifth-generation etch-and-rinse dentin bonding agent (DBA) and a dual cure resin cement. One problem with this approach is if the DBA is cured prior to cementation, the thickness of the cured DBA may prevent complete seating of the crown. Alternatively, if the DBA is cured after seating the crown at the same time as the dual cure cement, the hydraulic pressure generated during seating will likely collapse the collagen bundles and significantly compromise the hybrid layer and dentin bonding.

A second approach is to use Paul and Magne's immediate dentin sealing technique.[109] With this technique the dentin is sealed with a fourth-generation DBA immediately *after* tooth preparation, prior to making the impression. Thus the film thickness of the DBA is of no importance, and the tooth is sealed during the provisional phase. The sealed preparation must be isolated with Vaseline when making the provisional restoration to prevent bonding of the provisional to the dentin. The oxygen-inhibited layer of the DBA can also distort impressions, and the surface of the DBA must be carefully cleaned prior to cementation. This immediate dentin sealing technique results in much higher dentin bonds than are created when bonding at the cementation stage.

The third approach to bonding etched IPS Empress restorations in place is to use a self-adhesive resin cement. These cements do not produce a hybrid layer and strong bonds to dentin, but do provide an excellent seal. The long-term effect on crown survival with use of self-etching cements is not known.

Leucite-reinforced ceramic systems similar to IPS Empress (Optec HSP, Authentic) have appeared on the market. Optec HSP was a spectacularly unsuccessful product, which illustrates another important principle: *Every ceramic system, even if similar to another successful system, must generate its own clinical data to support use of the system.*

In the early 1990s, Vita introduced the In-Ceram system, utilizing slip-casting with infiltrated glass. Three different forms exist: In-Ceram Spinell, In-Ceram Alumina, and In-Ceram Zirconia. Spinell is the most translucent but has not performed well in clinical trials. Zirconia is the least translucent and has the same translucency as metal; it therefore has limited esthetic potential. In-Ceram Alumina has decent esthetic potential and has been used successfully for single crowns. These systems have not captured a major share of the market to date.

Nobel Biocare introduced Procera Allceram to the market in the 1990s. This system used a milled alumina or zirconia core veneered with low fusing ceramic. Crowns made with this system had inferior marginal integrity and a high incidence of chipping of the veneering ceramic because of lack of support of the ceramic by the core.

The next major system to appear on the market was IPS Empress II. It was introduced as a lithium disilicate reinforced material that was so strong it could be predictably used to fabricate three-unit fixed partial dentures. In this regard, IPS Empress II was a complete disaster. One local dental laboratory (Drake Dental Lab, Charlotte, NC) fabricated 360 IPS Empress II fixed partial dentures using Ivoclar's specification for connector height, and at 3 years, the failure rate due to fracture was 75% (personal communication; David Avery, Drake Dental Laboratory).

It should be noted that IPS Empress II was a lithium disilicate material and was not really related to the original IPS Empress, which was leucite reinforced. Part of the problem with IPS Empress II was that the veneering ceramic was not properly matched to the lithium disilicate core. This was corrected in a later version called IPS Eris. IPS Eris has shown promise for single crowns but should *not* be used for fabricating all-ceramic fixed partial dentures. IPS Eris has largely been made irrelevant with the introduction of IPS e.max, which is an optimized lithium disilicate material.

IPS e.max is a lithium disilicate material that was introduced in 2009 (Fig. 13.71). It can be pressed or machined and can be used as a monolithic material or layered. Monolithic IPS e.max has relatively high flexural strength of 360 to 400 MPa compared to IPS Empress, which has a flexural strength of approximately 160 MPa. Layered IPS e.max has approximately half the flexural strength as the monolithic form. No long-term clinical trials have been published at this point, but information gleaned from remake data from three commercial laboratories and over 21,000 IPS e.max monolithic and layered single crowns indicates the fracture rate at about 4 years is only 1%. As would be expected, layered IPS e.max crowns have better esthetic potential than monolithic crowns, but the monolithic crowns are stronger.

It should again be noted that higher strength does not automatically result in improved clinical performance, primarily because all-ceramic crowns do not fail because they are not strong enough. They fail because it is difficult to fabricate them without



• **Fig. 13.71** Lithium disilicate (IPS e.max) ceramic fixed dental prosthesis.



• **Fig. 13.72** Monolithic full contour zirconia restoration.

incorporating microscopic defects called Griffith flaws, and it is the propagation of these defects that results in catastrophic failure.

In the late 1990s, zirconia-based materials were introduced to the profession. The original restorations were cored with yttria-stabilized zirconia oxide and veneered with a compatible ceramic. Zirconia is a very strong material with flexural strengths as high as 1400 MPa. In clinical trials with these materials, core fracture was extremely rare even with posterior fixed partial dentures. However, the "Achilles heel" with these layered restorations was chip fracture of the veneer. The fractures did not occur at the interface between the core and the veneer, but rather were cohesive fractures in the body of the ceramic veneer. One clinical trial of over 900 crowns estimated that layered zirconia crowns fractured at a rate five times higher than that of metal-ceramic crowns.

There are two potential causes for this high incidence of veneer fracture. Because the core is computer designed and manufactured from a scan of the tooth preparation, it is highly likely that adequate support of the veneering porcelain is not provided by the core. This problem may have been solved by creating software that produces a virtual full-contour wax-up and then cuts that back to provide a core with optimal support. The second cause is the extremely poor thermal conductivity of the core material. Basically the core cools very slowly on removal from the porcelain oven and the veneering ceramic cools past its glass transition temperature while the core slowly continues to cool. This creates stress in the veneering ceramic, which causes it to fracture under occlusal loads. This problem may have been solved by using prolonged cooling times.

The reason core fracture is so rare with zirconia is that zirconia possesses a property called transformation toughening. When a defect begins to propagate, the material surrounding the defect undergoes a transformation from the tetragonal to the monoclinic phase, and this transformation is accompanied by a slight expansion, which effectively pinches off the defect and prevents it from propagating.

A relatively new zirconia restoration has been introduced in recent years. It is called monolithic zirconia or full contour zirconia (Fig. 13.72). This is basically a core of zirconia milled into a full crown and then either glazed or polished. Based on studies of layered zirconia, where core fracture is extremely rare, these restorations may reasonably be expected to have high survival rates. One major advantage of monolithic zirconia is that very conservative tooth preparations can be used because of the strength of the material. On molars, preparations similar to those used for cast gold crowns will suffice.

Full contour zirconia restorations are also relatively inexpensive. This is partially because they are designed and manufactured digitally. In an era where the price of gold is over or close to $1600/ounce, this low cost benefit is a real advantage.

There is no question that the esthetic results achieved with these restorations is compromised, but they are acceptable for molars and perhaps mandibular incisors. Research has shown that the best surface for full contour zirconia crowns is a polished surface. This polished surface is relatively kind to opposing enamel.

Recall that the preceding information is not meant to be either all-inclusive or a detailed description of available ceramic systems. Rather it is an overview of how these systems evolved with a brief description of the important features of the major players. More detailed descriptions will be given in subsequent chapters.

THE BOTTOM LINE

1. Many different ceramic systems have been introduced to the dental profession over the past 30 years.
2. All have been brought to market with little or no clinical testing.
3. Ceramic materials do not fracture because of a lack of strength. They fracture due to defect propagation and fatigue.
4. Because of #3, physical properties testing and load-to-failure studies yield little useful information on potential clinical survival rates.
5. Clinical trials or fatigue testing under water are the only appropriate means of predicting clinical survival of ceramic crowns.
6. Most of the ceramic systems introduced since 1980 have been clinical failures.
7. The current cost of gold is a major factor in the cost of restoring teeth with metal-ceramics.
8. Many all-ceramic systems, especially full contour zirconia, are cost-effective compared to PFM.
9. Monolithic crowns have good strength but relatively poor esthetics.
10. Layered restorations have good esthetic potential but reduced strength.
11. Current available systems that clinicians should consider are:
    a. Metal-ceramics: Reasonable esthetics, longest life span
    b. IPS Empress: Esthetics, anterior teeth only
    c. IPS e.max: Esthetic potential, potential longevity
    d. Layered and full contour zirconia: Potential longevity, cost

Contemporary Ceramic Materials

Since the early 1980s, many different ceramic systems have been brought to market. None of the systems had advanced clinical testing before coming to market and most systems failed primarily due to premature catastrophic fracture (Fig. 13.73). The driving force behind these new systems was twofold: Many clinicians were disappointed with the esthetic results they obtained with metal-ceramic restorations and hoped to obtain better esthetic results by using metal-free restorations. With some systems, improved esthetic results were obtained, but the price that was paid significantly reduced longevity. The second driving factor was the cost of noble metals. In 2016, the cost of gold ranged from $1200 to $1600 an ounce. This resulted in skyrocketing laboratory costs in a market that was not favorable to raising compensatory fees. Some of the newer systems offer significant savings in laboratory costs, partially due to nonuse of metal but also because many of these restorations can be fabricated digitally thus reducing labor costs.

There are a number of reasons why dentists fail to obtain predictable esthetics with metal-ceramic restorations and most of them are operator related. Most dentists tend to underprepare teeth that are to be restored with metal-ceramic crowns, and do not provide the laboratory technician sufficient room for the metal substructure, opaque layer and body, and incisal porcelain. Many use outdated cervical margin designs, and soft tissue management is generally poor, resulting in a high incidence of inadequate impressions.[110,111] Many dentists opt to send their laboratory work overseas and accept low-quality restorations in the name of cost reduction.

It is clear that with proper tooth preparation and soft tissue management, coupled with working with a decent ceramist, very esthetic restorations can be fabricated using metal-ceramics. Metal-ceramic restorations remain the gold standard for esthetic indirect crown systems because they provide a combination of reasonable esthetics with maximum longevity. Data from a number of clinical trials indicate that 95% of metal-ceramic restorations are in service 10 to 15 years after placement.[101]

As mentioned, early ceramic systems suffered from premature catastrophic fracture. This led Schärer and McLean to postulate that before a clinician uses a new ceramic system, he or she should be provided data from independent clinical trials that demonstrate a survival rate of no less than 95% at 5 years.[112] This demonstrates that they are not holding ceramic restorations to the same survival rate as metal-ceramic restorations, but they expect a reasonable survival rate for the economic survival of the restorative dentist.

It is reasonable for a clinician to adhere to Schärer's criteria, but the truth is that in order to publish a properly conducted 5-year clinical trial, it requires almost 10 years (Fig. 13.74)!

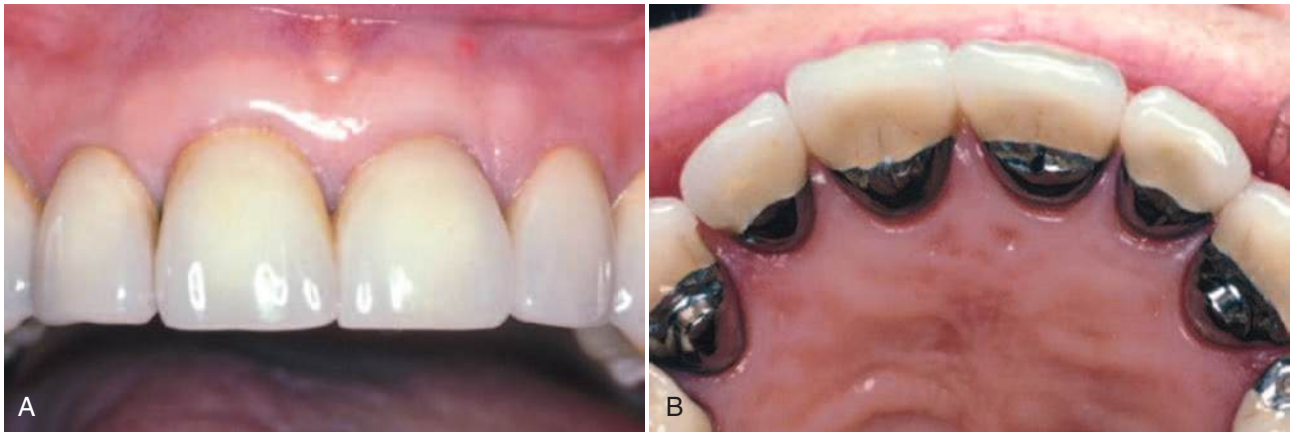
Thus even though manufacturers are diligently conducting such trials, information on investigated materials does not get into the hands of practicing dentists until many years after the materials are brought to market. An interesting approach to bring clinically relevant data to the market in a timelier manner will be described later in this section.

There are three major materials that have captured a significant portion of the ceramic market in 2017 that have not had properly conducted clinical trials published before they were widely used clinically. These three materials include leucite-reinforced IPS Empress (Ivoclar), lithium disilicate–reinforced IPS e.max (Ivoclar), and zirconia-based systems (many manufacturers). All of these materials can be used in the monolithic state or layered. Zirconia materials can be used layered in the partially stabilized state and monolithic in both the partially stabilized and fully stabilized state.

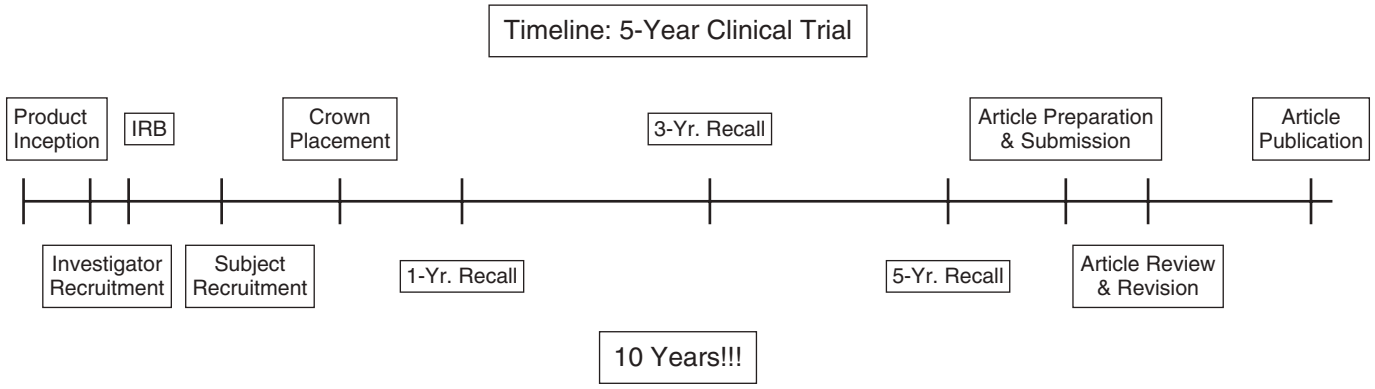
Monolithic ceramic restorations have relatively poor esthetic results but seem to provide longer clinical survival rates. With monolithic ceramic restorations, all of the color is on the surface, resulting in high surface reflectivity and specular reflection, a high incidence of metamerism, and no depth effect. Monolithic crowns are *not* suitable for maxillary anterior teeth. Layered restorations provide significantly improved esthetic results but lower survival rates.

The leucite-reinforced IPS Empress system was really the first successful ceramic system on the market appearing after an imposing list of failed systems. Layered IPS Empress should only be used on anterior teeth as clinical trials have demonstrated high failure rates on posterior teeth.[113] Monolithic Empress should not be used. IPS Empress can be pressed or milled, and the usual procedure is to press a leucite core and veneer it with the matched ceramic. Layered IPS Empress crowns must be etched with hydrofluoric acid and bonded in place with either dual-cure resin cements/bonding or self-adhesive resin cements. Fewer and fewer dental laboratories are fabricating these restorations today, and even though the esthetic results are excellent and survival on maxillary anterior teeth is satisfactory, their use in contemporary clinical dentistry is limited.

The material named IPS e.max is a lithium disilicate reinforced ceramic that can be used both in the monolithic state and layered.



• **Fig. 13.73** A and B, Metal-ceramic restorations after 21 years.



• **Fig. 13.74** Timeline outlining time required for publishing properly conducted 5-year clinical trial.

TABLE 13.12 Mechanical Properties of Leucite Reinforced, Lithium Disilicate, and Zirconia

Material	Flexural Strength (MPa)	Elastic Modulus (GPa)	Hardness (VHN)	Reference
Leucite reinforced glass ceramic (Empress)	160	96	550	(Aboushelib et al., 2007a)
Lithium disilicate (IPS e.max)	400	95	5800	(Ivoclar, Vivadent)
Zirconia (Y-TZP)	800–1500	210	1200	(Piconi and Maccauro, 1999)

TABLE 13.13 Thermal Conductivity of Zirconia Compared to Other Material

Material	Thermal Conductivity W/(mK)
Silver	429
Gold	318
Gold alloy	200
Base metal	40
Alumina	30
Zirconia	2

Most of what is known about the clinical performance with IPS e.max is with monolithic restorations. Monolithic e.max has a significantly higher flexural strength than IPS Empress at 400 MPa versus 160 MPa (Table 13.12). The flexural strength of layered e.max is about half of the monolithic form, which has been validated with both fatigue studies and clinical data.[114,115] It is important to remember that ceramic crowns fail by defect propagation, and it is likely that the clinical success of e.max restorations is a result of incorporating fewer inherent defects through pressing and milling the restorations. As impressive as the gain in flexural strength of monolithic e.max versus IPS Empress is, it pales in comparison to the flexural strength of monolithic zirconia, which has a flexural strength of 1500 MPa versus 400 MPa for e.max.

Zirconia restorations were first introduced to the profession as partially stabilized (yttria) zirconia oxide cores layered with a veneering ceramic. Initial clinical data established that the cores rarely fractured, but there was a high incidence of chip fractures of the veneering ceramic.[116] It was estimated that the incidence of veneer fracture with layered zirconia was five times that of metal-ceramic restorations. The low incidence of core fracture could be attributed to the property of transformation toughening (Fig. 13.75).

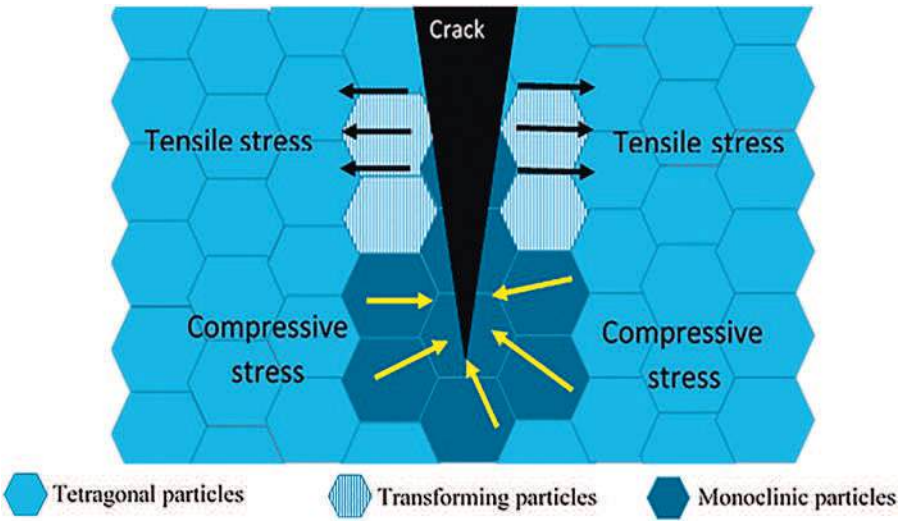
When a defect begins to propagate in a zirconia core, the materials immediately surrounding the defect change from the tetragonal to the monoclinic form of zirconia. This results in a slight expansion, which effectively "pinches off" the propagating defect.

Veneer chipping is likely caused by two issues. First, the high strength cores were originally fabricated by scanning the die of the tooth preparation and then a core of uniform thickness was milled. This core would follow the shape of the die and does not in any way account for the thickness of the veneering ceramic. The industry standard for support of the veneering ceramic is that the veneering ceramic should never exceed a thickness of 2 mm. New software that first produces a virtual full-contour image of the anticipated final restoration and then cuts back that restoration, to provide a core with optimum support of the veneering ceramic, should solve the support problem.

The second issue with veneer chipping is the thermal conductivity of zirconia. It is very low, perhaps 100 to 200 times lower than the conductivity of metals used to bond to porcelain (Table 13.13).

The net result of this is, as the layered ceramic unit exits the porcelain oven, the veneering ceramic cools first, below its glass transition temperature while the core very slowly continues to cool and shrink. This creates stress in the veneering ceramic, which is expressed clinically some time later by chip fracture. Slower cooling protocols have been introduced for firing layered zirconia restorations, and it is anticipated this will significantly reduce the incidence of chipping.[8]

One other solution to veneer chipping is being used by many dental laboratories. There is a substantial volume of anecdotal evidence that veneer chipping with Noritake veneering ceramic (Kuraray America, Inc.) is almost nonexistent. Clinicians would be wise to specify use of Noritake veneering ceramic when prescribing layered zirconia restorations.



• **Fig. 13.75** Schematic illustration of the transformation toughening phenomenon.

A relatively new form of zirconia has become extremely popular in recent years despite a complete lack of clinical data on survival. Monolithic partially stabilized zirconia (also called full contour zirconia) has grabbed a huge share of the world market for indirect esthetic crowns. The rationale for this restoration is that in clinical trials of layered zirconia restorations core fracture is a relatively rare occurrence. A monolithic zirconia crown is essentially a core with a glazed or polished surface, and thus is expected to display good survival rates.

One major advantage of partially stabilized monolithic zirconia is that crowns can be fabricated with significantly more conservative tooth preparations compared to those needed for metal-ceramic and lithium disilicate crowns. The exact tooth reduction required is unknown at this time, but a 1-mm occlusal reduction with axial reduction of 0.75 mm and a well-defined chamfer margin seem appropriate. A second advantage is that numerous fatigue wear studies indicate that polished zirconia is extremely kind to opposing tooth structure. A third and apparently compelling advantage of these restorations is that they are very inexpensive.

One significant disadvantage related to partially stabilized monolithic zirconia is that it is relatively opaque. The color can be improved using glazes and external colorants, but the overall esthetic result is mediocre at best (Fig. 13.76).

The esthetic results are likely acceptable in the molar regions for all but the most demanding of patients, but they are likely contraindicated for routine use in the maxillary anterior area.

The relative opacity has resulted in the introduction of a new zirconia-based system: fully stabilized monolithic zirconia (Bruxzir Anterior, Glidewell Laboratory). These materials consist of zirconia oxide fully stabilized with yttria (8–12 wt%). They also contain cubic zirconia of up to 53% and are slightly less opaque than partially stabilized zirconia. This improvement in opacity is likely not within the realm of human perception and is accompanied by a 50% reduction in flexural strength. Clinicians are advised to adopt this new technology with extreme caution. There is no clinical track record, improvements in translucency are minimal, and the reduction in flexural strength is significant.

In the absence of properly conducted clinical trials on the previously discussed ceramic systems, another approach has been used to generate timely clinical data on the survival of these restorations. Dental laboratories often warranty their crowns for 5 years and have meticulous data on remakes with these new ceramic systems. Assuming that most patients would return to their original dentists when experiencing premature failure with a ceramic crown to take advantage of the warranty, and assuming the dentists will return that restoration for remake to the same laboratory, examination of remake data provides insights of short-term to medium-term performance of large numbers of ceramic crowns.



• **Fig. 13.76** Monolithic zirconia fixed dental prosthesis. (Case courtesy Dr. Vilhelm G. Ólafsson.)

This "laboratory survey" approach has provided 4-year data on over 21,000 lithium disilicate (e.max) restorations and 5-year data for over 71,000 monolithic and layered zirconia (partially stabilized) restorations.[117,118] The surveys found all three types of restoration had survival rates that would easily meet and exceed the Schärer criteria. Monolithic e.max single crowns fractured at a rate of 0.91% at 4 years, with layered e.max single crowns exhibiting double the fracture rate at 1.81%, which is still well below Schärer's criteria. Monolithic zirconia single crowns had an even lower rate of fracture (0.71%) at 5 years. Layered zirconia single crowns had a four to five times higher fracture rate at 3.24% at 5 years. These restorations were fabricated prior to initiation of the new protocols expected to reduce the chip fracture rate of layered zirconia restorations. One other result of note was the fracture rate of e.max inlays/onlays, which was 1.01% at 4 years.

A note on bonding zirconia. Zirconia cores cannot be effectively etched with any known acid. Zirconia restorations are

best used with tooth preparations with adequate resistance and retention form, and should be cemented with resin-modified glass ionomer or self-adhesive resin cement. A number of different protocols have been developed for "bonding" zirconia restorations. All of these protocols involve air particle abrasion, use of a primer with MDP monomer and silane, and use of dual-cure resin cement. The bonds achieved with these protocols are initially weak and get weaker with artificial aging. There is a definite risk in use of air particle abrasion with zirconia restorations as some studies have demonstrated significant conversion of the core to the monoclinic form and substantial weakening after air particle abrasion.

Zirconia seems to have a strong affinity for proteins found in saliva and blood. It is likely the internal portion of most zirconia crowns will be contaminated during try-in. These proteins cannot be removed with phosphoric acid. A routine procedure for cementing zirconia crowns is the use of a sodium hydroxide solution (Ivoclean, Ivoclar) for 20 seconds after try-in before cementation to effectively remove these proteins.

Indications

Clearly metal-ceramic restorations can continue to be recommended for indirect esthetic crown restorations. However, the promising short-term results demonstrated with lithium disilicate and zirconia based systems permit their rational use in specific clinical situations.

It would seem reasonable to consider use of monolithic partially stabilized zirconia crowns for a molar tooth, in part due to their substantially reduced costs, and also because a more conservative tooth preparation can be used. These restorations can be recommended for mandibular anterior teeth because they are usually not highly visible and a more conservative tooth preparation can be used.

Monolithic lithium disilicate crowns can be recommended for routine use on premolar teeth due to their excellent survival rates, adequate esthetic results, and reduced costs. Layered lithium disilicate crowns can be recommended for maxillary anterior teeth. Fully stabilized zirconia crowns should not be used at this time until clinical information is available on clinical survival.

THE BOTTOM LINE

1. Metal-ceramic restorations continue to provide an impressive combination of reasonable esthetic results coupled with excellent clinical longevity.
2. Most ceramic systems introduced to the market prior to 1988 were clinical failures.
3. Ceramic systems should be documented with independent clinical trials that demonstrate 95% survival at 5 years.
4. It takes about 10 years from product inception to publication of a 5-year clinical trial.
5. The "laboratory survey" approach is a good one for obtaining clinical short-term survival rates.
6. Monolithic zirconia and lithium disilicate materials have adequate short-term survival rates.
7. Early results with layered zirconia showed unacceptable levels of chip fractures of the veneering ceramic. These problems may be solved with new approaches for support of the veneer and new firing cycles.
8. Fully stabilized zirconia crowns marketed for anterior applications have minor improvements in translucency and significantly reduced flexural strength.
9. It is not practical with current protocols to "bond" zirconia restorations. Zirconia restorations require traditional resistance and retention for and should be cemented with RMGI or self-adhesive resin cement.

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